Acronyms

AMGA: Annotated Model Grant Agreement
B&C: Building & Construction
B2B: Business-to-Business
B2C: Business-to-Consumer
CE: Circular Economy
CFP: CIRCULAR FoodPack
D4R: Design4Recycling
DSS: Decision Support Tool
eBeam: electron Beam
EAB: External Advisory Board
EFSA: European Food Safety Authority
ELV: End of life vehicles
EoL: End of Life
EVOH: Ethylene Vinyl Alcohol
FCM: Food Contact Material
(FGO) PCRs: Food grade origin (FGO)
GDPR: General Data Protection Regulation
HDPE: High Density Poly(ethylene)
HPC: Home and Personal Care
HSI: Hyperspectral imaging
IAS: Intentionally added substances
IP(R): Intellectual Property (Rights)
IRD: Infrared Dryer
KPI: Key Performance Indicator
LCA: Life Cycle Analysis
s-LCA: social LCA
LCC: Life Cycle Cost Assessment
LCSA: Life Cycle Sustainability Assessment
LIBS: Laser-induced breakdown spectroscopy
ML: Multi-Layer
MLC: Multi-Layer Composite
MLP: Multi-Layer Packaging
MS: Milestones
NIAS: Non-intentionally added substances
NIR | MIR: Near-infrared and mid-infrared
OTR: Oxygen Transmission Rate
PA: Polyamide
pc: post-consumer
PCR: Post-consumer Recyclate
PDEC: Plan for Dissemination and Exploitation of results and Communication activities
PE: Polyethylene
PE PCR: Polyethylene Post-consumer Recyclate
PET: Polyethylene terephthalate
pi: post-industrial
PO: Polyolefin, cnsisting of only PP and PE.
PP: Polypropylene
PVD: Physical Vapour Deposition
r: prefix for Recycle
RTO: Research & Technology Organisation
SbD: Safe by Design
SBS: Sensor-Based Specification
SME: Small & Medium-sized Enterprises
SO: Specific Objective, SOA: State-of-the-Art
SRM: Secondary Raw Material
SSH: Socio-economic Science and Humanities
TBS: Tracer-Based Sorting
WEEE: Waste Electrical and Electronic Equipment
WVTR: Water Vapour Transmission Rate

1. Excellence

Plastic food packaging solutions are of high value to the consumer market, as their set of intrinsic properties (food protection, light weight, ease of processing, cost, carbon footprint, information exchange efficiency, and functional excellence) allows them to outperform any alternative material in this market, leading to a plastic share of 64% in the European packaging market[1]. In flexible packaging applications (70% of plastic packaging in Europe), the use of plastic-plastic multi-layer composites (MLC) grew significantly over the last decades due to their outstanding functional excellence (oxygen, moisture and light barrier, printability, food compliance, etc.). As these properties are of special importance for food preservation, 87% of MLC packaging films are applied in the food packaging sector. However, at the end-of-life (EoL), the various layers of which MLC materials are composed of, cannot be separated by industrial recycling technologies and hamper any material recycling from waste fractions that contain MLC. Whereas the European multi-layer films industry produces almost 2 Mio tonnes of food packaging annually[2], waste fractions affected by the presence of MLC make up a majority of the European post-consumer (pc) flexible plastic packaging waste, that accounted for 14.3 Mio tonnes in 2018 and ~49% of the total collected European plastic waste.[3] Consequently, the current **recycling rate** of plastics from pc **food packaging waste[4] is very low,** partly because the **quality of recycled polymers is insufficient** for high-level applications, e.g. in flexible packaging.

Therefore, European EoL practices for plastic food packaging waste are incineration and/or landfilling (in total > 58%), which counteracts the circular economy concept. In reaction, the **European strategy for plastics in a circular economy**[5, 6] calls for (i) improvements of products' design for recyclability as well as collection, sorting and recycling technologies and (ii) boosting the demand for recycled plastics. The **use of pc recycled (PCR) plastics in food packaging solutions is, however, a critical issue,** which is strictly regulated by EC Regulation No 282/2008[7]. Due to consumer and health safety aspects, PCR application in food packaging is restricted to waste plastics coming from only food applications. This practically rules out any of today's PCR, because the state of the art sorting technology (Near Infrared (NIR)), cannot differentiate between food and non-food packaging. This deficit is one major dilemma of the circular economy for plastics.

1 All4Pack: Market and challenges in 2016.
2 GVM Flexible Plastic Packaging Market in Germany and in Europe, 2016.
3 Plastics – The Facts 2019, An Analysis of European plastics production, demand and waste data, Plastics Europe.
4 Post-consumer (pc): Material generated by households or by commercial, industrial or institutional facilities in their role as the end users of the product, which can no longer be used for its intended purpose. This includes returns of material from the distribution chain. CEFLEX Guidelines.
5 The European Commission, Strategy for Plastics in a Circular Economy, 2018
6 New Plastics Economy, Ellen MacArthur Foundation, 2016
7 https://eur-lex.europa.eu/legal-content/en/ALL/?uri=CELEX%3A32008R0282

1.1. Objectives

CIRCULAR FoodPack (CFP) enables the transition of plastic food packaging solutions from a linear economic model to a closed loop Circular Economy by targeting some of the most critical enablers in the value chain through technological innovations: (i) Selective tracer-based **sorting** of food packaging items from bulk end-of-life plastics of industrial and municipal waste keeps those packaging items in the circle that are considered to be recyclable and distinguishes safely between food and non-food packaging; **(ii) effective separation and purification of plastics** by innovative **pre-treatment and recycling** processes to enable a circular economy of post-consumer recyclate plastics at a very high quality level; **(iii)** the consortium partners will perform **food contact compliance tests** with recycled polymers as well as migration modelling to validate innovative functional barrier concepts; **(iv)** the development of high-barrier mono-material packaging designs that ensure functionality (gas and migration barrier) and recyclability, enhancing effects of (i) and (ii), supported by a comprehensive sustainability assessment considering environmental, economic and social impacts, as well as circularity aspects, food waste and compliance to health and safety legislation and a target-oriented consumer communication by considering gender related issues. Based on this feedback and iterative technological improvements, partners will finally provide new **food packaging designs including CIRCULAR FoodPack Post-Consumer Recyclates (PCR) with a validated and recyclable functional barrier**. The same value chain will apply to **current sorting fractions** which are not tracer-marked and cannot be separated into food and non-food fractions. Thus, premium quality home and personal care instead of food packaging applications will be targeted for this input due to legal constrains of EU Regulation (EU) No 282/2008.

In Europe the total estimated market in **consumer households** flexible packaging market is about 3.9 Mio tonnes per year, of which around 70% (3 Mio tonnes) is made up of mono-polyethylene (PE), mono-polypropylene (PP), or combination of both PE/PP.[8] Although the ability to sort and mechanically recycle these polyolefin based[9] materials at industrial scale has already been proven, the processes and infrastructure required to collect, sort and recycle pc-flexible packaging are not yet widely established across the European continent. Moreover, the quality of the recyclates is not yet sufficent for their implementation in flexible food or premium quality home and personal care packaging for circular use due to color, odour, contaminants and low mechanical performance. The remaining 30% of pc-flexible packaging includes MLC structures consisting of paper, aluminum, polyethylene terephthalate (PET), polyamide (PA), or ethylene vinyl alcohol (EVOH), and many of these MLCs also consist of a polyolefin layer in addition, at about 35 to 40%, mainly for sealing purpose. As such, PE, mainly linear (LLDPE) and low density (LDPE) is among the most widely used polymers, followed by PP in the packaging sector[10], both as mono-materials and also in MLC. Currently, it is not possible to recycle these at the purification levels and yields, required for their return to economy. CIRCULAR FoodPack proposes an ambitious, multi-disciplinary and comprehensive technical approach to make the recycling of MLC and mono-material flexible packaging industrially feasible, and to deal in a reliable way with their sorting, designing and reuse. PE, being light and versatile, is the first choice material in the manufacture of films today. Hence it is the most reasonable feedstock to work within the project and the foreseen design will be effectively recycled. PP, the second most used polymer is not even collected at all in some countries. This might change in the future, but that is still uncertain. Input packaging waste will cover post-consumer and Business-to-Business (B2B) return food packagings, in order to maximize the flexible packaging waste recovery outcome and to ensure that the sustainability and circularity is fully realized (Fig 1). One of the long-term goals will be to adapt the project technologies to other packaging waste streams in the future that are currently not recycled or with low value recycling, such as PET-trays or PP films. All technologies applied in this proposal will be subjected to a dedicated comprehensive sustainability assessment, taking the **entire lifecycle** into account. First, the economic added value will be evaluated, e.g. by analyzing the impact in terms of change in employment, or economic costs with life cycle costs (**LCC**). Secondly, a systematic analysis of environmental gains is performed, based on a life cycle approach (**LCA**), encompassing the full value chain. Finally, the contribution of the technologies in function of setting up circular economy material flows will be evaluated. Social LCA and market analysis to identify business plans and marketing strategies will further assist the developments by surveying stakeholders' interests in order to include such findings in a suitable exploitation strategy. To achieve these ambitious goals and to maximize the impact on the upcoming Circular Economy, the following specific objectives (**SO**) were defined:

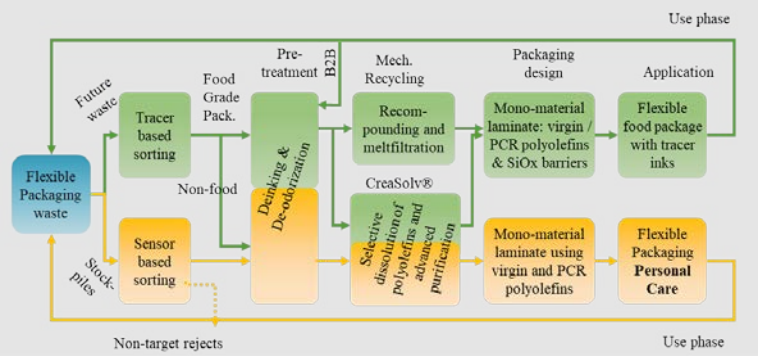

Fig 1. CIRCULAR FoodPack Concept with technological innovations

8 CEFLEX Technical report, designing for a circular economy, June 2020
9 Polyolefin: Group of polymer thermoplastics consisting of only PP and PE. Htttp:/www.plasticsrecyclers.eu/glossary
10 Plastic Recyclers Eruope, Flexible Polyethylene Recycling in EU

SO1. To develop and apply a food compliant tracer portfolio and to design efficient Tracer Based Sorting (TBS) devices that enable the automatic sorting of pure food packaging plastics from pc-waste streams.

Within CIRCULAR FoodPack (CFP), TBS based on downconversion fluorescence in the NIR will be applied, in order to enable the use of existing NIR sorter with minimal or no adaptions. Likewise food grade materials, which have higher chemical safety specifications than other plastics, are marked with an innovative fluorescent tracer system, which allows subsequent sorting out from the general waste stream, reaching a purity of **> 99%** for the traced fractions. This is considered as the first crucial step to facilitate circularity, because such food-grade plastic waste fractions fulfil sorting and collection requirements of EC Regulation (EU) No 282/2008. In the course of the project, current sorting fractions, that have not been marked by tracers yet, will be characerterized using novel photonic and hyper spectral measurement technologies (Sensor-Based-Specification SBS) to select such packaging waste fractions for downstream recycling that a contain multi-layers and a high share of PE. Furthermore, the potential to identify (and later sort out) materials that hamper recycling or process economy (e.g. low share of target polymers or non-deinkable printing) will be studied. The target will be to extract **>90%** PE-rich flexible films (incl. PE multi-layers). Both TBS and SBS approaches for post-consumer waste will also be benchmarked with Business-to-Business (B2B) return logistics of food packing with >99% purity from professional customers (e.g. large scale coffee packs).

SO2. To develop pre-treatment processes that enable improved purification of sorted pc flexible packaging and adapt the solvent-based CreaSolv® recycling process to achieve certified purification levels of recyclates.

The technological concept includes waste treatment and purification processes that enable surface cleaning and removal of volatile waste components: **(i)** Reactor technologies that are suitable for liquids to remove product residues, inks and lacquers, for improvement of washing processes for an effective deinking, **(ii)** Moderate thermal treatments (below 100°C) combined with temperature coupling to remove volatile contaminants and to improve the materials circularity by an effective deodourization. In addition. advanced purification will be performed by the solvent-based CreaSolv® technology. It dissolves target polymers (in CFP: PE) from pre-treated inputs selectively, separates other constituents of MLC and applies a patented purification process that will further reduce contaminants (ink and glue residues, additives). Inputs with varying degrees of pre-treatment by prior deinking and deodourization will be tested to identify the best purifying and most cost-effective combination of processes that allows a high overall purification and re-application of PE PCRs in high level applications as personal care packaging and new food packaging items. Combining and iteratively improving the specific cleaning operations will reduce the amount of **IAS and NIAS by > 99%** (proven by analytical screening).

SO3. To assess, validate and ensure food compliance of newly developed packaging throughout the project.

It is intended to re-use PCR polymers from the above-mentioned sorting, purification and recycling processes in new food packaging concepts by adding an effective **migration barrier**. Based on experimental **migration tests with model films** containing barriers and based on related **migration modelling,** maximum acceptable levels of NIAS and IAS in recycled polymers will be assessed and controlled in products from several process concepts and process cascades. Feedback to process developers will iteratively improve the quality level of recycled polymers and IAS and NIAS will reach levels **below half of the calculated maximum acceptable levels**.

SO4. To redesign multi-material flexible packaging to mono-materials "Design for circular food packaging".

At least **6 innovative packaging demonstrators** are to be designed, which will be (i) **completely recyclable**[11] due to the use of just one major basic polyolefin polymer type, PE, (ii) **functional** (adequate food-protection and prevention of food waste) with approved positive packaging tests for damage resistance, sealability, machinability, gas and water vapor barrier and adequate shelf life analysis with selected sensitive dry food types. Consumer appeal Key Performance Indicators (KPIs) will also be tested at standard level including checks of print quality, package gloss, thickness, modulus, sealability, (iii) **food safe**. Food contact will be enabled by an effective, consumer-safe and recyclable **migration barrier** based on a very thin, dense inorganic layer, e.g. silicon oxide (SiO_x, ~60 nm), optionally with a thin organic layer. This will allow the use of **> 35%** PCR polymers initially and **> 50%** later on.

SO5. To scale-up new mechanical and solvent-based recycling processes to pre-commercial stage and to demonstrate and validate new packaging development.

After successful elaboration and testing of sorting and recycling technologies at TRL 3-4, processes will be up-scaled and implemented in process cascades at TRL 5-6. That allows the production of sufficient and representative samples. These are required to test-wise incorporate recycled polymers (PE PCRs) in **6** new packaging demonstrators with 3 case-studies, in the field of food, home and personal care packaging: 3 food packaging demonstrators from TBS sorted waste, 1 personal and 2 home care packaging demonstrators using current sorting concepts governed by SBS.

Recyclability of new packages will be experimentally validated at **TRL 5-6** by use of the CFP value chain.

SO6. To assess the comprehensive sustainability of the newly developed value chains (with new technology and packaging design), including environmental, economic and social impacts, as well as circularity aspects, food waste and compliance to health and safety legislation.

The newly developed technologies and packaging will be subjected to a comprehensive Life Cycle Sustainability

11 AAA and AAA+ ratings of the HTP Cyclos approach. These ratings say, the packaging can be recycled completely, but the quota refers to the PE share only.

Assessment (LCSA), including environmental (LCA), economic (LCC) and social impacts (s-LCA), as well as considering circularity aspects, food waste and compliance to health and safety legislation. The assessment will account for the quality and market size of recyclates in order to reveal the benefits of recycling with high-end purposes compared to downcycling. The sustainability assessment will be executed in an iterative 2-stage way in order to give and receive intermediate feedback to/from the technology and packaging developers and to ensure that the new value chains perform at least 30% better in terms of GHG emissions, 32.5% better in terms of energy efficiency, compared to baseline scenarios (open loop recycling or incineration). Finally, to support waste managers and processors running recovery processes, a digital Decision Support System (DSS) tool for sustainability optimisation and evaluation of the most feasible process cascade and sector of application for a certain secondary material will be developed.

SO7. To guide and route commercializing of the project results to be prepared for a short term market implementation of successful developments

CFP targets to provide technical solutions for tomorrow's food packaging market, which has claimed substantial changes by 2025, **including complete recyclability of food packaging**, and requested short-term developments. CFP solutions are widespread over the fields of waste sorting, plastic recycling and packaging developments and are considered to exhibit important linkages. Each technology itself (tracer based sorting, deinking, deodourization, solvent based purification) and most suitable combinations thereof will be challenged by industrial project partners, experts in the advisory board and by experts' interviews **to provide and iteratively improve economic data and business plans** in order to facilitate industrial implementation. Technology providers, industrial partners and multi-stage interviews with packaging experts and consumers will define most suitable stakeholder-driven business models for a fast implementation and prepare strategies for a streamlined and stepwise market entry (e.g. defining a B2B circular model for packaging demonstrators; defining test markets/regions for tracer branding and tracer based sorting concepts to be aligned with downstream recycling; etc.). Stakeholder involvement and gender related marketing strategies will be additionally taken into account, both for project dissemination and exploitation.

1.2. Relation to the work programme

CIRCULAR FoodPack answers directly to challenges and scopes mentioned in the call Topic CE-SC5-24-2020.

Table 1: How CIRCULAR FoodPack (CFP) relates to the scope and challenges of the CE-SC5-24-2020 topic

Scope	CIRCULAR FoodPack (CFP) response
Develop new or improve existing innovative processes for the **sorting, disassembly/separation, recycling and/or the introduction into manufacturing process of materials from products made of composite and/or multi-layer materials** and **assess the potential barriers** for their implementation.	SBS and TBS Sorting will be applied to existing plastic packaging waste and to newly proposed tracer marked packaging items, allowing for effective separation of food packaging waste from non-food packaging waste. Furthermore, pre-sorted materials will be purified and recycled by combining existing shredding and washing steps with innovative deinking and thermal deodourization processes as well as the solvent-based CreaSolv® Process and/or mechanical recycling (re-melting, melt filtration and re-compounding). Thus, process cascades will apply to state-of-the-art (SOA) multi-layer packaging as well as to newly developed mono-material multi-layer concepts. Recycled polymers will be part of newly developed flexible food packaging and therefore re-enter the manufacturing process. Barriers for the implementation of PCR polymers exist as special legal constraints must be met to ensure food and consumers' safety. We will show that tracer systems with effective pre-sorting and purification techniques together with effective barriers in new flexible packaging solutions will overcome such hurdles.
Deal with **used products, production rejects or existing stocks such us material recovered from industrial and municipal landfills**.	CFP addresses mainly post-consumer (pc), post-industrial (pi) waste, and also B2B waste, and especially future packaging waste, which is highly important due to current disruptive changes in the production of plastic packaging. The project includes real products with large market sizes such as flexible food, and home and personal care packaging. Most of the intended innovative processes can also be applied to existing stocks, including pre-treated plastics from landfills, and will improve quality of recycled polymers produced thereof significantly due to the targeted high level of purification. This will be exemplified in CFP by production of personal and home care packaging from existing stockpiles.
Optimize value retention in the economy, especially for **applications with high performance requirements**. … **recommendations for design**…, to enable an increase in volume and **quality of reuse** and	CFP will significantly increase purity and mechanical properties, while reducing odours and other NIAS and IAS to an acceptable minimum, generating a value retention that currently does not exist. It will enable the (i) application of recycled polymers in their original field of application (most sensitive food packaging market) at highest quality level, fulfilling circular economy requirements and meeting the legislation-driven needs of the market for food-compliant PCR and (ii) design of ink structures that are more easily deinked, thus allowing true closed loop recycling as transparent material. (iii) The sustainable design for recycling will not only be recommended, but be also implemented in demonstrators. For the upcycling

recycling …. requirements for product information … **effective identification** and **management after use.**	of PE PCR for flexible packaging, inline e-beam deposition after PCR polyolefin extrusion process will be performed. (iv) Quality specifications for recycled polymers will be driven by final food packaging application and new design concepts. (v) Recommendations for messages and, where appropriate labels, will be developed to appropriately communicate and inform consumers and waste managers, enabling effective collection, identification and management after use.
The **Environmental impact** (e.g. substitution of virgin plastics, …), social impact (e.g. related to health and safety legislation) and cost of innovative processes implemented (e.g. recycling processes) be assessed in a holistic way, taking the entire lifecycle into account.	Increasing the (holistic) sustainability of packaging is the overall goal. The comprehensive Life Cycle Sustainability Assessment (LCA, LCC ,s-LCA) will steer technical developments in order to provide technologies and products that are economically, environmentally and socially viable and bring added value in the Circular Economy. The assessment will consider the substitution of virgin plastics, thereby accounting for the quality and market size of recyclates to reveal the benefits of upcycling compared to downcycling. Also, the assessment will consider food waste generation, the compliance to health and safety legislation. Health and safety legislation will be addressed by comprehensive migration and food compliance testing of recovered materials and by adequate assessment of new technological processes.
Provide evidence of the **potential market impact** that the proposed solutions could bring, including quantitative information on current economic actors in the chain and anticipated consumer acceptance and changes of consumer attitude (taking into account gender issues, when relevant). To this end, quantitative information on the size of the targeted market is expected.	Expected targeted market is the entire plastics packaging industry, i.e. 26 million tonnes, which is approximately 40% of Europe's plastics production[12]. AMCOR is the largest flexible packaging convertor worldwide, serving all relevant markets (with a rough estimate of several million tonnes of flexible food packaging in the market), while NESTLE is the largest global food company. Together they have a large impact on the current and future European consumer markets. Both AMCOR and NESTLE made the commitment to have their packaging portfolio being fully recyclable or reusable by 2025[13]. Similar commitments were made by other food market players. If NESTLE will use the new laminate structures for its products, it may have a global impact on future laminate market. Gender aspects will be considered among others in consumer perception studies (Task 8.1), the outcome of which will inform the communication and dissemination context to ensure our message is appropriate to reach both genders. This may have a global impact on laminates used by competitors and may define the future laminate market. SUEZ is one of the most successful organizations specialized on the collection and treatment of plastics waste throughout Europe. Their widespread network of waste treatment facilities as well as a well-established intra-corporate collaboration with SUEZ R&D will support a fast implementation of successful technologies developed in the project on the one hand, and will enable an establishment of more comprehensive waste management structures by e.g. combining B2B concepts together with suitable recycling facilities or linking enhanced TBS with adapted recycling technologies. SIEGWERK will take up results from Design4Recycling (D4R) concepts and deinking studies to implement and strengthen future ink and printing concepts with improved recyclability. POLY is a leading company in the field of TBS and holds an outstanding market position to implement its physical information coding system in future logistic and waste management systems. Results of the project will assist POLY establishing the TBS concept in a more comprehensive "circularity management approach" aligning waste management and PCR based laminate production. KREYEN has developed a suitable deodourization technology, which is intended to play a substantial role in future pc packaging recycling. Thus, the market position of this green technology will be increased and, in addition, by supporting a widespread use of such technology, better PCR qualities can be expected in the future, enabling a wider field of application of well purified recycled polymers, even in the challenging market of food packaging raw materials.
Participation of **relevant industrial partners** (technology providers, end-users etc.) is considered important. Activities are expected to	10 industrial partners (5 SMEs), POLY, IRIS, SUEZ, KREYEN, SIEGWERK, ECOZEPT, BayFOR, NESTLE, AMCOR (2 locations/legal entities), enable an assessment along the whole value chain: from the world's largest food enterprise (NESTLE), to a machine manufacturer (KREYEN) specialized in bulk material handling, an ink formulator (SIEGWERK), which is one of the global market leaders for printing inks for all kinds of packagings, a packaging material producer

12 GVM Flexible Plastic Packaging Market in Germany and in Europe, 2016.
13 NESTLE (www.nestle.com/csv/global-initiatives/zero-environmental-impact/packaging-plastic-pollution) & AMCOR (2018) (www.amcor.com/sustainability/2025-pledge)

achieve TRL 5-6 by the end of the project.	(AMCOR), inline sorting system developer (POLY), waste management (SUEZ), including technology provider ECOZEPT for market analysis and identification of business opportunities and marketing strategies for market acceptance of the CFP developments in parallel with the LCSAs. BayFOR (non-profit) will support the consortium with management, exploitation, dissemination, communication. We aim to upscale several techniques with TRL3-4 towards TRL5-6, see Table 2.
This topic is in support of **EU Strategy** for **Plastics** in a **CE.** Selected projects ... encouraged to participate in activities [like] … clustering, workshops, common exploitation dissemination etc. Applicants should plan … budget to cover those activities without concrete common actions at this stage.	We directly contribute to EU Plastics Strategy by a better design of plastic products, higher plastic waste recycling rates, and a better quality of recycled polymers. The results will be widely communicated and disseminated on various relevant events, e.g. EU Circular Economy Stakeholder Conference, World's Leading Trade Fair for Water (IFAT), Sewage, Waste and Raw Materials, Plastics Recycling Show, Europe. Partners will communicate the project on national and regional thematic events of relevance. 1.5 PM of the coordinator's budget is foreseen for clustering activities in WP8 with other projects also related to Plastics Strategy. Some consortium members are currently part of project clusters (Table 3) that participate the Plastics Circularity Multiplier Initiative with 21 projects and more than 200 partners. Fraunhofer Institute IVV is part of the Fraunhofer cluster of excellence "Circular Plastics Economy"[14].

1.3. Concept and methodology

(a) Overall concept underpinning CIRCULAR FoodPack:

The basic idea is to enable a circular economy of flexible food packaging plastics via collection and sorting of packaging waste into food and non-food packaging, based on tracer systems, followed by the separation of plastics, and subsequently a high level of valorization of food grade PCRs by means of advanced recycling processes. Food grade PCRs will undergo reprocessing and upcycling by means of consecutive film extrusion, and mono-, or bi-axial film orientation processes before entering a new production cycle of novel mono-material based packaging with new specialized **inorganic tracer materials** dispersed **in deinkable printing inks** as part of the layer of the flexible food packaging films. A thin inorganic functional barrier concept based on electron-beam deposition will prevent a migration of possibly present contaminants from the PCR based film into the filling good. If needed, inline electron-beam surface pre-treatment is to be applied for additional heat stability and topography optimisation.

CIRCULAR FoodPack concept follows three technology pillars: **(i) Sorting:** Food packaging items are marked by a tracer that enables Tracer-Based-Sorting (TBS) of food packaging plastics by use of adopted state of the art NIR sorters and/or more dedicated TBS sorting machines. In addition, rejects from TBS and current sorting fraction are characterised by SBS and selected for production non-food PCR if sufficiently high in PE content. **(ii) Separation / Recycling:** Separated waste food packaging plastics are subjected to pre-treatment and recycling processes that further purifies the target polymer PE from other materials and eliminates intentionally added substances (IAS) including inks, and NIAS that occur non-intentionally mainly as a result of degradation processes. **(iii) Design / Validation:** Recycled and cleaned polymers become part of a mono-material multi-layer packaging concept that includes a very thin (< 60 nm) recyclable barrier (e.g. SiO_x), effectively separating the PCR polymer from the food, as well as a virgin plastic layer that will be in direct food contact. End-user validation for at least three well-identified packaging use case studies will ensure the industrial approval of the developed solutions. The comprehensive sustainability assessment and the business modelling will provide iterative feedback for the right decisions and upscaling paths up to industrial validation.

Interdisciplinary aspects: Our trans-disciplinary team combines all relevant skills and organisations needed to address the full extent of the challenge of improving the sorting, separation and recycling of multi-layer materials used in flexible packaging, and their EoL valorisation solutions (different scientific and technical disciplines, engineering, social sciences and humanities (SSH)). The participation of a number of leading industry players is testimony to the soundness of the concept for generating broad impact and will maximise the consortium's ability to transfer the results of the project into broadly available market ready solutions. CIRCULAR FoodPack will enable know-how transfer through its value chain of partners: from technology institutes & academia (5 RTOs: Fraunhofer and UGENT both with several different research departments dedicated to project tasks, UM, KIT, and NTUA) to industry participants (5 SMEs: POLY, KREYEN, IRIS, ECOZEPT, BayFOR, and 5 large enterprises leaders in their field: SUEZ, SIEGWERK, AMCOR with its two different entities; AMCOR-K, AMCOR-G, and NESTLE). The combined expertise and knowledge from several disciplines allows to tackle the inter-disciplinary challenges required for circularity approaches (Fig 2).

Waste Management, Collection and new Sorting technologies (POLY, KIT, IRIS, SUEZ), engineering, novel pre-treatments for separation and recycling process cascades for the production of post-consumer recyclates (PCRs) for

14 https://www.fraunhofer.de/en/institutes/institutes-and-research-establishments-in-germany/cluster-of-excellence/circular-plastics-economy.html

their implementation in flexible food packaging (UGENT, UM, KREYEN, Fraunhofer), plastic/packaging converters to design novel packaging concepts with PCR for circular food packaging harnessing the potential of circularity in flexible packaging (AMCOR-G, AMCOR-K, Fraunhofer, SIEGWERK), end users packers and health&safety (Fraunhofer, NESTLE, and customers of AMCOR, e.g. Elidose, Persans SL, Ubesol SL for home and personal care products), providers of technological solutions, comprehensive life cycle sustainability assessment through LCA, LCC, s-LCA, business plan through Business Model Canvas tool and decision support system (UGENT, IRIS), and innovation, market analysis, dissemination specialists (BayFOR, ECOZEPT, NTUA).

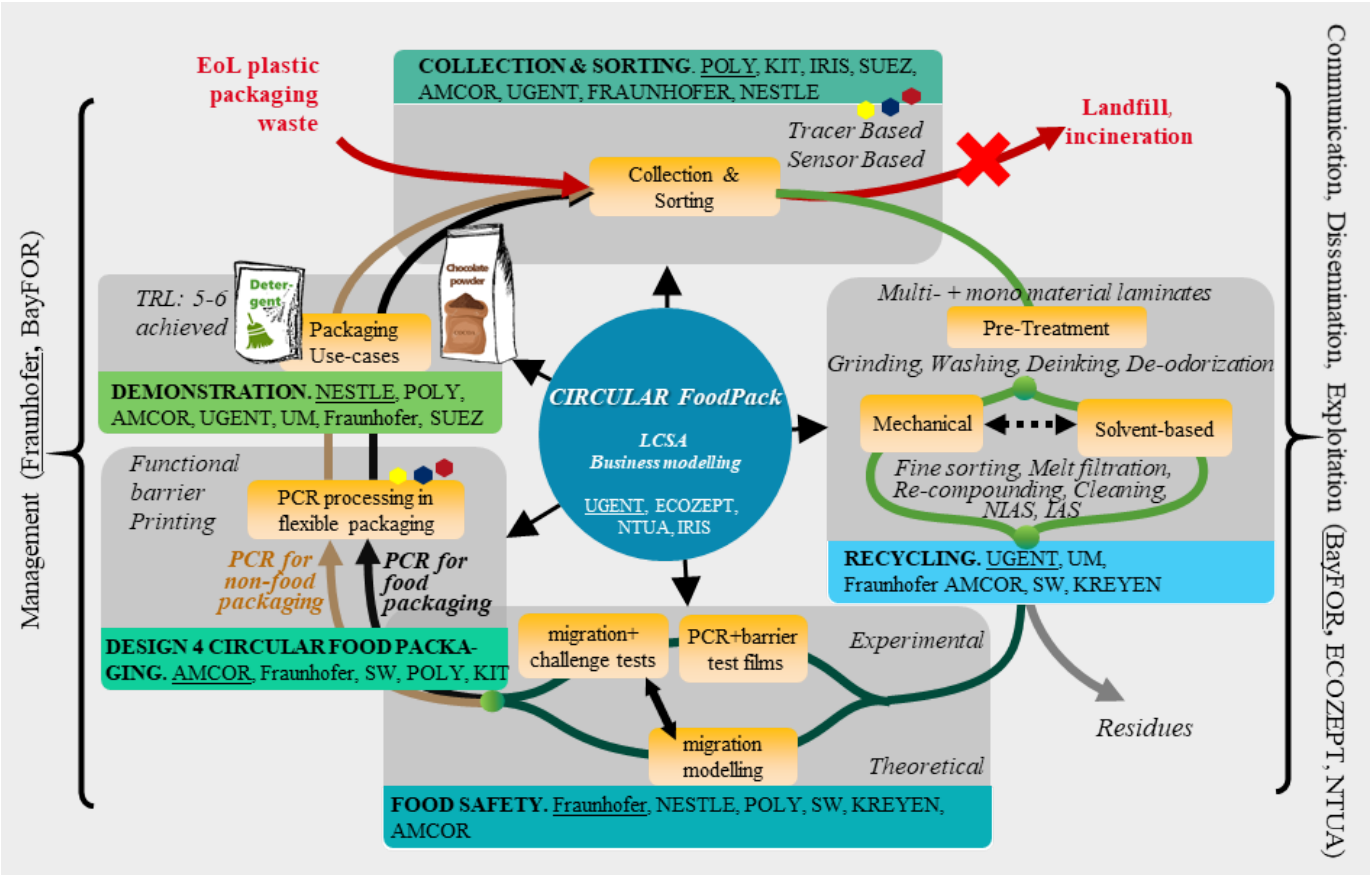

Fig 2. CIRCULAR FoodPack Concept with value chains

Stakeholder involvement: To complement our consortium, further stakeholders will be involved in an external advisory board (EAB) for complementary technical, policy and business-related insights, especially outreaching other sectors using packaging as well as other potential adopters of PE PCRs and even retailers (which are relevant actors for stimulating market pull). Relevant stakeholders will be selected from the following areas: Virgin material suppliers (e.g. DOW, Borealis), sorting and recycling technology providers (e.g. **STEINERT,** HOLYGRAIL, PELLENC), waste managers (e.g. **ISWA, PRE**, KIDV), packaging companies (e.g. P&G, **ELIDOSE, PERSANS** SL, **UBESOL SL**) packaging associations (e.g. **CEFLEX, IVLV, ZLV, FPE**), and **JRC** (EU science and knowledge service providing independent scientific research-based advise and support to **EFSA** and EU policy). Partners in bold already indicated their valuable commitment to be possible members of EAB. Their involvement will promote plastic packaging circularity, and support dissemination as well as policy work. Market analyses will be conducted by interviewing major experts of the European packaging industry (e.g. packaging producers, food manufacturers, retailers, waste management and recycling facilities) and also consumers to ensure a stakeholder-driven development.

Societal engagement: Project results are expected to create a new market of circular flexible packaging, which will lead to job creation in low and higher level income classes. Appropriate communication activities (such as Social Media and dedicated events, led by BayFOR) will involve the broad public and inform about collection schemes and sorting, as well as the concept of circular economy in general.

Positioning of the project: The three technology pillars with their initial and targeted TRLs are specified in Table 2. In summary: We start from technologies at TRL 3 (-4) and target TRL 5 (-6). Existing conventional re-processing technologies will be integrated, which eases scale-up. We will continuously monitor the performance of the technologies along the project to ensure reaching TRL 5, as a first step towards future commercialization.

Table 2: Summary of the Technology Readiness Level advancements that will be brought about as results of CIRCULAR FoodPack

Technology	TRL	Comment on current status and improvements
(i) Sorting:		
Tracer-Based-Sorting (TBS) for food packaging waste	4 ➔ 5-6	TBS was pioneered by POLY and KIT in the last years based on developing and optimizing both fluorescent tracers and corresponding detection processes. The first industrial application was a TBS separation process for glass fiber reinforced PVC flakes. After this, POLY and KIT have focused on plastic packaging by increasing the portfolio of efficient sorting tracers and optimizing the detection and sorting process to a high throughput of dirty and deformed packaging articles highly variable in size, and color artwork. Application in the strict food packaging market has not been targeted yet, so the TRL is at 4. Improvement in TBS will inlcude implementation of tracers in deinkable inks and high purity positive sorting of tracer branded food packs (99%) and implementation at adapted NIR sorting lines at TRL 5-6
Sensor-based Specification (SBS)	4 ➔ 5-6	Furthermore, besides fluorescent based sorting, IRIS tailors photonic solutions (e.g. NIR, MIR, HSI, or LIBS) to different packaging items not equipped with tracers. TRL is > 6 for mono-materials, and currently at 5 for the specific MLC in MultiCycle[15] Project (Table 3). Whereas physical penetration of IR radiation is limited to around 20µm, it could be shown in MultiCycle project that enhanced photonic solutions and machine learning approaches gather important information on laminate structures, which are not used by downstream recycling processes so far. Here the concept will be extended specifically to CIRCULAR FoodPack needs identified by recycling technology providers. Whereas monitoring of such spectroscopic information is developed in principle, upscale from TRL 4 to 5-6 is required to be adapted in CIRCULAR FoodPack.
(ii) Pre-treatment & Recycling		
Thermally induced decontamination of sorted waste plastics	3-4 ➔ 5-6	Thermally induced volatilization of volatile contaminants for deodourization in waste plastics has been a research topic for the last 5-6 years. The technology by KREYEN proposes an innovative temperature coupling that reduces treatment times and has a high economic potential. Since it has not been applied to waste food packaging yet, the TRL is assessed at 3-4 and will reach 5-6 by the end of the project.
Solvent-based recycling for multi-layered food packaging materials	4 ➔ 5-6	CreaSolv® has been developed by Fraunhofer for more than 10 years and can treat a number of polymers (PE, PP, PA, PET, PS, ABS, EPS, PLA, PBT, PVC) from a large variety of waste fractions (pi- and pc- packaging, ELV, WEEE, Building & Construction, pi- composites). For PE recovery from multi-layer materials, TRL is already at 6-7, however, the cleaning operation unit proposed in this project has not exceeded TRL 4 yet. For the recovery of several main plastics from MLC, the TRL will reach 6-7 in MultiCycle (Table 3), but in CIRCULAR FoodPack, we concentrate on PE as a single target polymer and improve the purification of IAS and NIAS substantially and at TRL 5/6.
New ink designs for de-inking with washing techniques	4 ➔ 5	UGENT and SW developed several approaches of deinking and delamination using solvent- and water-based technologies on the basis of typical print structures on the market and ink technologies like water- and solvent-based inks as well as UV inks at TRL 4. Tests in different ranges of existing and new detergent and solvent based deinking processes, feed into design of ink formulations with alternative binders/additives/pigments that can be considered as being easier to deink. The effect of different special "deinking" primers is also examined in order to facilitate an easy deinking in case of cross-linked systems like an UV or solvent-based 2-component ink, which can be considered as not being deinkable and thus requiring a special approach. Generally the special "easy-to-deink" ink systems developed by SIEGWERK consider the low-migration requirements for food packaging. Additionaly in the newly designed mono-material based laminates, it might be required to use primers and coatings to protect the barrier layers during life-time of the packaging and finally facilitate a proper delamination and deinking during the recyling process. The most promising tests are to be piloted at TRL5, which allows for sufficient deinking for further product testing.

15 H2020 MULTICYCLE Advanced and sustainable recycling processes and value chains for plastic-based multi-materials (SPIRE 820695)

(iii) "Design for circular food packaging": Mono- material based multi-layer packaging concept for food packaging, sortable, recyclable		
Mono-material multi-layers for food packaging w. PE PCR: • Mono-axial stretching • electron-beam (e-beam) • deposition of printing inks with tracers • deinkable printing inks	3-4 ➔ 6	Fraunhofer and AMCOR have been testing mono-material multi-layer packagings for about 3 years. AMCOR has a Biaxially Oriented PP (BOPP) based film in their portfolio (BOPP/SiO_x/Adhesive/PP), however, without any PCR. Structures with PE PCR films, which will be one product focus here, are still technologically not validated in lab (TRL 3). E-beam and mono-axial stretching technologies exist at AMCOR, upcycling of PE PCR by mono-axial stretching before barrier deposition will reach a TRL of 6. Silicon oxide (SiO_x) coating by e-beam deposition will be implemented on PE film surface in order to reduce the potential mass transfer (migration) of pc compounds from PE PCR film into the food. Consequently, the SiO_x will allow the use of recycled PE PCR granulates for packaging of dry foodstuffs (chocolate powder, coffee, creamer) and will protect food from migration of unwanted contaminants from the PE PCR film. The oxygen and water vapor barrier performance is also attained by SiO_x layer. The project aims at TRL6 demonstration in pilot lines by using innovative steps to be able to reuse PE PCR in an all PE based flexible packaging material efficiently. The "easy-to-deink" inks of SW will be incorporated into laminates. These will be dispersed with tracers before printing. We will follow design for recycling guidelines, PE based laminates will reach a TRL of 6.
Migration modelling and assessment of suitable highly effective functional barriers	4-5 ➔ 6-7	Fraunhofer will combine present experimental set-ups: (i) Migration, (ii) permeation, (iii) lag time experiments with advanced mathematical migration modelling approaches in order to evaluate the performance of highly effective functional migration barrier concepts, in order to consequently assess if contamination of PE PCRs are reduced down to levels below any toxicological concern qualifying these grades to be used in specific food contact applications.

Linked (inter)national research and innovation actions: The project proposal bases on background knowledge, IP belonging to several project partners, which has been gathered within prior and running research projects. A selection of the most relevant projects that CIRCULAR FoodPack partners (have) contribute(d) to are listed in Table 3 highlighting the derived expertise, results and processes that will feed into here. They will also be liaised upon in clustering activities described in section 2.2.

Table 3: (Inter-)national, European and regional research and innovation activities linked to CIRCULAR FoodPack (C: Coordinator)

Project Acronym (type of project) - Link with CIRCULAR FoodPack (CFP)	Partner
Circular Flooring (H2020) recycles PVC flooring with solvent-based CreaSolv® Technology using a new process demonstration capacity (20 kg/h) at Fraunhofer IVV.	Fraunhofer (C), BayFOR
Multicycle (H2020) recovers PE, PA, PET and PP from pi- and pc packaging waste, re-use in non-food applications selected the loop of pc high-grade plastics. Optical sorting by SBS HSI and modeling, LCA/LCC for circular value. Synergies in sorting technologies, PCR processing in flexible packaging, CreaSolv® upscale to a 25 kg/h capacity. IRIS developed an hyperspectral imaging in the SWIR region (HSI-SWIR) coupled with a chemometric-machine learning model working symbiotically to enhance the polymer recognition in layered plastic films.	IRIS (C), Fraunhofer, AMCOR
Circular packaging (German BMBF) builds and tests a TRL 8 facility of the CreaSolv® process to recover non-food PE from waste packaging. The expected quality level is far higher than current SOA flexible plastic recycling, but will not achieve food grade levels. The facility is of high importance to CFP exploitation activities as Circular packaging handles and solves up-scale issues that are highly important for later on implementation of a circular food pack strategy.	Fraunhofer (C)
Decoat (H2020) Recycling of coated, painted textile and plastic materials. Debonding of coatings in textiles, electronics. Fluorescent markers in coatings for sorting. Synergies with CFP for ink, adhesive deinking concepts and sorting.	IRIS, Fraunhofer
NonTox (H2020) increases the recycling rates of plastics waste containing hazardous substances by developing recycling processes to produce safe and high quality secondary plastic materials, and optimizes process economics by integration. Synergies in processing, and clustering activities.	Fraunhofer
FlexFunction2Sustain (H2020) is an open innovation Ecosystem for sustainable Nano-functionalized Flexible Plastic and Paper Surfaces and Membranes. Synergies with CFP will be clustering and participation to common workshops via Fraunhofer, and AMCOR, and also synergy to eco-design	Fraunhofer (C), AMCOR
GLOPACK (H2020); focuses on market uptake of packaging materials from agro-food residues, identifying feasible businesses, environmental impact assessments for different foods, sustainability	Fraunhofer, ECOZEPT,

assessment methodology. Synergies in eco-design of package, LCSA, packaging business strategies.	UGENT
ORIENTING (H2020): Operational Life Cycle Sustainability Assessment Methodology Supporting Decisions To-wards a Circular Economy. Takes up the challenge of developing a holistic Life Cycle Sustainability Assessment (LCSA) methodology that overcomes the key deficiencies of existing schemes to assess complex production-consumption systems, including product circularity	UGENT, Fraunhofer
C-PlaNeT (Circular Plastics Network for Training): European Joint Doctorate training project. It brings together 15 international young PhD researchers working on circular economy for plastics.	UGENT (C), UM, Fraunhofer
PLASTICITY (INTERREG 2 Seas): Compositions and quantities of industrial and commercial plastic waste in urban environment. New solutions for collection logistics, sorting equipment, designing final products from waste. Engaging stakeholders in urban platforms.	UGENT (C)
MaReK (Marker based sorting and recycling systems for plastic packaging) (BMBF, Germany) develops a marker-based sorting technology for plastic packaging. New inorganic fluorescent materials, new optical detection, new handling, synergy to TBS.	POLY, KIT
Photonic Markers (National Fundings in Germany): Develops new up-conversion fluorescence markers and determines their suitability for sorting.	POLY, KIT

(b) Methodology

CIRCULAR FoodPack plans developments at lab, pilot and demonstration scale, assisted by desk top methods (see Fig 3). The starting material are representative end-of-life plastic packaging items containing MLP.

Lab scale and pilot scale R&D:

1 R&D at **IRIS** on **Sensor Based Specification** systems **(SBS)** using NIR, Teraherz, LIBS sensory, for the intensive characterisaton of waste streams as a key learning. The information will feed the recycling processes to set recycling process parameters, to increase yield, and the quality of PE post-consumer recyclates (PE PCRs), according to waste characteristics on the one hand and specifications requested by downstream recycling process on the other.

2 R&D on **Tracer Based Sorting system (TBS)** to design, optimise and synthesize tracers (markers) by **KIT** for detection via existing NIR sorting machines. The novel specialised tracers will be employed to optically monitor and sort non-food and food packaging items at **POLY**. Tracers that are food compliant will be tailor-designed in their formulations and sizes to be compatible in novel ink formulations for printability in flexible packaging.

3 Laboratory, and pilot scale **purification and recycling** processes to apply to SBS and TBS fractions including **washing (UGENT), de-inking (SIEGWERK), delamination (UGENT), deodourization (KREYEN), dissolution based purification (Fraunhofer), and mechanical re-processing (UM)**. After investigation of purification potential of each single process, combinations of processes will be applied and optimised.

4 **Physicochemical analyses** at **Fraunhofer** of PE PCRs by gas and liquid chromatography, mass spectroscopy**,** to detect impurities present in PE PCR and their migration potential. Furthermore, molecular weight distribution, density, rheological properties, as well as organoleptics, smell, colour, defects for converting processes will be characterised at **UGENT** and **Fraunhofer**.

5 **Migration and diffusion models** by **Fraunhofer** to formulate the functional barriers in terms of layer composition, and thickness for inorganic and organic thin layer combinations, will be developed. Migration modelling parameters will be derived from migration / permeation experiments and will take into account the effects of material storage prior to the packaging process, as well as after packaging of the respective filling good. The architecture of the functional barriers will be designed, accordingly.

Pilot scale process developments:

6 Research on PE PCR processing by **(co)-Extrusion technologies for PE PCR** conversion into a flexible mono-material based laminate: First extrusion production trials will be performed at small pilot scale at **Fraunhofer** with 2 to 5 kg of PE PCR pellets for tailoring of PE PCR formulations, and then larger scale with >20 kg will be performed for laminate production with functional barriers at **AMCOR-G**.

7 Research on pilot processing of **functional barriers** by e-beam deposition of thin SiO_x layer (~ 60 nm) at **AMCOR-K** on (co)-extruded PE PCR film by **AMCOR-G**, with or without wet-chemical primer and top coating to ensure maximum blocking or substantially delaying the possible migration of contaminants in mono-material laminates intended for food packaging. The laminates developed for flexible food packaging applications will also include the tracers dispersed in the **printing inks** of **SIEGWERK**. The printability of marker inks on printing machines will be studied at **KIT** and **AMCOR-G**. Novel designs of fully recyclable packaging laminates will be developed for flexible packaging with PE PCRs, including tracers to guarantee the materials can be identified and guided to the fitting recycling technique for food originated packaging waste.

8 A monitoring prototype by **IRIS** for **Sensor Based Specification (SBS),** based on LIBS, NIR and/or MIR HSI technologies; combining the sensor developments with digitalisation. SBS will properly characterize, and identify multi-layer materials, with those that are negatively sorted (currently rejected), under relevant industrial conditions.
9 **TBS technology** of **POLY** will be installed in-line at **SUEZ** at the sorting line used for sorting experiments under realistic conditions.
10 **Pilot-scale pre-treatment processes** for **waste flexible packaging laminates** in process units of **UGENT**, **KREYEN**, grinding, washing, deinking, and deodourization, and pilot-scale solvent-based and mechanical recycling units will recover purified PE PCR from pre-treated waste, using technical equipments at **Fraunhofer, UM** and **UGENT**. New high level cascade recycling processes for purified **PE PCRs** will be available.

Demonstration of circularity in food and non-food flexible packaging applications:
11 The novel PE PCRs originated from non-food and food mixed waste will be applied in **laminates for non-food packaging** applications by **AMCOR-G** and **AMCOR-K**, reduction of virgin PE content by > 50% during laminate production.
12 The food grade PE PCRs originated from food packaging waste, will be applied in laminates by **AMCOR-G** and **AMCOR-K** (with functional barriers) for food packaging, with a reduction of virgin PE content by at least 50% during laminate production
13 **Personal and home care packaging demonstrators** to assess the processability, and applicability of PE PCRs at the customers of **AMCOR-G**
14 **Food packaging demonstrators** by **NESTLE** for PCR technical quality assessments as well as food safety, the demonstrators will undergo the recycling processes at **UGENT, UM** and **Fraunhofer**, the recovered materials will be characterised and reused by **AMCOR-G** and **AMCOR-K**, and evaluated at **NESTLE**.

Supporting/advising desk top work:
15 **Life Cycle Sustainability Assessment (LCSA)** by **UGENT**, environmental (LCA), economic (LCC) and social impacts (s-LCA) will identify sustainability optimisation potentials and advise technical developments.
16 **Business models and cases** are planned developed by **NTUA** and **ECOZEPT** to assess and optimise the economy of the process chain and prepare exploitation of R&D results towards market.

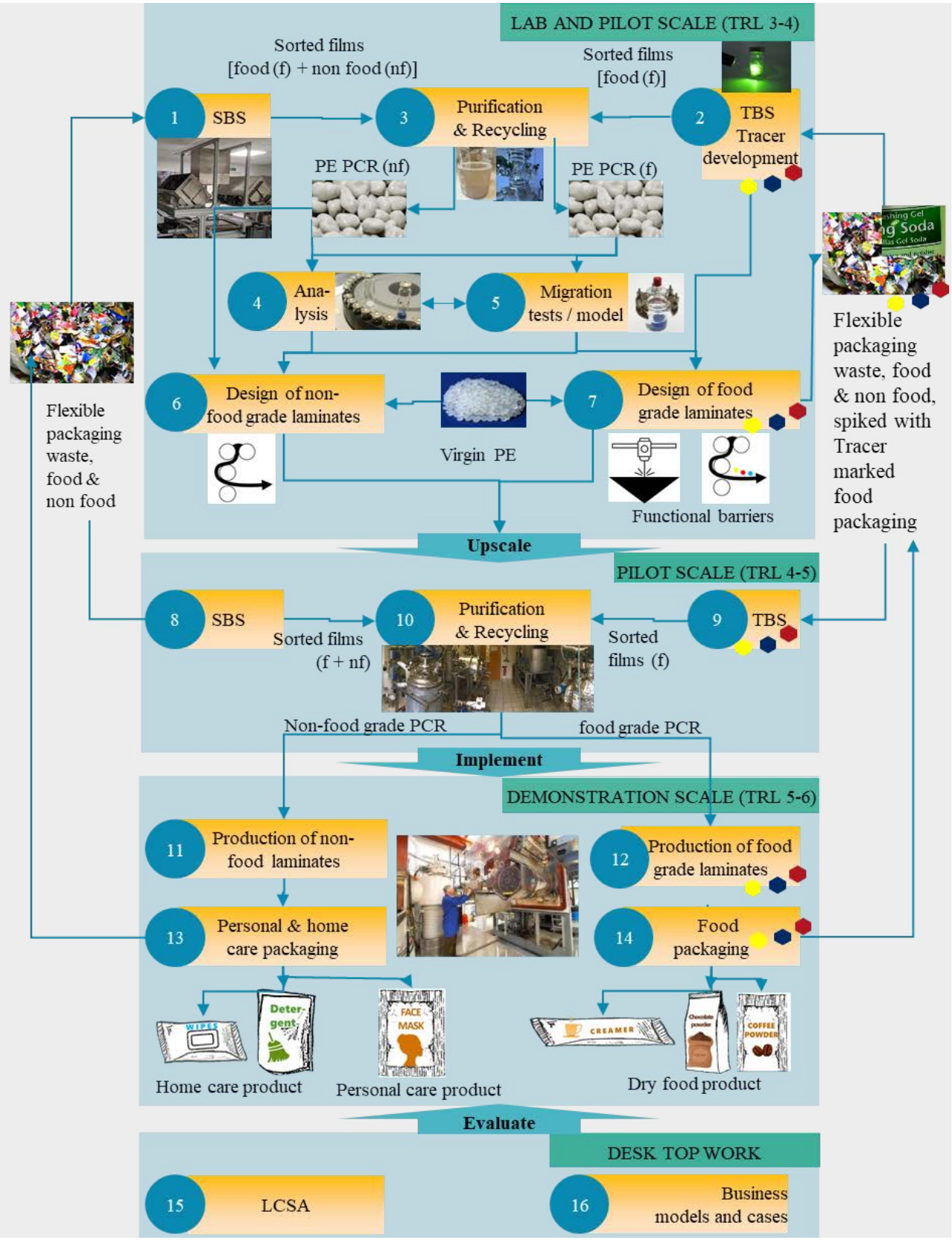

Fig 3. CIRCULAR FoodPack methodology (nf: non food, f: food)

Transversal activities:

Environmental, safety and standardisation issues: Detailed assessment of environmental (full LCA) aspects for post consumer recyclate (PCRs) production and preparation of corresponding guidelines for their applications in food packaging materials are planned. CIRCULAR FoodPack will provide scientific evidence for the safe integration of PE PCRs in food packaging applications taking into account current EU legislative requirements and safety demands (set out by Framework Regulation (EC) No 1935/2004 and Recycling Regulation (EU) No 282/2008) and will give input for further regulatory provisions and standardisation specific for the production and use of PE PCRs.

Market analysis including gender issues: Existing marketing studies and results will be taken into account and multi-stage surveys will be undertaken at two points of the project with packaging experts and consumers to include stakeholders' feedback in the CIRCULAR FoodPack product development. The project innovations will in particular consider gender dimension and cater for the needs, motivations and differences of different genders, where applicable. There are multiple studies on consumer behavior. One major factor influencing purchase decisions is gender[16]. Since shopping for household maintenance is still the responsibility of women[17], gender aspects have definitely need to be taken into account when designing new packaging materials or products. The consideration of environmental factors on packaging selection and the purchase decision with CIRCULAR FoodPack products is also likely to vary depending on consumer's gender. The same is true for the perception of designs of new packaging. The gender dimension is also considered for communication & dissemination activities to ensure target-oriented communication messages.

1.4 Ambition

(i) Sensor Based Specification (SBS) and Tracer Based Sorting (TBS) – mainly related to WP2 work

State-of-the-Art (SoA) and drawbacks: SoA sorting of packaging uses SBS and relies mainly on the detection of different near-infrared (NIR) absorption bands of the main polymers such as PE, PP, PS, PET and subsequent separation by air nozzles bars. However, NIR sorting cannot reliably differentiate between different polymer grades of the same type, e.g. food grade PE and non-food PE. In both cases the absorption of PE would give the same response. Furthermore, state of the art sorters do not identify (and sort) different materials present in multi-layer films or packaging containing metalised or carbon black pigments. In the MultiCycle project[18] IRIS proposes Hyperspectral Imaging (HSI) in different spectral ranges for detection of MLC and solutions like Laser Induced Breakdown Spectroscopy (LIBS) for black polymers or the identification of aluminum, AlO_x or SiO_x.

SoA research options to provide sorting by pre-defined fractions include TBS (Fig 4) and Watermark based optical sorting. Watermark recognition relies on picture recognition, precise colour and geometric pattern measurement of the artwork of a packaging article. However, industrial waste sorting means that the packaging articles to be sorted are dirty, dusty, deformed, folded and small, so that the recognition of a specific watermark is uncertain[19]. TBS in contrast, uses robust fluorescent sorting tracers that are printed on packaging items and emit an unidirectional signal as soon as the packaging article is moving through laser light. Past R&D of POLY and KIT has proven the feasibility to synthesize suitable inorganic fluorescent materials with different emission spectra which can be combined to provide a sorting code system. In addition, the detection and sorting technology development of POLY has shown that both, article and flake sorting by tracers are efficient sorting concepts. As the TBS sorting principle is positive, i.e. tracer marking indicates pushing into the sorted fraction, resulting purities of ejected items is high (>95%). Furthermore, POLY and KIT have started to develop new fluorescent tracer materials that allow the use of existing NIR sorting machines [20,21].

Step beyond SoA: CIRCULAR FoodPack (CFP) will adjust the high potential of the above mentioned innovative sorting concepts to the needs of circular food pack approach, including (i) the increase of target polymers in sorted fractions, (ii) separation of specific components of the flexible packaging waste hampering recycling processes and - with a strong focus – (iii) separation of non-food from food packaging items which is considered as a "door opener" to enable a reuse of PCR in food packaging. As a bridge technology for current non-tracer-marked waste, SBS will enable to detect and monitor different types of packaging films including MLP (multi-layer packaging) and will gain in-line information on the polymer composition of a waste stream. Data analytics based on chemometrics and machine learning will be applied to allow the identification of materials under real-time industrial conditions that are currently not identified by the use of standard chemometrics (Fig 5). This makes use of the current findings that

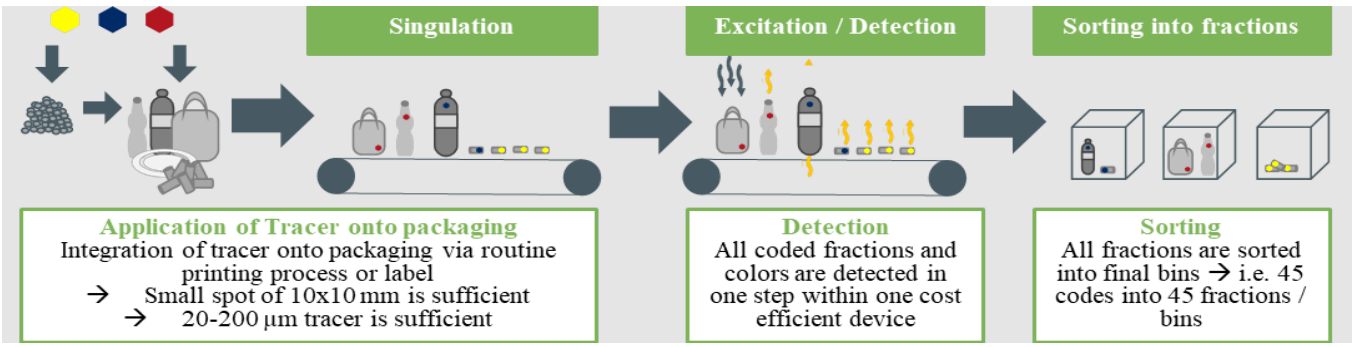

Fig 4. Scheme of marking and sorting process with the TBS "complete" concept

16 V. V. Iakshmi, D. Aparanjini Niharika, G. Lahari, IOSR J.Business and Management (IOSR-JBM), Volume 19, Issue 8. Ver. V. (August 2017), PP 33-36
17 Vivien Procher, University of Wuppertal and RWI; Colin Vance, RWI and Jacob University Bremen, Ruhr Economic Papers #393
18 H2020 MULTICYCLE Advanced and sustainable recycling processes and value chains for plastic-based multi-materials (SPIRE 820695)
19 https://us.pg.com/blogs/HolyGrail/ called on 31.01.2020.
20 A.Turshatov, er al. (2018), Opt. of up-conversion photonic markers based on a La2O3 host for plastics recycling and anti-counterfeiting applications 2018.
21 B.S. Richards (2020) Applications of up-conversion: from PV to plastic recycling and anti-counterfeiting, Invited seminar at Univ. Aarhus, Denmark, 2020.

appropriate sensors can distinguish between PE films, PE based MLP and other flexible packaging items, e.g. PP monofilms or PP MLP. This in-line monitoring helps identifying those sorting streams that are highly qualified for downstream PE recovery due to high shares of PE and PE multi-layers. In such a way SBS selects suitable film fractions for the CIRCULAR FoodPack recycling technologies and will enable production of pure PE fractions for new flexible personal care packaging.

In CIRCULAR FoodPack, TBS will step beyond state-of-the-art by **(i)** increasing the number of sorting codes by further R&D in both tracer materials and detection technology; **(ii)** development and validation of efficient tracers for TBS sorting in NIR sorting machines and within the NIR spectrum; **(iii)** transfer tracer integration via print onto food packaging enabling removal of tracers by deinking; **(iv)** optimization of an existing sorting machine platform for handling of flexible food packaging articles. The tracer development will include synthesis i.e. of chemically-stable oxide-based fluorescent tracers co-doped with lanthanide ions. As material specifications for food-grade packaging plastic are more strict than for other plastics, it is considered a crucial first step to already eliminate non-food compliant materials. Within CIRCULAR FoodPack, the tracers will be dispersed in a light-colored or white printing ink and applied onto the outer packaging layer. Thus, strongly fluorescent, printable, chemically-stable and non-toxic markers detectable via existing NIR sorting machines are required. The inks used will be tailor-designed to deinking and delamination, to guarantee the high efficient removal of the tracer during the recycling step, and to guarantee that there is no remaining marker residue in PCRs from the previous tracers, PCRs will be subjected to migration analysis.

In more detail, the amount of light absorbed by existing tracers will be enhanced by increasing the range of wavelengths absorbed such that the brightness of the downshifting phosphor emission in the NIR goes up. Ideally this is performed in a way that the light source in the NIR machine or a bank of LEDs can be used (i.e. no lasers required) (Fig 6). It is furthermore foreseen to **enhance the number of unique NIR emission wavelengths, finally providing** a "NIR palette" of emission wavelengths. That will increase the number of targets (whether polymer types, specific ML structures, food vs non-food use) that can be marked. The number of individual emission wavelengths targeted is 5 (Fig 6c), which assuming combination of markers (2^5-1) would allow 31 combinations of these emission signatures. The novel markers will be based on the following emitters: Yb^{3+}, Ho^{3+}, Pr^{3+}, Tm^{3+} and Er^{3+}. In addition the host material of markers will be optimised **in order to increase the quantum yield of the NIR emission.** A range of host materials will be investigated to fulfil the criteria as phonon energy, solubility and size control (to ensure compatibility with printing processes). The latter relates strongly to a tailoring of synthesis pathways including solid-state reaction method, sol-gel synthesis, and co-precipitation methods.

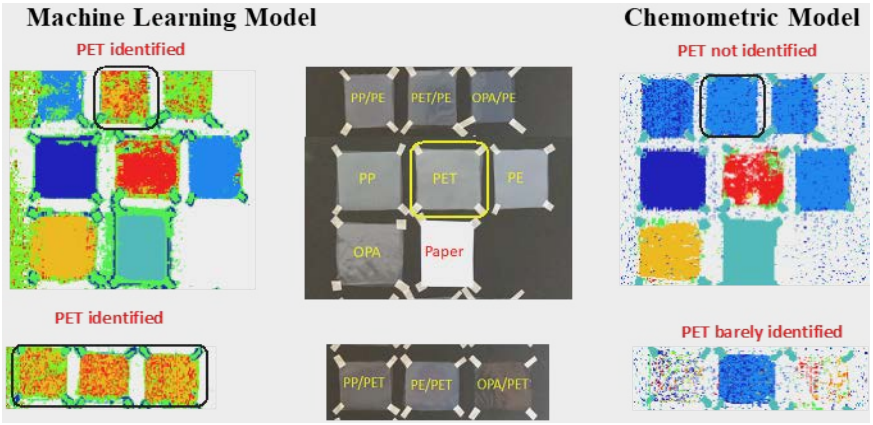

Fig 5. Data elaboration of multi-layers materials identified with IRIS technologies

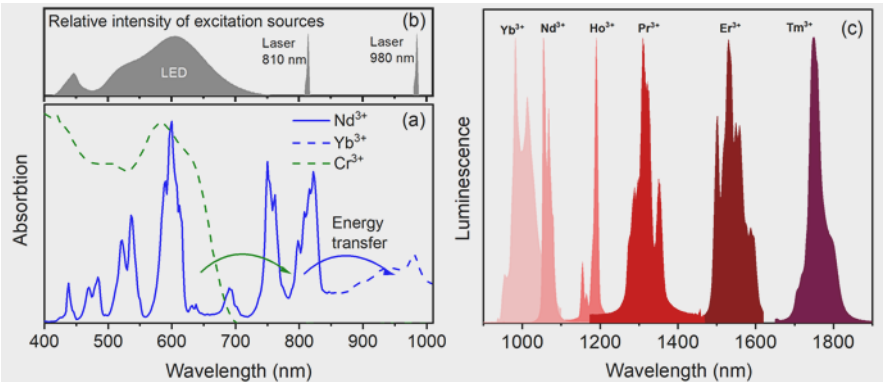

*Fig 6. **(a)** Absorption spectra of Nd^{3+}, Yb^{3+} and Cr^{3+} ions. When three ions are mixed in one host, the energy transfer $Cr^{3+} \rightarrow Nd^{3+} \rightarrow Yb^{3+}$ is expected; **(b)** Emission spectra of excitation sources used in the project. Yb^{3+} can be excited with 980 nm laser, Nd^{3+} with 810 nm laser, Cr^{3+} and Nd^{3+} with white light emitted diode (LED); **(c)** excitation of Cr^{3+} or Nd^{3+} results in NIR emission of Yb^{3+}, Nd^{3+}, Ho^{3+}, Pr^{3+}, Er^{3+}, Tm^{3+} due to energy transfer. As luminescence peaks are well resolved, delivery up to 6 different markers is feasible, Nd3+ however will not be considered due to a slight overlap.*

Innovation potential: While the marking system is being developed Sensor-Based-Specification (SBS) will be considered as charactersiation tool for packaging items not yet marked to recycle materials for non-food applications, with a high potential for implementation in industrial sorting lines. TBS will contribute to the circularity of food packaging and provide cost-efficient sorting solutions for packaging items according to pre-defined criteria. This enables positive sorting of food-grade plastics and/or the separation of MLC films with of specific MLC structures.

(ii) Recycling of Flexible Packaging including Multi-layer Materials – mainly related to WP3 work

SoA and drawbacks: State-of-the-art treatment of waste packaging separates bulk waste stream into defined fractions using a multi-stage concept that combines mechanical units (sieve drums, hoppers,..) with magnetic, eddy-current and optical sorters for separation of glass, metals, aluminum and different plastics. Rigid plastics are separated by NIR targeting at PE, PP, PS and PET streams departed from flexible packaging, which are in turn divided in a fraction of large pieces (>200 mmx300 mm) and a fraction of smaller items. The flexible streams are usually not sorted very efficiently by polymer type and contain PE and PP mono films and – with a majority in the fraction of smaller items – MLC films. Subsequent treatment of flexible plastics comprise shredding, post-sorting, washing, drying and re-compounding steps with or without melt filtration[22]. As a result, recycled polymers with a purity of around 90% (in terms of polymers) and significant amounts of NIAS and IAS are produced, including odourous volatiles, which can be used in best case in applications such as garbage bags.[23] A particular problem of post-consumer plastic films is the presence of odourous constituents. Such odourous constituents typically have a broad diversity in terms of origin and chemical properties. Furthermore, they are typically not only found on the outer surface, but are also adsorbed deep into the polymer, which make them hard to remove[24]. Despite the broad chemical variety of odour components, currently implemented washing processes in recycling plants are mainly based on cold or hot water, with or without addition of caustic soda and detergents. These washing media are effective to remove visible dirt and organic contamination present on the plastics, but they are not able to efficiently remove all odour components, especially since these components typically have a broad range of chemical properties. Both Fraunhofer, as well as UGENT have elaborated analytical procedures to identify odourous constituents and quantify odour removal[25, 26, 27].
Furthermore, the mix of colors present in flexible packaging waste results typically in production of grey or brown recyclates for dark applications, often by adding carbon black to the recyclate. These properties hinder the reuse in high value applications and lead to downcycling to low quality products. More sophisticated process cascades use sink/float or optical sorting processes to further refine and/or color-sort initial fractions or apply stripping processes at elevated temperatures, e.g. in compounding processes.[28] These measures improve qualities partly, and especially also are still difficult for film fractions. Even if sorted, (odourous) NIAS are still present and a loss of mechanical properties is observed. Interest in deinking and delamination procedures is growing rapidly, although the industrial application of effective deinking procedures is limited, or even does not exist, for post-consumer waste. In general a distinction can be made between water/detergent based deinking, as proposed for example by Cadel Deinking[29]. Other approaches such as the NOREC process propose solvents such as ethyl acetate[30]. Recently UGENT developed and patented a deinking procedure as well. Similarly, delamination of multi-layer films can be achieved by withdrawing the adhesive that is used to attach two layers by using a specific solvent[31]. The use of acids (e.g., acetic acid, propionic acid, formic acid, butanoic acid) to delaminate is already relatively widely described in scientific literature[32]. Other endeavours have attempted delamination based on water, with e.g. acetone, or acids[33]. However, again implementation at larger scale is limited. At the moment, there is very little generic insight in which type of delamination and deinking procedures work on which kind of laminate and ink structure, and especially in the reasons why and how these procedures work or don't work on a plastic film mixture. Similarly, liquid (solvent) based technologies have been developed in the past years. They enable a separation of multi-layer films either by dissolution/weakening of low volume intermediate layers (e.g. adhesives) or by physically dissolving major layers (e.g. LDPE), with the latter being more effective in cleaning the target polymer. This is due to the effect that delamination separates laminate structures by weakening the adhesive power, but cannot clean adhesives from the substrate completely as most adhesives are duroplasts and do not dissolve. In contrast, dissolution of the target polymer (e.g. LDPE) allows separation of non-target materials (pigments, adhesives, metals) from the dissolved polymer based on solid/liquid and liquid/liquid separation processes. Such a solvent-based polymer recycling technology (CreaSolv®) has been implemented at semi-industrial scale in Indonesia for the recovery of polyolefins from multi-layers, however at a quality level not sufficient for food grade packaging[34]. In other research projects CreaSolv® proofed its ability to effectively elimiante NIAS (e.g. brominated dibenzofurans) and IAS like brominated flame retardants from plastics in electronic waste.[35]

Step beyond SoA: CIRCULAR FoodPack intends to implement purification unit operations in mechanical and solvent based recycling process cascades that enable in total a sufficient reduction of contaminants present in EoL

22 K. Ragaert et al., Waste Management 69 (2017) 24-58
23 Strangl M. et al. Journal of Cleaner Production 181 (2018) 599-607
24 A. Cabanes, et al., A review on VOCs from recycled plastics, Sustain. Mater. Technol. (2020).
25 M. Strangl, et al., Charac. of odourous contaminants in pc plastic packaging waste, J. Sep. Sci. (2017).
26 M. Strangl, E. Ortner, T. Fell, T. Ginzinger, A. Buettner, Odour characterization along the recycling process of pc plastic film fractions, J. Clean. Prod. (2020.
27 R. Demets, et al, Dev. and app. of an analy. method to quantify odour removal in plastic waste recyc. proces., Resour. Conserv. Recycl. 161 (2020) 104907.
28 Strangl M. et al, Resources, Conservation and Recycling 146 (2019) 89-97
29 http://cadeldeinking.com/en/, patent EP2832459B1
30 https://www.norec-info.com/en/ patent DE4122277A1
31 Horodytska, O., Valdés, F. J. & Fullana, A. Plastic flexible films waste management – A state of art review. Waste Management 77, 413–425 (2018).
32 Kaiser et al., Recycling 3 (2018) 1-26.
33 Example patent nrs: DE 4122705, WO 9304116, EP 599905 , US20040054018A1
34 https://www.unilever.com/news/news-and-features/Feature-article/2018/our-solution-for-recycling-plastic-sachets-takes-another-step-forward.html
35 Schlummer et al. (2006). Waste Management & Research 24:573-583.

packaging plastic, such as NIAS and IAS including printing inks and tracer compounds of TBS concepts (see above) technologies, with innovative purification methods.

Related to deodourization , a method of KREYEN is proposed, which relies on a highly effective energy transfer system earlier applied in food processing. The process is split up in two phases: First the material is moved through Infrared Dryer (IRD). After this process, the material is moved into a silo, where it is continuously flushed by hot gas (Fig 7 left). The IRD heats up the product in very short times to the necessary temperatures. In this case, temperatures near to the melting point are reached. The infrared technology heats up the material from the inner to the outer surface (Fig 7 middle). Because of this, a partial pressure difference occurs that brings the odours very quick from the inside to the outside. This ensures a quick diffusion of the Volatile Organic Componds (VOC)'s from the Material to the surrounding atmosphere. Due to the continuous movement of the material, the surface area is exchanged frequently in the rotating drum (Fig 7 right). This accelerates the odour removing, because each particle of the material is brought very close to its melting point, accelerating the odour removal.

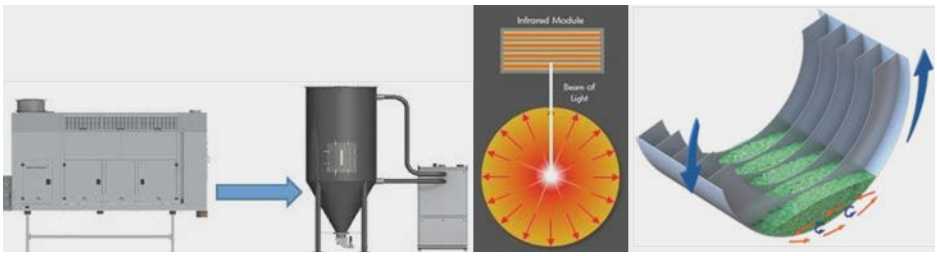

***Fig 7. Deodourization** (KREYEN): Silo, the material heated up by IRD, rotating drum*

Related to deinking, an innovative deinking technology of UGent (Fig 8) is proposed[36]. This is the first deinking method able to remove all types of inks, including UV crosslinked inks. Within the project, the UGENT method and existing deinking methods based on detergents and solvents are applied on a set known and unknown samples, which will deliver insight for the first time towards design for deinking guidelines. This allows to adjust printing inks of SIEGWERK to be applied on the PCR products. Also, the delamination and deinking is explored as pretreatment towards CreaSolv®. Potentially robust and cost-effective deinking and delamination can already clean up to a large extent a whole set of NIAS, even without having the ambition to achieve at that point a 100% deinking efficiency. Furthermore, the existing CreaSolv® Technology will be upgraded by a shortly patented cleaning device that improves the elimination of gel droplets (cross-linked PE) and smallest particles from the polymer solution and reduces critical NIAS and IAS from the polymer.

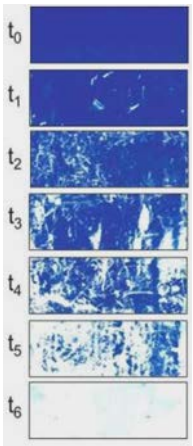

***Fig 8. Deinking** (UGENT)*

Innovation potential: Two post-sorting processes, namely deinking and deodourization will be implemented in existing mechanical treatment cascades and produce pre-cleaned flakes that are for the first time subjected to solvent-based recycling (CreaSolv®) due to its share of MLC. The integrated new purification unit will further reduce deteriorating materials, including gel droplets or volatile and non-volatile NIAS and IAS. These new brick stones in the process concept will boost the current quality level of recycled polymers to a high-purity grade that is intended to comply with food grade specifications.

(iii) Assessment of food packaging safety based on effective functional barriers for PCRs' contaminants – mainly related to WP4 work

SoA and drawbacks: For all recyclates in direct food contact, compliance of the recycled materials has to be shown according to the Recycling Regulation (EC) No 282/2008. This compliance evaluation is based on three main parts: **(i)** the input contamination levels of the recyclates, **(ii)** the cleaning efficiency of the applied recycling process and **(iii)** the exposure of potential hazardous substances to the consumer. The latter is based on the application of the food contact material (mainly depending on types of contacting food, filling and storage conditions), resulting in a specific exposure scenario. Part **(i)** and **(ii)** however, are not fixed and can be approached by the applied recycling technologies. Reduced input contamination and increased cleaning efficiency of the recycling process will significantly improve the quality of the recyclates and will highly contribute to the safety of the recycled polymers and their re-usability. In contrast to PET (where closed loop recycling from bottle to bottle is well established in Europe), recycled polyolefins from post-consumer (pc) origin cannot be used in direct food applications as these recyclates currently do not meet the safety levels and regulatory demands applying to food contact materials. Up to now, none of the recycling processes for pc polyolefins that were evaluated by the European Food Safety Authority (EFSA) were compliant to the set safety criteria[37], required to re-use the material in its original direct food contact application ('closed loop')[38]. To date, such recycled plastic materials may only be used behind a functional barrier layer, which must ensure that migration of any contaminants will remain below the levels of toxicological concern

36 EP 20167745 Method for deinking plastic material
37 EFSA Panel on food contact materials, enzymes, flavourings and processing aids (CEF); Scientific Opinion on the criteria to be used for safety evaluation of a mechanical recycling process to produce recycled PET intended to be used for manufacture of materials and articles in contact with food. EFSA Journal 2011
38 EFSA CEF Panel (Food Contact Materials, Flavourings and Processing Aids), 2015. Scientific Opinion on the safety assessment of the processes 'Biffa Polymers' and 'CLRrHDPE' used to recycle high-density PE bottles for use as food contact material. EFSA Journal 2015;13(2):4016, 25 pp.

during the lifetime of the packaged good. Such levels, e.g. 0.15 µg/kg food required for (potentially present) genotoxic substances, as described by EFSA[39],[40] can hardly be established using analytical-instrumental techniques. For the migration evaluation, the tool of mathematical migration modelling is established to predict transfer of molecules from packaging items into packed food[41]. Implementing Fick's Law together with molecules and packaging material properties, modelled transfer rates reflect experimental migration data with reliable precision. Therefore, migration modelling has been accepted as a powerful tool in assessing risks of human packaging related dietary exposure towards contaminants[42]. The modelling approach is also considered to be a suitable and powerful tool for assessing the performance of functional barriers, but has not yet been widely applied for this purpose.

Step beyond the SoA: CIRCULAR FoodPack will address the food safety issues of new packaging designs by (i) applying well-defined (TBS and SBS sorted) input streams. (ii) Furthermore, improved sorting and cleaning procedures need to support this goal. However, due to the physico-chemical properties of polyolefins as high-diffusive polymers, migration of substance from the polyolefins into contacting food is by nature rather high; Regulation (EU) No 10/2011 defines stringent migration threshold limits for NIAS of 10 µg/kg food for chemicals that are *not classified* as mutagenic, carcinogenic or toxic to reproduction and of 0.15 µg/kg food for genotoxic compounds. Therefore, (iii) highly effective barrier solutions need to be applied in order to reduce contamination down to levels below of any toxicological concern. Performance of such highly effective functional barrier concepts will be evaluated within the project using experimental set-ups such as migration, permeation and lag time experiments. For the assessment of migration barrier properties and effectivness, modell laminates with an intentionally contaminated PE layer and different barrier options will be used. One model structure is presented in Fig 9, showing a PE film spiked with representative model contaminants, laminated to the functional barrier coated film. These analytical methods will be combined with advanced mathematical migration modelling approaches. As a result, maximal acceptable contaminant levels can be calculated for PCR plastics for given barrier materials, barrier thicknesses and storage conditions (temperature and time). Thus, migration modelling will substantially support and guide the development of the food contact materials with PCR content. As a decision and design tool, it well recommends maximum contaminant levels in PCR (as a benchmark in recycling) and a suitable barrier concept. Based on Fraunhofer's experience on migration and compliance assessments, including petitioning to US-FDA and EFSA for regulatory approval processes on food contact materials, food safe design concepts will be integrated. Recommendations and guidelines will be established for the assessment of the performance of functional barrier layers related to the intended food application, which shall support both food packaging industry to fulfil the safety demands as well as the risk assessor (EFSA) and risk manager (EU Commission) to establish safe procedures for the use of pc polyolefins.

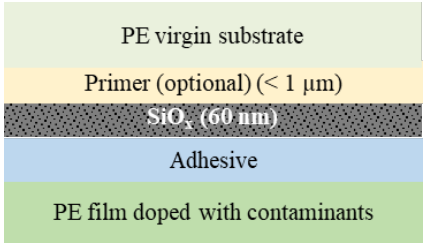

Fig 9. *Model structure of test samples with well-defined contamination levels in PE film for migration tests*

Innovation potential: CIRCULAR FoodPack addresses the main challenges related to the use of PE PCRs in food grade packaging that currently prevent their re-use in such applications. Accepted methodologies will quantify the food safety benefits of TBS sorting (sort food grade packaging only), of the improved purifying concept in the recycling cascade and of the functional barrier concepts developed in the new packaging designs. Migration modelling will define maximum acceptable levels of NIAS and thus deduct high percentages of purified PCR materials in such new food packaging items. Based on specific migration data for demonstrators built in this project, test methods will demonstrate that food safety concerns are not to be expected.

(iv) Implementation of Circularity into Food Packaging Value Chain – mainly related to WP5 and WP6 work

SoA and drawbacks: A typical plastic-plastic multi-layer composite (MLC) structure consists of LDPE applied as a sealing layer, PA or PET for mechanical strength and printing, and (Ethylene vinyl alcohol) EVOH layer (~ 5 µm) as oxygen barrier. Depending on the application and compatibility of the different materials seven or more layers are implemented in a single composite film. Such highly complex and non-recyclable MLC films are subject to progressively increasing EPR fees and general taxes as legislation is becoming more stringent. Some (new) structures are however entering the market that strictly speaking are recyclable, but in fact are not recycled back into food grade packaging materials as these are not sorted in mechanical recycling processes. It is necessary to design flexible packaging films suitable for collection, sorting and recycling after use to create a circular economy for flexible packaging. Polyolefin based consumer flexible packaging makes up the largest proportion of pc flexible packaging waste-stream (70-80%) and the ability to sort and **mechanically recycle polyethylene (PE) is already proven in industrial scale in Europe**. Current developments by DOW and Borealis at K-Fair 2019 showed that mono-PE films for stand-up pouches can be produced through the combination of monodirectionally oriented PE films combined

39 EFSA CEF Panel (Food Contact Materials, Enzymes, Flavourings and Processing Aids), 2016. Scientific opinion on recent developments in the risk assessment of chemicals in food and their potential impact on the safety assessment of substances used in food contact materials. EFSA Journal 2016;14(1):4357.
40 EFSA (European Food Safety Authority) and WHO (World Health Organization), 2016. Review of the Threshold of Toxicological Concern (TTC) approach and development of new TTC decision tree. EFSA supporting publication 2016: EN-1006. 50 pp.
41 E.J. Hoekstra, et al. Practical guidelines on the application of migration modelling for the estimation of specific migration; EUR 27529 EN; doi:10.2788/04517
42 Welle, F. A new method for the prediction of diffusion coefficients in poly(ethylene terephthalate). J. Appl. Polym. Sci. 2012, 129, 1845–1851.

with low melt PE sealing films[43]. They have no PCR content. One of the biggest issues is their processability on packaging lines (wrinkling, curling, slow line speeds…) at economically viable processing speeds. Furthermore, the current generation adhesives used for PO lamination causes yellowing of recyclates, while the printing inks are very hard to remove once applied on the package, causing strong discoloration. No proper technological solutions for these issues are available to date. The use **of PE recyclates** from food packaging for reimplementation into packages and the creation of a closed loop for packaging material to packaging material is therefore not yet possible at scale. In terms of barrier, EVOH, aluminum and metallization are the 3 most common applied barriers, all non-compatible with the PE recycling stream. Inorganic barrier layers are slowly entering the market at high performance alternatives to Aluminum (AlO_x, SiO_x) and have found commercial applications when biaxially oriented BOPET and/or BOPP are used as the deposition substrates. For PE, not a lot is available and those materials that are, are suffering from poor barrier functionality due to the mismatch in stiffness and topography between the oriented PE film and the inorganic barrier layer. Electron beam treatment is one possibility to enhance the processability of extruded films from PE PCRs[44], further enhancing the mechanical stability brought by machine direction orientation or biaxial oriented stretching of the outer PE films, while at the same time providing the proper surface topography required for inorganic barrier layer deposition[45], a critical building block needed for strong permeation barrier without disturbing the recycling streams, thus avoiding the need for thick EVOH layers that are considered incompatible with SOA mechanical recycling. In order to be able to reuse the PE PCRs in food packaging, or even in personal care packagings, the migration of unwanted pc compounds into the packed stuffs must be prevented. SiO_x is in addition to its oxygen and water barrier performance has also functional barrier[46,47] potential as it is known from its use in PET bottles againist potential contaminants from pc-recycled PET, but has not been applied on PE PCR films yet as functional barrier.

Step beyond SoA: At large scale, processing of polyolefin (e.g. PE) on single-layer extrusion lines, will be possible, by mono-axial stretching process, and thereby their vacuum deposition with a very thin inorganic barrier layer of SiO_x (~ 60 nm). The SiO_x coated mono-axially oriented PE PCR film (MDO PE PCR) will be laminated onto the extruded PE, resulting in a final structure: **(outside) deinkables (with markers) / MDO PE PCR / SiO_x barrier** / Adhesive / **PE (virgin) sealing (facing the food).** In case the barrier performance is not sufficient for the selected packaging application, an additional organic gas barrier layer < 1 µm will be considered either before as a primer or after the SiO_x coating as a top-coat, or both (Fig 10 (a), top-coat not shown, optional). This has the advantage that only very thin layers are used, and recycling is not negatively affected. Appropriate conformity and safety assessments are required for all materials (Fraunhofer). As off-set migration during storage (as a reel) is a known issue, an additional functional barrier layer facing the outside of the package will be considered, effectively sandwiching the PCR. AMCOR-K has much experience in the e-beam assisted deposition of SiO_x layers and will use its knowledge to improve the gas and functional barrier performance when deposited on a PE base. A lacquer applied as a primer onto the PE substrate before and/or after SiO_x deposition is reasonably expected, to further enhance the migration barrier performance by sealing off potential defects/pinholes in the SiO_x layer, while, when applied as a primer, it will planarize the PE surface for SiO_x deposition, minimizing the presence of any surface defects or contaminants. In case there would be issues to process the PCR on the machine direction orientation (MDO) equipment, an alternative structure is proposed in Fig 11 (b), where the PCR layer remains unstretched and stifness will be provided by a virgin (MDO) PE base layer. To overcome the possible machinability problems by packaging, during sealing, co-extrusion of virgin PE is foreseen, further sandwiching the PCR, improving the overall film quality (but reducing the %PCR in the film).

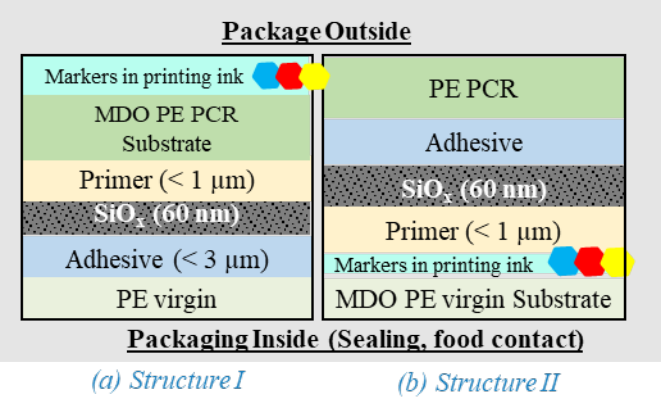

Fig 10. Generalised type of a laminate for flexible packagings based on D4 Recyclability

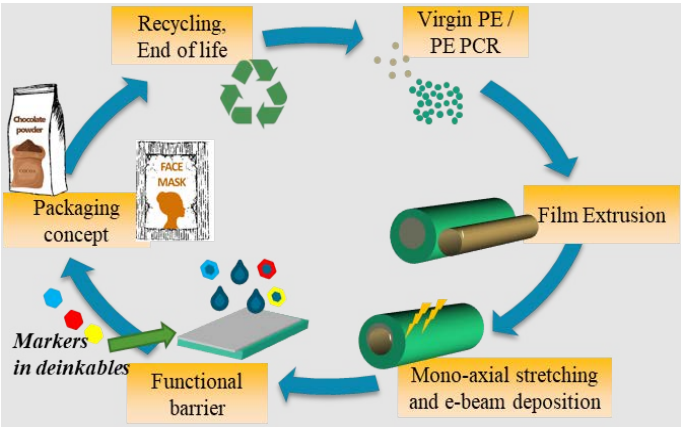

Fig 11. Design for Circular Food Packaging

43 EP2585298A1, "Single polymer film structures for use in stand-up-pouches", DOW and EP1961558A1, „Laminated multi-layer films", Borealis
44 Woods R. J., Pikaev A. K., Applied Radiation Chemistry: Radiation Processing, John Wiley Sons, INC., 1994, 288-307, 349-392
45 Biaxial stretching of film, Principles and applications, Edited by Mark T. DeMeuse, Woodheald Publishing Limited, 2011.
46 Functional barrier limits the migration of a component into food or food simulant in amounts below an accepted threshold or a migration limit of no concern
47 Welle, F. SiOx layer a functional barrier in (PET) bottles againist potential contaminants from pc recycled PET. Food Additives and Contaminants. 25, (2008)

Further alternatives such as thin AlO_x, and also EVOH will be tested as (additional) migration barriers to cover potential off-set issues.

SIEGWERK has developed a series of delamination primers for different applications, which are for the time being not yet commercialised, but already tested on a pilot scale (TRL 4) with very good results in terms of matching all requirements. By using deinking conditions as discussed above, the delamination and deinking is taking place in one step during the recycling. It can be applied using a gravure or flexo printing process. SIEGWERK will further develop this primer formulation, as a primer layer for SiO_x deposition, in addition, if required also as a top coating. The primer formulation will match (i) to the requirements like adhesion on the used films, and (ii) the requirements of the recycling processes (by deinking).

One other innovation will be the application of tracers from KIT and POLY in printing inks of SIEGWERK, that are deinkable. Some possible locations for the printing inks with tracers are presented in Fig 11. The project will chase all the possibilities for finding an all PE laminate, which is completely recyclable, functional, sortable and consists of at least 50% PE PCR in it, by providing design guidelines for circular food packaging (Fig 11).The packaging types are diverse since various food and non-food types are to be packed. This will lead to differences in the laminate, such as the layer thicknesses, barriers used, PE types, inks included, still all to be sortable and recyclable. A matchmaking is to be performed between the new materials and the final package demonstrators for each targeted packaging product type. The effects of laminate layer sequence on their recyclability will be rated and provided here as guidelines.

Innovation potential: We will provide 3 use cases with **mono-material multi-layered packaging films** that can be completely recycled and are suitable for food packaging, guaranteeing a sustainable use of resources. The project will demonstrate the full recyclability of this structure, including deinking and its possible reproduction using **PCR** coming from its own recycling process. The very thin functional barrier, e.g. SiO_x, will prevent migration and thereby will allow direct food contact, due to the PCRs being from food grade origin flexible packaging.

(v) Comprehensive sustainability assessment of circular value chains- mainly related to WP7 work

SoA and drawbacks: Circularity on its own does not equal sustainability. Developing sustainable value chains in a Circular Economy requires a holistic sustainability assessment or Life Cycle Sustainability Assessment (LCSA)[48], taking into account environmental, economic and social impacts. Although life cycle-based standards and guidelines exist to assess the environmental (LCA) (e.g., ISO 14044[49], PEF method[50]), economic (LCC) (e.g., SETAC guidance[51]) and social (s-LCA) (e.g., Handbook for Product Social Impact Assessment; upcoming revised Guidelines by the Social Life Cycle Alliance) dimensions, their practical implementation is often still challenging (e.g., data availability, lack of reliable methods for certain impact categories, assessment of emerging technologies). Although a holistic standard integrating the three dimensions is not yet available, a tailor-made sustainability framework covering the three sustainability dimensions focused on waste management in European cities was developed in the H2020 REPAIR project[52]. Moreover, taking into account Circular Economy priorities is an additional challenge for holistic sustainability assessment. LCA-based indicators were developed to evaluate the circular economy performance of recycling value chains[53]. Recent reviews[54] pointed out that despite the growing number of circularity indicators, important aspects such as value retention over different cycles and product lifetime are not yet fully captured and need further research. LCA of plastic waste management has been widely performed, but most studies focus on mixed plastic waste, mainly rigid plastics, instead of plastic film waste. Generally, these studies identified recycling as the preferred treatment over non-recycling options when the quality of the recycled material is good enough to ensure high virgin material substitution rate[55]. However, in order to reveal the environmental benefits of recycling with high-end purposes compared to downcycling, the potential market size of the recyclates, as well as the recyclate quality, should be considered[56]. Finally, the relationship between packaging design and food waste (and consumer behavior) is an important aspect when developing new packaging concepts[57].

Step beyond the SoA: CIRCULAR FoodPack will evaluate the sustainability of the newly developed circular value chains in a holistic way, by applying a comprehensive sustainability assessment framework tailor-made by UGENT, taking into account environmental, economic and social impacts, as well as Circular Economy priorities, food waste and compliance to health and safety legislation. To support sustainable decisions by waste managers and processors

48 UNEP-SETAC (2011) Towards a LCSA: Making informed choices on products. United Nations En-vironmental Program. DOI: DTI/1412/PA.
49 ISO, 2006. ISO 14044: Environmental management - LCA - Requirements and guidelines. International Organization for Standardization, Geneva.
50 Zampori L and Pant R (2019) Suggestions for updating the Product Environmental Footprint (PEF) method. Luxembourg: JRC Ispra. DOI: 10.2760/424613.
51 Hunkeler, D., Lichtenvort, K., & Rebitzer, G. (2008). Environmental life cycle costing. https://doi.org/10.1201/9781420054736.
52 Taelman, S.E., Sanjuan-Delmás,D., Tonini, D. & Dewulf, J. (2019). An operational framework for sustainability assessment including local to global impacts: Focus on waste management systems. Resources, Conservation and Recycling, 162. https://doi.org/10.1016/j.rcrx.2019.100005
53 Huysveld, S., Hubo, S., Ragaert, K., Dewulf, J, 2019. Advancing circular economy benefit indicators and application on open-loop recycling of mixed and contaminated plastic waste fractions. Journal of Cleaner Production 211, 1-13.
54 Moraga, G., Huysveld, S., Mathieux, F., Blengini, G. A., Alaerts, L., Van Acker, K., ... & Dewulf, J. (2019). Circular economy indicators: What do they measure?. Resources, Conservation and Recycling, 146, 452-461. https://doi.org/10.1016/j.rescon-rec.2019.03.045
55 Horodytska, O., Valdes, F.J., Fullana, A., 2018. Plastic flexible films waste management - A state of art review. Waste Management 77, 413-425.
56 Nessi S., et al., Comparative Life Cycle Assessment (LCA) of Alternative Feedstock for Plastics Production – Part 1, European Commission, Ispra, 2020a
57 Yokokawa, N., et al. (2018). Framework for analyzing effects of packaging on food loss reduction by considering consumer behavior. J. of Cleaner Production, 174, 26-34.

running recovery processes, a digital Decision Support System (DSS) tool for sustainability optimisation and evaluation of the most feasible process cascade and sector of application for a certain secondary material will be developed. A combination of AI-based techniques with special focus on white label approaches, capable of providing understandable feedback to the users will be developed. IRIS previous experience on these systems for other sectors will be applied, combining Rule Induction algorithms for Expert Systems applications.

Innovation potential: The comprehensive sustainability assessment framework developed specifically for CIRCULAR FoodPack will evaluate the sustainability of the newly developed technology and packaging in a way that is: (i) systematically taking into account the entire life cycle, (ii) performed in a holistic way by considering environmental, economic and social impacts, (iii) addressing also circularity aspects, food waste and compliance to health and safety legislation. The DSS tool will provide crucial insights in relation to the potential value of the currently unsorted material and will support the uptake of AI-based digital tools as backing for waste managers decision making.

(vi) Market analysis, business and organisational models to ensure commercial uptake - related to WP8 work

SoA and drawbacks: More and more large multinational food and packaging companies commit themselves either individually or collectively in public or private initiatives (EU Circular Plastics Alliance or Ellen MacArthur Foundation's Global Commitment) to pledges to making their packaging more recyclable or to a percentage of recycled plastics content in packaging products by already 2025. However, market actors emphasize that these pledges are ambitious and only feasible if material is available in sufficient quality and quantity and at competitive prices[58]. Studies provide additional evidence that business models for a better plastic recycling are still facing multifaceted market barriers, e.g. uncertainties regarding supply, higher production costs, initial investment costs, material quality differences compared to virgin plastic and lack of waste segregation by waste generators.[59] The bottleneck for successful business models most often lies in the transfer from pilot feasibility to roll-out market feasibility. To overcome these market barriers for an improved sorting, separation and recycling of post consumer food packaging wastes, close cooperation and joint activities of current economic actors in the value chains are necessary that also consider market needs and therefore require specific business models.

Step beyond the SoA: CFP will develop business models for implementation of the CIRCULAR FoodPack value chain, suggesting actors and most suitable technological interfaces for sorting, recycling and production of food grade packaging films from post-consumer recycled plastics. These models will be jointly discussed and improved by process providing R&D and industry partners as well as current global/European players. The CIRCULAR FoodPack technologies will be developed based on a stakeholder-driven approach by integrating stakeholders' preferences and expectations. Economic actors of the whole value chain as well as consumers will be surveyed by qualitative and quantitative market research methods to include stakeholders' feedback in the development of the CIRCULAR FoodPack technologies and to ensure a quick commercial uptake of the technologies. For the development of business models, it is particularly important to focus not only on the purely economic feasibility of processes, but rather to include future scenarios e.g. effects of future EPR taxes and innovative business models that coincide with lower circular value chains costs. In CIRCULAR FoodPack, these scenarios will be included in the SWOT analysis to identify strengths, weaknesses, opportunities that also will identify actions to be taken to improve the market acceptance. Further on, exploitation plans for 2-3 most promising scenarios will be developed through a cost and process optimization approach provided by the Business Model Canvas (BMC).

Innovation potential: CIRCULAR FoodPack will identify suitable markets and material flows where the critical size for a market roll out can be easily reached. The project will define products that best fit to the CIRCULAR FoodPack technologies and where an improvement of the recycling processes is most necessary from an environmental point of view. Markets and sectors will be identified where the CIRCULAR FoodPack technologies create a value added for business customers and consumers and a possible price premium for the packaging can be generated. The activities in CIRCULAR FoodPack will ensure the coherence of pilot feasibility with market roll-out feasibility by connecting (1) technical and scientific excellence at pilot scale with (2) the access to large markets via the consortium partners to ensure a market roll out and (3) the definition of business models to exploit the potential of technologies generally applicable to other cases and companies in the market.

Freedom to operate within CIRCULAR FoodPack

The consortium conducted a patent survey to identify possible opportunities for knowledge and technological advancements (Table 4). There are no patents that hinder the development and innovation ideas based on the knowledge of the partners. Several project partners own their key patents related to the developments (Table 5).

58 European Commission 2019: Assessment report of the voluntary pledges under Annex III of the EU Strategy for Plastics in a Circular Economy.
59 Dijkstraa, H., et al., 2020. Business models and sustainable plastic management: A systematic review of the literature. J. of Cleaner Production.

Table 4. Results of the patent search focused on main innovation areas of CIRCULAR FoodPack (CFP)

Technology (Patent No) – Title & Short Description	Innovation potential
Method for sorting and recycling PP articles (EP1966295A1) - Use in PP compositions suitable for production of injection-stretch-blow-moulded bottles. UV tracer to improve the ease and accuracy of sorting PP and PET bottles.	CIRCULAR FoodPack aims to sort food and non-food flexible packagings, base on different marker concepts.
Deodourizing method of regenerated plastics (JP2007210199A) - Calcium hydroxide (~ 5 wt.%) is added to waste plastics, crushed, melted, extrusion molded into a strand, cut to small lumps, pellet manufactured	The technology needs additives, which is not preferable in circular use, and has several processing steps. CFP focuses on the removal of VOC by means of Infrared Dryer
Method and apparatus for processing of plastic material (US20110251368A1) - Extrusion system for thermoplastics incuding a suction device for removing volatile components from plastics	The technology requires a long residence time depending on the dimensions of the degassing unit. Plastics need to be melted prior to decontamination.
Method For Purifying Reclaimed Polypropylene (US20180171095A1): The plastic is brought in contact with a first extracting solvent and then dissolved in a second solvent. The second solution is purified by adding a solid media to produce a third solution comprising purer polymer	Different solvents are applied in series, which makes it a technical and energy-intensive technology.
Method and apparatus for recycling printed plastic films (EP1419829A1): Consecutive and/or simultaneous soaking phases with non-ionic surfactant, organic solvent, an aqueous mineral base are used as a deinking medium.	The recovery of deinking medium is difficult due to use of various solvent and surfactant mixtures. The technology is limited to deink surface printed plastic films.
Mult-ilayer Packaging films with recycled PA content. (WO2019231442) - No offensive odour, useful for packaging organoleptically sensitive food.	CFP will develop (i) recyclable all PE based mono-material multi-layered laminates for flexible packaging, (ii) including PE PCRs (50% of laminate) with functional barriers, (iii) tracers in "deinkable" inks for sortability. (iv) The laminate will fulfil all the functional requirements, e.g. oxygen, water vapor transmission, sealability, machinability, (v) The functional barrier will profide migration and gas and water vapor barrier performance.
All Polyethylene laminate film with barrier adhesive layer (WO2020074688 (A1)) - Recyclable, suitable for use in a flexible packaging.	
Gas barrier transparent film (JP2002321301A) - flexible, gas-barrier with a thin film layer of an inorganic oxide, better in recyclable properties	
Single polymer film for use in stand-up-pouches (EP2585298A1) - A mono- or a multi-layer film, has acceptable water vapor transmission rate and good tear resistance.	

The project consortium is convinced that the **sorting, deinking, delamination, deodourization,** and **recycling processes** used in **flexible packaging** is still largely unknown**.** Especially, in the area of **mono-material based laminates with PCRs** including **functional barriers**, the opportunities for patented IP are multiple, including options for expanding and perfecting partners' existing patents.

Table 5. Overview of the most important patents of the CIRCULAR FoodPack Project partners

Scope	CIRCULAR FoodPack – Relevant patents of the partners
Tracer based Sorting (TBS)	**POLY:** "Material having a marker and a method for sorting a mixture of materials:", EP2828053(B1); "Method for recycling materials from used objects", EP2019731 (B1); "Method for identifying materials", WO2017220079 (A1) **KIT:** "New Composition With Enhanced Luminescence", EP3431570(B1)
Sensor Based Specification (SBS)	**IRIS: "**Device for detecting foreign matter in an edible substance or in a container containing the edible substance and method employing the device" - EP19382101; "An integrated optical system for the examination of sample materials", EP18198458.
Deodourization	**KREYEN:** "Method for decontaminating bulk food and/or feed material", 10440978. "A new patent for deodourization of recyclates is under application."
Deinking	**UGENT: "**Method for deinking plastic material", EP 20167745; The present invention relates to a method to deink plastic material, for example plastic waste. The present invention also relates to a method to delaminate and deink plastic material.
CreaSolv®	**Fraunhofer:** "Method for treating waste containing plastic", WO2011082802 A1; "Verfahren zur Anreicherung von mindestens einem Polymer aus einem Polymer-haltigen Abfall und Polymer-Recyclat", DE102013212813A1; „Method for separating and recovering target polymers and their additives from a material containing polymers and use therefore", EP1311599b1; "Method for recycling polyolefin containing waste", EP3305839A1.
Recyclable laminates	**AMCOR** has an extensive know how on the production of poly propylene based mono-material based laminates. "Recyclable Films for Product Packaging", WO19132954; "Recyclable Films and Packaging", WO20050860; "High-Barrier Recyclable Film", WO20112148; "Packaging Films with Recycled Polyamide Content", WO19231442
Recyclable, Sortable packs with PCRs	**NESTLE:** No patents on this topic filed currently, but there are patent applications in preparation.

2. Impact

2.1 Expected impacts

CIRCULAR FoodPack will become a key accelerator in the transition of the plastics industry to a circular economy by generating (1) solutions that will create impact in terms of structured knowledge and standardisation; (2) solutions that are technological or product innovations with a TR L of 5–6, which will be moved further towards market maturity and exploited commercially; and (3) significant advances in food-compliant recycling.
CIRCULAR FoodPack will deliver results that achieve meaningful impacts for stakeholders in the plastics value chain, and for society and the environment in general. The core of the concept and work plan is the involvement of all partners along the value chain. This direct involvement is needed to enable co-creation and acceptance on the one hand and the timely uptake and implementation of project results on the other. The results, outcomes, and impacts of the CIRCULAR FoodPack project are summarised in Fig. 12.

2.1.1 Expected impacts mentioned in the work programme

(a) Increased yield and quality when sorting products made of composite or multi-layer materials (← SO1, SO2, SO5)

Current sorting plants for post-consumer plastic packaging typically include several steps to separate mono-material plastics from mixed packaging waste, leading to a residue rich in MLCs. This fraction can represent 20–30% of all post-consumer plastic waste[60] and is destined for **incineration** (Fig. 13). CIRCULAR FoodPack will take on this rejected fraction (including existing stockpiles) and apply innovative sorting technologies to bring large amounts back into circular economy rather than keeping the linear model. The project's **sensor-based specification (SBS) for mixed waste** will bring in a better inside into these rejects and help to select those that can significantly contribute to an increase of the net yield of such facilities. In addition, materials counteracting high yields in downstream PE recovery like PP based mono-films and MLP are detected. Although industrial implementation of innovative SBS in existing sorting lines is highly challenging, this is clearly targeted in the exploitation of CFP results.

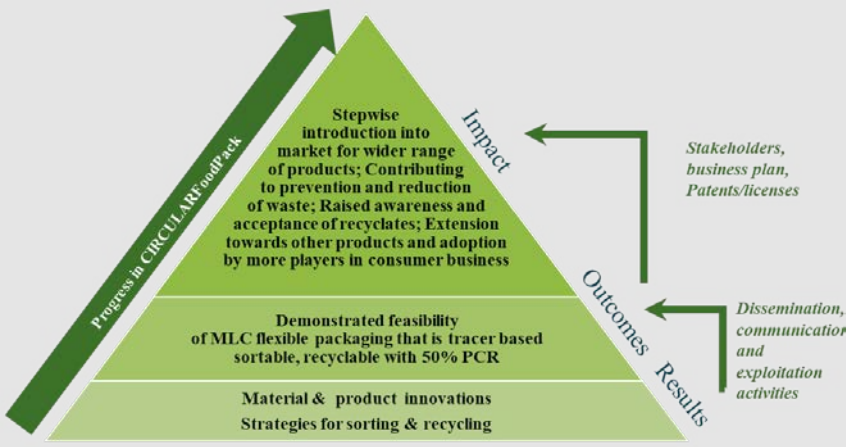

Fig. 12. CIRCULAR FoodPack – Results, Outcomes, and Impact.

In terms of quality, one major challenge is that it is not yet possible to sort food packaging from non-food flexible packaging items, thus hindering the transition to circularity for 'return to food contact' applications. CIRCULAR FoodPack will overcome this obstacle by applying innovative **tracer-based sorting (TBS) for tracer marked packaging items, such as the food-grade mono-material multi-layer films developed and demonstrated in this project.**

At TRL 6, we expect to recover more than 90% (yield) of newly designed and tracer-marked food-grade packaging items from bulk EoL packaging at a sorting purity (quality) of < 1% non-food-plastics, which is a requirement to achieve EFSA approval for non-PET recycled FCMs.[61] For existing stockpiles of untraced food packaging, SBS technologies and post-sorting treatments will allow more efficient separation and up-concentration of the MLC fraction.

Fig. 13. Mixed flexible plastics residue.

From post-consumer packaging waste entering sorting facilities, we will target fractions that are currently not well sorted due to the absence of advanced sorting technologies such as TBS, and also due to the lack of separate collection, purification, pre-treatment and recycling technologies to recover **valuable PCRs**. When launched on the market, such technologies will boost the recycling of flexible packaging waste from the current value of ~15% to > 50%.[62] Materials from TBS will be suitable for any application (although food packaging is a primary target), whereas the CIRCULAR FoodPack SBS process cascades will provide high-quality raw materials for demanding non-food flexible packaging markets, including films and larger bags in the B&C industry, and packaging products for home and personal care.

60 Communication with the project partner, SUEZ
61 EFSA CEF Panel, 2015. Scientific Opinion on the safety assessment of the processes 'Biffa Polymers' and 'CLRrHDPE' used to recycle high-density polyethylene bottles for use as food contact material. EFSA J. 13. https://doi.org/10.2903/j.efsa.2015.4016
62 Personal communication with CEFLEX. Boost is mostly related to enabling materials recycling from MLC rich flexible packaging waste < DIN A4

(b) Increased recycling of raw materials from products made of composite or multi-layer materials, in terms of volume and/or quality (← SO2, SO3, SO5)

For the first time, CIRCULAR FoodPack will achieve the recycling of formerly 'unrecyclable' food-grade EoL MLC packaging waste by developing innovative plastic recycling approaches that recovers > 90% of PE from the input material, including multi-layered packaging films. In this respect polymer purity is not a single determining property but a more complex concept encompassing:

(i) polymer-pure composition, achieved by the separation of non-target polymers by delamination, solvent-based purification and melt-filtration/re-compounding, and supported by upstream grinding;

(ii) lack of odour, achieved by washing and deodourization (e.g., as applied to flakes after de-inking and/or pellets after regranulation); and

(iii) the lowest levels of IAS and NIAS, achieved by washing, de-inking and solvent-based purification and deodourization.

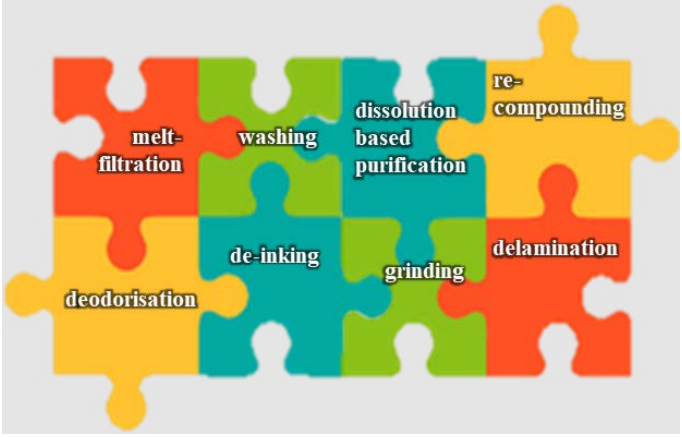

Fig. 14. Effective polymer purity as the cumulative result of multiple synergetic treatement steps.

As such, polymer purity cannot be attributed to a single treatment step, but is instead achieved by synergistically combining and optimising several aspects of waste treatment (Fig. 14). In CIRCULAR FoodPack, the resulting polymer purity will be > 99% and the technical quality of the recovered materials will be comparable to at least 90% of original raw materials. This is because CreaSolv® treatment also eliminates downgraded products (oligomers) from use-phase and post-consumer waste treatment. Odour, IAS and NIAS levels will fall to a range that allows food-safe re-application in mono-material multi-layer films produced during the project, mainly in combination with functional barriers. Accordingly, multi-layer polymers from food packaging will, for the first time, contribute to a circular economy for plastics at the most sensitive quality level. This will achieve a transformation from a linear take-make-dispose model to the holistic inclusion of different recycling technologies.

(c) Reduced use of virgin raw materials (← SO3, SO4, SO5)

MLC plastic waste from food packaging is currently considered unrecyclable, but CIRCULAR FoodPack will render these MLC plastics recyclable, thus removing them from the materials destined for incineration (or even landfill) and injecting them into the volume of materials available for the production of new packaging. By using the recovered materials in this manner, the use of virgin raw materials will be reduced drastically. If only a conservative 5% of LDPE is exchanged in this way, it would avoid the use of 450,000 tonnes of virgin LDPE per year,[63] in turn preventing the emission of 714,500 tonnes CO_2 and thus contributing not only to circularity but also to the EC's climate goals.[64] CIRCULAR FoodPack aims to replace at least 70% of one single bulk PE layer in the final flexible laminate with PCR, avoiding the use of **6,300,000 tonnes of virgin PE per year**. Plastic packaging has a better weight/functionality ratio than glass and paper, and fulfils the main function of packaging: the protection of goods according to different requirements. The project's results will help to reduce the amount of oil needed for packaging applications, thus making a valuable contribution to the sustainable use of resources. This means the project will replace the linear approach (production – consumption – waste) to achieve the first real circular economy for these materials (Fig. 15).

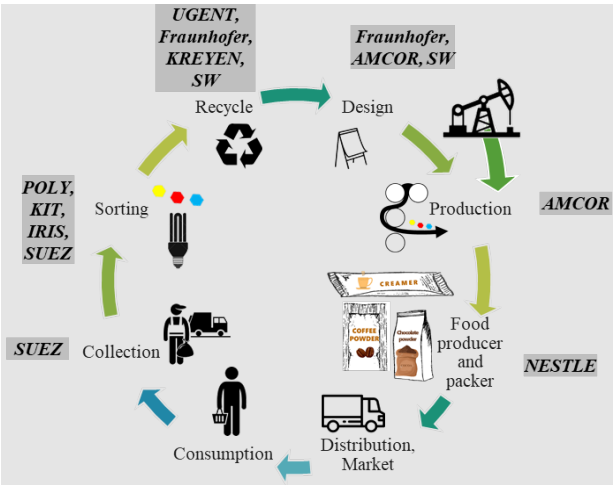

Fig. 15. Circularity of food packaging.

(d) Increased knowledge on the design for reuse and recycling ("circular design") of products currently made of composite or multi-layer materials (← SO3, SO4, SO5)

The packaging industry must urgently improve the sustainability of its products. In Europe, three quarters of all products are packaged in plastics, and ~40% of all plastic applications are found in the short-life packaging sector, which produces two thirds of all plastic waste. The changing legislation pertaining to packaging materials and increasing pressure from other market participants (see ALDI Packaging Mission)[65] have increased industry and end-

63 Based on annual virgin LDPE tonnage of 9 Mtonnes, Plastics-The Facts 2019, Plastics Europe
64 Based on EcoInvent data: 0.69 kgCO2 eq for rPE versus 2.28 kgCO_2 eq for 1 kg virgin LDPE.
65 ALDI_National Packaging Policy_EN_ALDI Belgium and Luxemburg_July 2019

user demand for packaging R&D. In parallel, §21 of the new Packaging Act in Germany[66] creates incentives for the use of recyclable plastic materials. The legislator defines a minimum standard for this purpose, which dual systems use as a guideline for the license fees for marketers. Furthermore, the strategy for plastics in Europe is that all packaging should be reusable or recyclable by 2030.

CIRCULAR FoodPack will use 'design for and from' recycling approaches to build mono-material based multi-layer laminate structures that are completely recyclable and produce PCR plastics that can be reused in the same packaging design in compliance with food-contact regulations. EU Regulation No 282/2008 strictly governs the application of PCR in FCMs, so CIRCULAR FoodPack will perform migration tests and modelling to develop appropriate functional barrier concepts that ensure food-contact applications are safe. Significant knowledge will be gained to understand and define design concepts and to develop and validate suitable tests for the assessment of food safety in compliance with existing European legislation. In addition to functional barrier concepts for food compliance, the structures must also fulfil the same barrier properties against gases and water vapour to protect the packaged goods and thus ensure quality and shelf life. The mechanical and sealing properties of the new structures must also be confirmed. Tracers used for the sorting of novel mono-material based multi-layer laminates waste will be easily washable, because the ink formulations will be tailored to make them debondable during pre-treatment. These debondable and lighter-coloured printing inks will ensure efficient washing, enhancing the recyclability of the tracers. The tracers will not represent a significant change in printing requirements and will easily be taken up. Given the very short usage time of our target products we can implement the CFP technologies on a short term, generating an early impact on the circularity of food packaging.

The recovered PE PCRs will be used to produce at least three demonstrators for the flexible packaging of coffee, creamer and cocoa powder. These will be ground-breaking **European demonstrators of recycled flexible plastics being returned to food-contact applications** other than PET bottles. The demonstrators will be validated against virgin counterparts. The novel laminate design concepts will also be shown with three other demonstrators in non-food packaging for detergents, cleaning agents, and personal care (home and personal care (HPC) packaging), covering almost all of the non-food product sectors (Fig. 16)[67].

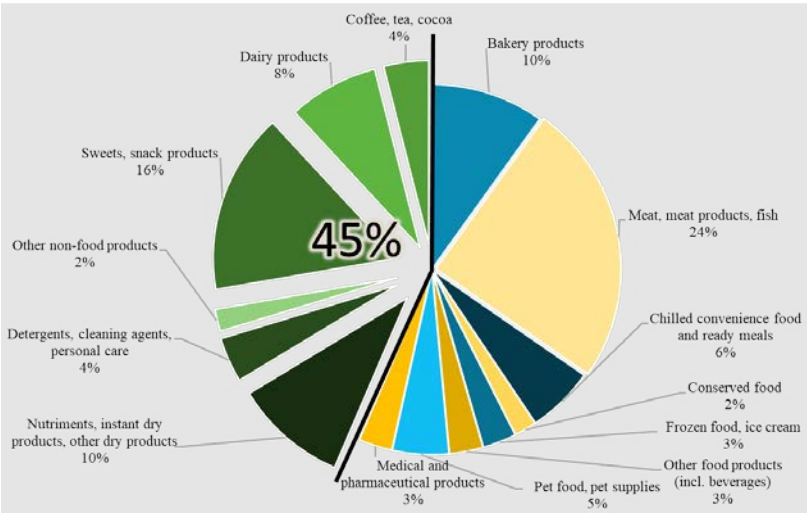

Fig. 16. Share of product categories for packaging in Europe.[64]

By comparing the performance (technically and in terms of circularity) of our demonstrators with current products handled in current EoL options and improved EoL processes, we will gain an in-depth understanding of the contribution of our advances compared to the state of the art. Our results will serve as flagships for the eco-friendly design of recyclable flexible mono-material-based multi-layer products and will be disseminated in targeted outreach and communication actions and events. The project website will show a simplified and public version of the decision support tool (DSS) to transfer relevant circular design knowledge to stakeholders. We will publish guidelines for the circular design of at least six demonstrators representing the packaging formats used in flexible food and HPC packaging. Based on a very rough calculation, these CIRCULAR FoodPack demonstrators represent up to 45% of the product categories that use plastic/plastic MLC materials in Europe, which makes them highly relevant as guiding examples for the whole industry.

The CIRCULAR FoodPack consortium includes highly competent partners spanning the value chain, starting from the sorters and recyclers through to packaging producers and food product suppliers. Interest groups from the B2C market (manufacturers of branded goods, and retail companies) are important drivers for the market launch of products, but the economic return is also seen in the B2B markets in cooperation with packaging manufacturers, which will be undertaken within the project by NESTLE.

(e) Increased knowledge on the process environmental footprint, including the net effects on greenhouse gas emissions, of improved sorting, separation and recycling of composite and multi-layer materials. (← SO6)

In addition to compiling results from the LCSA and simulating different pathways along the value chain, the best process cascades and applications will be recommended based on multi-parametric optimisation models. Iterative feedback will be delivered to process and packaging developers based on the LCSA results, ensuring that the new process is at least **30% better** in terms of greenhouse gas emissions, 32.5% better in terms of energy efficiency[68]

66 German packaging law, https://deutsche-recycling.de/en/packaging-regulation/
67 GVM Flexible Plastic Packaging Market in Germany and in Europe, 2016.
68 https://ec.europa.eu/clima/policies/strategies/2030_The%202030%20climate%20and%20energy,32%25%20share%20for%20renewable%20energy

than suitable baseline models (open loop recycling or incineration). Table 6 shows how we expect the CIRCULAR FoodPack technologies to lower the environmental footprint (including net effects on greenhouse gas emissions) of the sorting, separation and recycling of MLCs.

The assessment of the overall environmental performance of the novel CIRCULAR FoodPack technologies is one of the objectives of WP7. Holistic LCA will be applied to every step of the new processes and the resulting demonstrators according to ISO 14040/14044. We will aim to identify hot spots and select processes and products for upscaling based on the lowest projected greenhouse gas emissions. This will be coupled with recommendations for government and policymakers on the implementation of certification and bonus/malus schemes for more environmentally sustainable multi-layer products at an acceptable cost for society over their full lifecycles. Novel certificates (e.g. specifying the content of recycled materials) and labels for use by producers and retailers will be designed. Furthermore, recommendations in terms of product information, tracking and optimal use throughout the lifecycle will be developed. We will deliver at least two fact-based, scientific white papers on CIRCULAR FoodPack processes and products compared to current benchmarks and waste management approaches, which will be targeted at EU policymakers.

Table 6. Benchmarking of CIRCULAR FoodPack planned innovations compared to other solutions.

Aspect		Standard solution	Other emerging solutions	CIRCULAR FoodPack
Flexible packaging	EoL approach for PE	• Downcycling to low-quality products • Landfill or incineration for energy valorisation	• Chemical recycling is complex, expensive and uses hazardous chemicals	• Recycling processes can apply to standard mono-material, MLC, and novel mono-material laminates, also solving sorting issues of food packaging waste
	Reuse opportunity	• Limited to post-industrial waste, some downgrading • Recycled post-industrial waste in MLC is much less than 10% and no PE PCR used in flexible food packaging	• Chemical recycling can produce food-grade plastic for packaging at higher costs and higher CO_2 emission	• Reuse in similar high-quality applications, food packaging • > 50% PE PCR targeted in flexible food packaging laminates
	CO_2 emission	1.9-2.2 kg CO_2 / kg PE[69,70] (virgin production, cradle-to-gate[70])	1.2 kg CO_2 / kg PE[71] Reduction by 35%	0.5-0.7 kg CO_2 / kg PE PCR[69,70] Reduction by 70%

2.1.2 Further substantial impacts

Reducing the volume of plastics sent to landfill and incineration. In Europe, 7.2 million tonnes of plastics are sent to landfill annually and 12.4 million tonnes are incinerated for energy recovery.[72] The long-term persistence of plastics in the environment is exacerbated by their cumulative effect. Recognized as one of the major international environmental challenges by the European Environment Agency (EEA)[73], plastics account for 89–95% of marine litter. This currently includes all MLC packaging. The sorting and upcycling technologies developed by CIRCULAR FoodPack will shift this post-consumer packaging from linear, environmentally unattractive options to effective recycling, which is vastly preferable and also introduces a true circular economy for these products. The shift from incineration with energy recovery to material recycling could result in a reduction of 3.3 tonnes CO_2 equivalents per tonne of MLC flexible packaging waste. This includes the benefit gained from replacing virgin granules with recycled counterparts. Here, 90% of PE from currently un-used packaging waste sorting fractions entering the recycling facility will be mechanically recycled to granules while the residues of the recycling process are incinerated. This avoids the emission **of 3.1-5.0 million tonnes of CO_2 equivalents per year** corresponding to the entire estimated volume of the **MLC flexible** packaging market in Europe (1.5-1.8 million tonnes per year)[74].

Additionally, CIRCULAR FoodPack will significantly help to reduce the amount of plastic leaking from controlled waste management systems, thus reducing environmental pollution and improving ecosystem resilience. This will benefit consumers by providing access to a healthier environment and therefore food that contains lower levels of microplastic fragments and harmful substances leaking from plastic waste.

Reducing the volume of exported plastic waste. Despite the 2018 import ban in China, it has remained common practice to ship 'unrecyclable' plastic waste halfway across the globe, primarily to South-East Asia, where its ultimate fate is unclear. By increasing the value of MLC packaging waste, CIRCULAR FoodPack will make it more attractive to manage the waste regionally and recycle it within the EU rather than exporting it.

69 The Association of plastic recyclers APR,LCAs for post-consumer recycled resins: PET, HDPE and PP, 2018
70 Based on Ecoinvent Data: 0.69 CO2 eq. for rPE versus 2.28 kg CO2 eq. for 1 kg virgin LDPE
71 Values from the study of CE Delft on mixed plastic (DKR350) in NL (2019)
72 The Circular Economy for Plastics, A European Overview, Plastics Europe, 2019.
73 Source: State of Europe's seas, EEA Report No 2/2015, ISSN 1977-8449
74 CEFLEX, Designing for a Circular Economy: Recyclability of polyolefin-based flexible packaging, 2020.

Industry growth and job creation. The circular economy unlocks tremendous potential for the development of innovative new materials. The global paradigm shift from the take-make-dispose linear model towards a circular model represents a unique opportunity for Europe and the European process industries to kick-start a world leading competitive circular economy with significant economic, environmental and societal benefits. To increase the innovation capacity of the industry, the approach is to capitalise on a valuable opportunity for the EU plastic recycling industry and end-user industries to replace currently-used virgin materials. The closed loop will also provide the industry with an opportunity to obtain value from waste rather than paying for disposal. CIRCULAR FoodPack is a timely development that will enable the transition to a circular EU plastic industry. This will provide great business opportunities reinforced by increased visibility, and policy awareness and support for such solutions. Following current market trends for recycling and growth in the target end-use sectors, CIRCULAR FoodPack will achieve high market penetration to meet the targets of recycling 75% of plastics by 2030, leading to significant increases in revenue and more jobs on the waste provision and recycling levels of the value chain. Significantly reducing the volume of waste plastics sent to landfill, incineration and export creates an opportunity for the expansion of existing plastic-recycling facilities and the building of new ones across Europe, leading to local job creation and economic growth. Typical recycling plants for LDPE films have a throughput of 10,000–20,000 tonnes per year and employ 20–40 people. Based on the projected collection of 0.5 million tonnes of food-grade polyolefin flexible packaging feedstock in Europe, CIRCULAR FoodPack technology will lead to the construction of more than 25 new sites dedicated to food-grade recycled LDPE, directly creating ~500 new jobs as well as additional jobs in plant construction and the supply chain. To avoid transporting waste over long distances, the sorting and recycling facilities will be widely dispersed, including rural areas, also providing opportunities for SMEs. Further jobs will be created in the sector for new marking and sorting technologies.

Furthermore, convertors would gain access to food-compliant PCRs allowing large segments of the food market to be served with PCR-containing packaging solutions, not only allowing Fast-moving Consumer Goods (FMCGs) to meet their sustainability targets and comply with new national and European legislation, but also preventing a return to paper/rigid/metal/glass packaging (which would until then offer a more circular approach but at the cost of a larger carbon footprint). The introduction of PE PCR content into mono-material flexible packaging solutions would further strengthen the oncoming investment wave with regards to (co)extrusion, converting and packaging capacity suitable for the production of mono-material packaging, resulting in several billions worth of capital and the direct and indirect creation of tens of thousands of new jobs in the converting industry over the next 5–10 years.

CIRCULAR FoodPack will therefore support circular business models and associated employment. While contributing to the implementation of the Circular Economy Action Plan, the Seventh EAP, and the Commission Strategy on Plastics, the project will also help to implement priority goals in the field of green growth and resource efficiency in parallel with the overarching goals of EU 2020 Strategy to develop Europe in a smart, sustainable and inclusive way (Table 7).

Table 7. Impact of CIRCULAR FoodPack on the EU 2020 strategy.

EU2020 Priority	EU Flagship	CIRCULAR FoodPack Contribution
Smart growth	Innovation Union	Creation of new market within the recycling sector including creation of jobs; facilitating technology upscale and uptake; increase EU industrial competiveness; technological leadership in circular economy
Sustainable growth	Resource efficient Europe	Improvement of resource efficiency; essential energy and CO_2 savings in production of flexible food packaging; reduction of waste and landfilling/incineration of valuable resources in line with EU policies; reduction of dependency on petroleum imports by replacement of petrochemical products with SRMs
Inclusive growth	Agenda for new skills and jobs	Safeguarding and developing industrial safety to establish better working conditions and thus improving the quality of jobs in Europe

The recycling sector and business models proposed by CIRCULAR FoodPack will increase GDP, create jobs and stimulate local economies. Today, the recycling sector offers a diverse range of jobs ranging from low-skilled to highly-qualified positons, covering the collection, storage and handling as well as the processing and production of goods. Although Eurostat does not keep records of employment in the recycling sector, their conservative estimates indicated a continual increase in recycling-related employment between 2008 and 2015.[75] An EC impact assessment of a legislative proposal on waste and recycling targets to 2035 predicted that up to 178,000 direct jobs could be created in the recycling-related employment sector.[76] CIRCULAR FoodPack will contribute to this development, with a significant share of highly-qualified personnel as usual for this sector, reflecting the development of innovative technologies and life-long-learning programmes for existing staff.

Synergistic growth through scientific cooperation. CIRCULAR FoodPack is a Research and Innovation Action (RIA) and thus focuses on the technological aspects of realising the circular economy for plastics. Setting up a forum to exchange knowledge and experience with holistic projects also covering the social and political aspects of circular

75 https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=SWD:2018:17:FIN
76 EU Commission: Additional analysis to complement the impact assessment SWD (2014) 208 supporting the review of EU waste manag. targets. Brussels

plastics will embed the project within the value chain and broader developments in the field. This link will be made with the project MSCA-ITN-EJD C-PlaNeT (https://www.c-planet.eu/), which includes Fraunhofer as a partner and is coordinated by UGent. Likewise, CIRCULAR FoodPack will provide valuable input to existing industrial networks striving to increase the recycling of (flexible) packaging, such as CEFLEX, New Plastics Economy (Ellen MacArthur Foundation) and HolyGrail2.0. Several CIRCULAR FoodPack partners are also partners in or consultants to these global initiatives, providing a direct line for knowledge exchange. HolyGrail2.0 focuses on watermarking technology for food/non-food sorting, which is parallel to the envisioned developments in CIRCULAR FoodPack. HolyGrail2.0 technology is currently at TRL7 and will be advanced to TRL8–9. The greatest weakness of watermark technology (a more or less 2D surface is required for scanning, which is not guaranteed in 'crumpled' waste) would be solved by the CIRCULAR FoodPack TBS technology (which does not require flat surfaces for scanning), making the projects complementary rather than competitive. Exchange of knowledge between initiatives will help European industry to identify an optimal combination of technologies to further realise the circular economy for plastics.

Input for standardisation and guidelines. The EC states that a vital aspect of moving forward with the European Strategy for Plastics in a Circular Economy is the development and implementation of supporting standards and guidelines.[77] There is the broad standard on the recovery and recycling of plastics (ISO 15270:2008) and specific standards for sampling and sample preparation (CEN 16010 and 16011), and the characterisation of mono-material plastic waste (EN 15342 to 15348), but no standards for the characterisation or recycling of MLC plastics.

Likewise, existing design guidelines – often provided by industry associations such as PRE[78] or CEFLEX[79] – focus on obtaining as much mono-material waste as possible, based on the status quo that MLC is 'unrecyclable'. CIRCULAR FoodPack will lay the foundations for the expansion of standards and guidelines for the development, sorting and recycling of MLC plastic packaging. Although the guidelines will mainly apply in Europe, it is evident that many of the problems facing Europe are replicated in other parts of the world, and guidelines will be valid as long as local and regional differences are taken into consideration. In addition, there is a distinct contradiction in the objectives of waste management legislation (Directive (EU) 2018/851 on waste) obligating food packaging producers to increase recycling rates, while FCM Recycling Regulation (EC) No 282/2008 strongly emphasises food safety, when new FCMs are made from post-consumer recycled FCMs. The EC is aware of this conflict[80] and stakeholders agree that safety is more important than recycling targets. It is also clear that current recycling processes do not enable the purification of polyolefins and the removal of contaminants of concern, in order to guarantee food safety. CIRCULAR FoodPack will provide much-needed research in this area, offering scientific evidence for future legislation and standardisation covering the use of recycled polyolefins in food-contact applications. We will develop and validate novel tests for the compliance and safety assessment of polyolefin-based recycled materials in flexible food packaging, which will benefit EFSA (responsible for risk assessment at the EU level), national monitoring authorities, and the EU legislature. The EU food and packaging industry will also benefit from this support for compliance assessment.

Cross-cutting priorities. Socio-economic science and humanities (SSH), and gender aspects. The general public currently has a negative perception of plastic products in general, and it is often unclear which plastics provide different types of benefits for the protection of goods. Non-Governmental Organisation campaigns broadcast the dramatic effects of plastic waste and litter (especially media representation of marine litter and micro-plastics). CIRCULAR FoodPack technology fulfils all the requirements for circular material flow, addressing plastic pollution in the environment (land and sea) by implementing new sorting and recycling concepts. CIRCULAR FoodPack includes an SSH dimension, specifically market analysis, business planning, and gender-relevant marketing strategies, These have been applied from the conception of the project, which will ensure project implementation in a stakeholder-driven approach to ensure acceptance by actors spanning the value chain, including consumers. This will encourage the purchase of products derived from SRMs and the collection of EoL packaging waste. The influence of environmental factors on the selection of materials differs by gender as well as EoL concerns. We will address these issues in our s-LCA, as well as market research to gain insights into how consumers' environmental values are shaped, allowing us to develop creative ways to engage with diverse audiences and communities about the societal challenges linked to limited resources and unsustainable consumption patterns. In addition to consumer perception studies, the project will implement a stakeholder-driven approach by conducting interviews with packaging experts such as packaging producers, food manufacturers, retailers, waste management representatives and recycling facilities on the perceptions and acceptability of the CIRCULAR FoodPack products. Furthermore, the project will host co-creation workshops with scientific, industrial and consumer communities as a systematic approach for stakeholder cooperation to discuss eco-friendly design, standardisation, incentive schemes and DSS.

CIRCULAR FoodPack is a model project in terms of promoting women in science, and many female principal investigators (PIs) among the project partners already have prominent roles. Examples include (1) Prof. Kim Ragaert, leader of the CAPTURE platform and nominee for the Plastics Recycling Ambassador award; (2) Dr. Diana Kemmer, head of the "Product Safety and Analytics" department and a certified food chemist with more than 15 years of

77 European Commission, 2018. A European Strategy for Plastics in a Circular Economy.
78 "https://recyclass.eu/recyclass/design-for-recycling-guidelines/." [Online]. Available: https://recyclass.eu/recyclass/design-for-recycling-guidelines/.
79 "https://ceflex.eu/guidelines/." [Online]. Available: https://ceflex.eu/guidelines/.
80 EU Commission: Study supporting the Evaluation of Food Contact Materials (FCM) legislation - (Regulation (EC) No 1935/2004), July 2020

experience at the Fraunhofer Institute for Process Engineering and Packaging (IVV); (3) Dr. Esra Kucukpinar, also from Fraunhofer IVV, who will co-coordinate the project and control the development of new recyclable mono-material flexible packaging designs based on her > 15 years of experience in this field; and (4) Dr. Virginie Decottignies, head of the Material Recycling and Recovery Department at CIRSEE Material Cluster, and expert in waste characterisation and plastic recycling. When presenting the accomplishments of the project to industry and the public, their roles will be made prominently visible, without necessarily stressing their gender. This should contribute to the general public's appreciation that it is perfectly normal for women to have a leading role in the sciences. As such, our project will consider the relevance of the gender dimension in communication and training, to ensure the message of diversity in science reaches both males and females. This will help to educate both current professionals and the next generation of scientists and engineers needed to realise the EU circular economy and maximise its creativity and innovation. The gender dimension of our consortium and our measures to align with Article 16 of the H2020 programme are presented at the end of Section 3.3.

Contribution to improving innovation capacity and the integration of new knowledge; dissemination of project outcomes to the current and next generation of the waste, recycling, converting and packaging end-users.

The EU plastic strategy estimates that only 5% of the value of plastic packaging material remains in the economy, the rest being lost after a very short and single use phase.[81] As such, the CIRCULAR FoodPack technologies and innovations will allow the EU to reduce the annual €70–105 billion lost by not maintaining plastic within the circular economy. Although the generation of new knowledge is positive, industrial use increases its innovation capacity and realises impact. To ensure market-replicable innovation, the consortium therefore unites actors across the industry value chain, which is necessary for market success, and the project has integrated a number of transversal activities such as environmental, regulatory, safety and techno-economic assessments. Finally, business planning activities will maximise the post-project exploitation of the results. The project gives trans-sectorial industry and transdisciplinary research and technology players an opportunity to collaborate and share in the technical knowledge and market expertise. Effective knowledge transfer from the RTOs (Fraunhofer, UGENT, UM, KIT, NTUA) to the participating enterprises, especially SMEs (POLY, KREYEN, IRIS, ECOZEPT) often lacking their own R&D capacities, will encourage the industry partners to carry forward the results post-project, which will increase the likelihood of market uptake. We have included Task 8.2 to ensure this seamless knowledge transfer.

As discussed in the Ambition section, CIRCULAR FoodPack is founded on a unique patented process (with potentially patentable improvements), which together with the high added-value applications have great potential compared to existing processes and recycled products. The work plan includes all activities needed to maximise the post-project readiness of the process and to exploit the results in a short time, therefore making high-quality PE PCR available to feed the new value chains.

Although the consortium focuses on PE-based packaging solutions for sorting, recycling and reuse in flexible packaging, the partners are nevertheless aware that the technologies, methods, concepts and innovative processes arising from the CIRCULAR FoodPack project will bring tremendous value to the plastic packaging industry as a whole, including rigid packaging and thermoformed trays, encompassing multiple polymer families such as PP, PS and PET, covering at least 30 million tonnes of the plastics used for packaging (17.8 million tonnes of this were collected as post-consumer waste in 2018)[82].

Based on this unexploited potential, CIRCULAR FoodPack's objective is to increase joint research and innovation capacity within the industry, opening new market segments to replace currently used virgin materials. This closed loop will also change the perception of EoL plastics from a burden on society to a valuable Secondary Raw material (SRM) stream. This will be reinforced at the downstream level (SRM converters and end users), which will benefit from new products, revenues from their post-industrial wastes and lower landfill taxes. CIRCULAR FoodPack offers the European plastics processing industry an economically sound strategy to gain a strong foothold in the recycling of flexible packaging, which will lead to an invaluable head start in this global competition.

2.1.3 Framework conditions and barriers / obstacles

Framework conditions influencing the expected impact

Expected legislation on the content of recycled plastics. There is currently a strong drive for legislative measures that would – directly or indirectly – support the uptake of recycled plastics into new resins or products, which would be in line with the European Strategy on Plastics. It is currently unclear whether this would be enacted in the form of a taxation on virgin resins (as planned in Italy)[83], a taxation on packaging products not containing a minimum content of recycled plastics (as supported by Member States like Finland and Denmark), an obligatory recycled content in products (as already decided by the European Parliament for beverage bottles)[84], or a combination of the above. In any case, when such legislation comes into force, this will increase the demand for high-quality recycled plastics.

81 https://ec.europa.eu/commission/news/first-ever-europe-wide-strategy-plastics-2018-jan-16_en
82 Plastics – The Facts 2019, An Analysis of European plastics production, demand and waste data, Plastics Europe.
83 https://www.reuters.com/article/us-italy-tax-plastic-tampons-idUSKBN1Y31UP
84 EU Parliament, 2019. DIRECTIVE (EU) 2019/904 OF THE EU 1075 PARLIAMENT AND OF THE COUNCIL, June, 2019 on the reduction of the impact 1076 of certain plastic products on the environment. Off. J. Eur. Union.

CIRCULAR FoodPack will help to meet that demand by making MLC plastics effectively recyclable and providing validated concepts for food packaging containing substantial quantities of PCR plastics.

Food contact regulation for recycled plastics. FCM approval for recycled plastics – especially those other than PET – is currently difficult to obtain.[85] No guidelines exist for flexibles, but it is common practice to build on existing guidelines, like the ones for PET. For example, EFSA has stipulated that the input stream for HDPE must be 99% pure in terms of utilisation as FCM before becoming waste, which is stricter than the 95% requirement for PET[85]. This is mainly because the industry was able to state that all PET entering the market was for food packaging,[86] whereas this is not the case for HDPE. Flexibles are expected to fall under the same higher purity target because there are also flexible packaging products that are not FCMs. CIRCULAR FoodPack TBS technology can meet this 99% target, which has previously only been possible (via the WRAP consortium) by hand picking,[87] which is not economically viable in the EU. CIRCULAR FoodPack would therefore offer the first automated process that has the potential to achieve FCM compliant input streams for flexibles.

The market price of oil. The crash in the oil price (May 2020), caused by international market pressure and the COVID-19 pandemic, has placed immense strain on the recycling of plastics.[88] As the oil price goes down, the price of virgin plastics declines accordingly, with more than a 30% decrease for PET and polyolefins. This results in a very strong competitive disadvantage for recycled pellets, which have a more or less fixed price determined by the costs of waste collection, sorting and reprocessing. For example, PET flakes for the production of polyester fibres fell 50% in price in May and June 2020. Even so, driven by some of the framework conditions mentioned in this section (pledges, drive for recycled content), the prices of food-grade rPET – the only plastic recycled at scale for FCM applications – not only remained stable but so did the demand. The CIRCULAR FoodPack TBS technology will help build a similar economy, which does not currently exist, for rPE. This new circular economy for plastics will be strengthened by a new wave of Extended Producer Responsibility (EPR) fees and (inter)national taxes specifically derived from the arival of plastics containing non-recyclable/non-PCR materials on the market, compensating the pricepoint differential of virgin plastic (packaging) solutions. Over time, these new regulations may even be used to prevent non-sustainable packaging from entering the market in the first place.

Public awareness and industry pledges. The public has a generally negative perception of plastic-based packaging systems and often does not understand the functions and benefits of plastic packaging. CIRCULAR FoodPack can help to improve this image by advancing the concept that the packaging material can be recycled rather than discarded after single use. Driven by public awareness and genuine willingness to 'do better', many companies have made formal pledges to either make their products recyclable or to increase the uptake of recycled content in their products. These pledges may be individual, via the EU's Circular Plastic Alliance[89] or bundled via network organisations like the Ellen MacArthur Foundation, which has more than 450 signatories to its Global Commitment to start building a circular economy for plastics.[90] The signatories include companies representing 20% of all plastic packaging produced globally, some of which are well-known consumer businesses such as L'Oréal, MARS, NESTLE, PepsiCo, The Coca-Cola Company, Unilever, the world's largest retailer – Walmart, major packaging producers such as AMCOR and Berry Global, and two of the largest resource management specialists – Veolia and SUEZ. Approximately 60% of plastic packaging is reusable, recyclable or compostable today. The highest percentage of recyclable material is found in rigid plastic packaging, like bottles. The companies listed above are working towards the 2025 ambition level of 100% reusable, recyclable or compostable plastic packaging. This is increasing the demand for **high-quality recycled flexible plastics**, which CIRCULAR FoodPack will help to meet. Some companies, which are part of the New Plastic Economy alliance, are also members of the project consortium. (Table 8).

Table 8. Commitments and planned actions of some top companies[91] (NESTLÉ, AMCOR, SUEZ) partners in CIRCULAR FoodPack

SUEZ (Volume of plastic processed): 150,000 metric tonnes
Commitment: Triple the sorting and recycling infrastructure to reach 600,000 metric tonnes production capacity of high-quality recycled plastics. **Action:** Implement new sorting strategies on selective collection waste to extract valuable PE fractions of flexible packaging from the light plastic fraction at sorting plants. Increase the performance of sorting lines. Implement new recycling plants dedicated to flexible PE fractions from packaging, including production lines dedicated to food-grade PE.
AMCOR: (Plastic packaging volume): NOT DISCLOSED
Commitment: (1) a full transition to recycle-ready structures by 2025; (2) introduction of recycled content back into our packaging solutions as much as possible; and (3) collaborate along the value chain to increase recycling rates. **Action:** (1) More than 100 Full time equivalents (FTEs) are working to phase out multi-material packaging solutions divided among four global inno-streams (long-term R&D trajectories). AMCOR is part of EMF, Ceflex, New Plastic Economy, and Plastics Recyclers Europe, which support and implement "design for recycling" guidelines across the industry. (2) The implementation of recycled

85 De Tandt et al. Mechanical recycling of plastic food contact materials – a regulatory overview of the obstacles imposed by REACH and the FCM regulations for recyclers. In print, Waste Management 2020.
86 EFSA CEF Panel, 2011. Scientific Opinion on the criteria to be used for safety evaluation of a mechanical recycling process to produce recycled PET intended to be used for manufacture of materials and articles in contact with food. EFSA Journal 9 (7:2184).
87 Welle, F., 2005. Develop a food grade HDPE recycling process.
88 https://www.icis.com/explore/resources/news/2020/05/19/10509185/recycling-chain-facing-deeply-challenging-times-investment-opportunities-remain
89 https://ec.europa.eu/growth/industry/policy/circular-plastics-alliance_en, Webpage: New Plastic Economy
90 https://www.ellenmacarthurfoundation.org/our-work/activities/new-plastics-economy/global-commitment.
91 https://www.newplasticseconomy.org/assets/doc/GC-Report-June19.pdf

content back into packaging structures without compromising functionality and product safety. Both standard and advanced recycled PCR are being explored with several tens of kilotonnes of material to be processed from 2021 onwards already allocated to various Amcor packaging solutions.
NESTLE (Plastic packaging volume): 1,700,000 metric tonnes
Commitment: None of the packaging, including plastics, ends up in landfill or in oceans, lakes and rivers. To reduce the use of virgin plastics by one third by 2025. **Action:** 100% of packaging is reusable or recyclable by 2025, shifting from virgin plastics to food-grade recycled plastics and accelerate the development of innovative packaging solutions.

Barriers and obstacles

Despite the innovation potential that CIRCULAR FoodPack creates, some obstacles could hinder the uptake of its technology. Successful uptake is based on three main pillars: (1) materials with better design in terms of functionality and recyclability, (2) conformity of PCR for food-contact applications, and (3) improved sorting of waste via economic recycling technologies. These three pillars will be connected through the synergistic collaboration in CIRCULAR FoodPack, reflected in the WP structure and its possible interconnections. The value chain collaboration will guarantee that existing and acquired knowledge is shared among the consortium partners, resulting in a seamless exploitation. This will allow the consortium to meet increasing market and consumer demands for sustainable materials, as well as complying with regulatory obligations captured in various European directives and EU member states' legislation.

Potential competition with other TBS initiatives. CIRCULAR FoodPack is not the only consortium proposing research into TBS. Others include the PRISM initiative in the UK, supported by WRAP,[92] and HolyGrail2.0, supported by the Ellen MacArthur Foundation.[93] CIRCULAR FoodPack could be perceived as competing with these initiatives, but this is not the case. The CIRCULAR FoodPack and HolyGrail2.0 projects are complementary and this could help to overcome some of the weaknesses of the 'invisible barcode' technology. It is the consortium's intent to exchange information and exploit synergies where possible, rather than enter into competition. This can be achieved by hosting joint workshops, e.g. during packaging fairs. Synergies may include: (1) exchange of data on SBS system and TBS efficiencies, allowing each initiative to further optimise its technologies; (2) challenging the TBS approach (detectable regardless of object positioning) synergistically with the 'invisible barcode' (possibly unreadable on 'crumpled' items that are more prevalent in flexibles), to increase both yield and purity for the HolyGrail2.0 initiative; and (3) comparing cost, ease of implementation, LCA and yield/purity scores for both initiatives. Project partner SUEZ, also involved in HolyGrail2.0, will promote data exchange between the consortia and arrange a test trial if possible.

Chemical recycling. The rise of chemical recycling could be perceived as a potential barrier for the project. This alternative pathway for the recycling of plastics has gained momentum over the last few years because it is not subject to EFSA approvals with regard to food-contact applications. The second-generation feedstock derived from the chemical recycling process is put back into the cracker to make wholly 'new' polymers. However, none of the chemical recycling technologies have proven industrial-scale viability thus far, in terms of economics and energy consumption. Material recycling remains a preferred pathway as it is more directly circular (polymer to polymer rather than polymer to fuel/naptha to monomer to polymer), so we do not foresee insurmountable competition, but rather a potential for complementarity, depending on process yields and footprints.[94] Furthermore, there is a general consensus in various industrial consortia that material reycling should be the preferred option where possible and that chemical reycling should target only waste streams where material recycling is no longer an option.

Legislative uncertainty with regard to FCM approvals. European Recycling Regulation (EC) No 282/2008 emphasises food safety when new FCMs are made from post-consumer recycled FCMs. Even with CIRCULAR FoodPack's expectation of being able to work towards the 99% purity on PE PCR, it may still remain very challenging to obtain approval for plastics other than PET for re-use in direct food-contact applications. This is evident from the lack of positive EFSA decisions on polyolefin (PP or PE) recycling processes and the complete lack of follow-up draft decisions by the EC. The lack of a clear step-by-step framework to obtain FCM approval for polymers other than PET is an ongoing concern for the recycling industry. The lack of any legislative framework for the recycling of (PE) flexibles would be a barrier for the uptake of the (technically) food-grade recycled plastics produced by CIRCULAR FoodPack. A recent EC report[95] noted the contradictions between FCM regulations and waste management legislation: "Safety requirements on recycled plastics under Regulation (EC) No 282/2008 may place restrictions on the recycling of FCM into new FCM; these restrictions depend on the polymer, the process, and/or intended use. Such restrictions may hinder efforts to increase recycling rates for FCM." The report also stated that "**although polyolefins have an important market position, current recycling processes do not enable a purification of polyolefins and removal of contaminants of concern. Industry considers that research is needed in this area**." This report is a sure sign that the EC is not only aware of the current bottlenecks, but also willing to act upon them.

92 https://www.letsrecycle.com/news/latest-news/funding-boost-for-invisible-barcode-plastics-sorting-project/
93 https://www.newplasticseconomy.org/assets/doc/Holy-Grail.pdf
94 Schlummer e al. The role of chemistry in plastic recycling. Kunststoffe international 5/2020
95 EU Commission: Study supporting the Evaluation of Food Contact Materials (FCM) legislation - (Regulation (EC) No 1935/2004), July 2020

Scalability. Scalability has been a massive hurdle for recycling technologies other than conventional mechanical recycling. Economically and environmentally viable scaling is still the major current constraint holding back the rollout of chemical recycling technologies. However, the main technologies proposed in CIRCULAR FoodPack can be complimentary (as a pre-treatment) to mechanical recycling and belong in the separate category of solvent-based recycling, in which the polymer chain is not broken up (as in chemical methods) but contaminants do not persist (as in mechanical methods). Solvent-based recycling is typically less challenging to set up in terms of (reactor) equipment and also less expensive to operate. Some (demonstrator) plants already use the CreaSolv® technology, such as a demonstrator plant in Indonesia built in cooperation with Unilever, an industrial plant with a throughput of 3000 tonnes per year under construction in the Netherlands for the recovery of PS from expanded PS insulation panels (start of operation: 2021), and a demonstrator with 200 tonnes per year nominal capacity set up in the European Multicycle project in Germany in summer 2020.

Value chain readiness. The main commercial barriers for the successful uptake of CIRCULAR FoodPack are the readiness of the value chains for a circular economy. Success requires political ambition and a solid legislative framework to push markets and industries from linear to circular economies. With the recently launched Green Deal and the 2020 launch of the European Plastic pact, a clear European framework is being built on the foundations of existing directives, pushing the market towards a more circular approach (Fig. 17). **(1)** At the sorting stage, there needs to be industry willingness to standardise certain aspects and likewise create efficiency. Packaging products and sorting factories cannot be equipped with several possible sorting technologies just because the industry is not willing to align. Several different sorting technologies could be combined so that each of them handles the sorting task at which they are best. TBS is one of the most promising next-generation sorting technologies, which could handle and sort all packaging products, such as non-food and food products, in a reliable and efficient manner. TBS should therefore be tested and improved under real conditions. A prototype could be implemented in a sorting plant as a demonstrator. The development of this technology is also linked to the real use of the tracers on packaging so that sorting strategies would initially be possible for the sorting of food-grade flexible packaging. **(2)** De-inking and washing are also key steps to achieve high-quality PE PCRs, so the reactor technology would need to be adaptable to existing reactors or independently developed at least to TRL5–6 as a prototype for testing at an existing recycling plant with a dedicated process line. **(3)** The project is expected to deliver new and effective circular value chains with possible interconnections, which will supply PE PCRs to replace virgin plastics in flexible laminates for circular use in flexible packaging (given that packaging will be sortable as food and non-food waste). **(4)** From the perspective of packaging convertors, the potential negative effects of TBS tracers on the processing will be limited, because these substances can be mixed in into the ink/primer/adhesive system for application using current infrastructure. The main challenge will be industry alignment for the application of said tracers in practice and at scale, given the additional raw material costs. However, access to next-generation recycled materials could be granted exclusively to parties that apply the tracer during production, providing an excellent incentive for such a program. A clear bussiness case could be built on this basis, because the demand for such material would outweigh supply in the initial stages. The use of PCR material on existing lines will be challenging because the overall laminate might be food-contact compliant but the resin might not be. This means that production lines used for food production would need additional cleaning, or dedicated lines might be needed for PCR processing. Standard operating procedures may therefore be required across the industry to ensure product safety in all production runs.

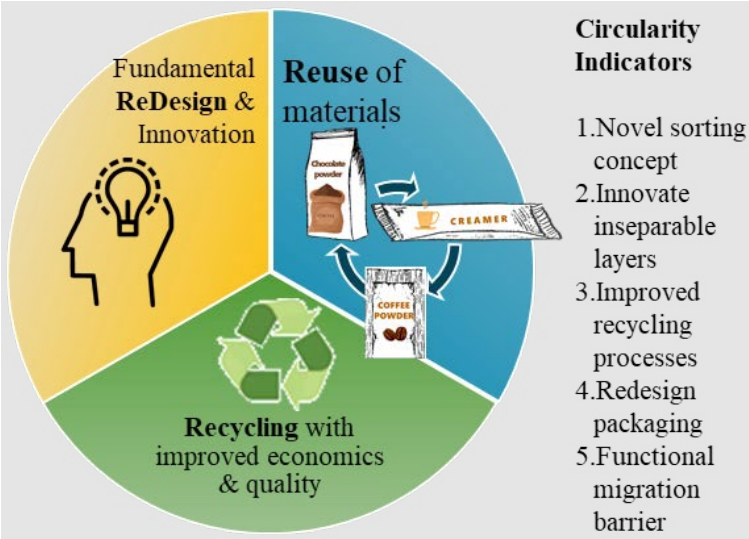

Fig. 17. CIRCULAR FoodPack – Successful Uptake.

(5) The food industry should actively support the use of the laminates proposed herein. For NESTLE, this might be accomplished through the Golden Rules. To achieve wider acceptance by the food industry, these developments should be promoted through consortia such as the Ellen MacArthur Foundation. Food compliance will be demonstrated for PE PCRs from food packaging with functional barriers, leading to more data on existing solutions, confirming feasibility, and supporting EFSA's food approval on PE PCRs. This point is necessary to develop the market for food-grade PE PCRs, as requested by recyclers and brand owners.

The value chain approach, supported by systematic training programmes, will also guarantee that selected knowledge sharing among the consortium partners leads to continuous exploitation to meet the increasing market demand. Cooperation among the partners will be important, especially in the post-project phase, to realise the applicable business models.

2.2 Measures to maximise impact

The impact of the CIRCULAR FoodPack project will be maximised by developing a tailored dissemination and communications package in WP8, targeting stakeholders in the European materials and recycling communities.

2.2.1 Dissemination and exploitation of results

Dissemination and communication plans

Fraunhofer, BayFOR and NTUA will lead the dissemination of knowledge generated by the CIRCULAR FoodPack project. The consortium will support these activities to increase international visibility and awareness. We will target each partner's network (Fig. 18), encompassing the scientific community, regulatory bodies, packaging producers, end-users, food/HPC brand owners, the pharmaceutical packaging industry, waste managers, recyclers, and process developers. Besides these, further main target groups of policy makers, regulatory bodies, the scientific community and relevant project clusters, the media and general public have been identified and will be further assessed in the Plan for Dissemination and Exploitation of Results and Communication activities (D8.2). The consortium will publicise the activities of CIRCULAR FoodPack and the key role of the EC. The preparation of **targeted promotional material** for different stakeholders, such as scientific organisations, regulatory bodies, financial investors, recyclers, producers and manufacturers, will lie at the core of our dissemination and communication activities (Table 9). This material will include press releases, rollups, posters, leaflets, regular newsletters, and a website and social media presence to showcase the project and its results. Moreover, the project outcomes will be presented at conferences, workshops, fairs and events that are open to the public to demonstrate the benefits of EU-funded research and its contribution to the challenges facing European society. The EC will be informed in advance about such activities. With these measures, we will emphasise the significance of the CIRCULAR FoodPack consortium in the circular use of flexible food packaging materials, and in the long term achieve a market uptake that is driven **not only by producers but also by consumers**. In order to achieve impact with all stakeholders, we will produce a detailed description of all dissemination and communication activities as part of the overall Plan for Dissemination and Exploitation of Results and Communication Activities (PDEC, D8.1/D8.2) at the beginning of the project (month 4). It will be updated throughout the project with a final version in month 42 that also includes outreach metrics. The main objectives of the **communication and dissemination** activities are summarised below:

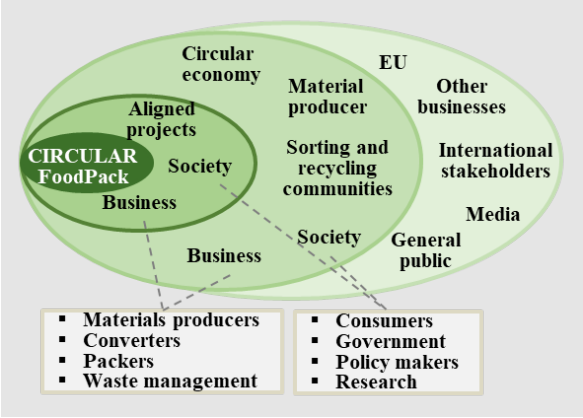

Fig 18. Overview of target groups

- Develop a strategy to **raise public and economic awareness** (main message, indicative timeline)
- Ensure bidirectional **communication flow** to and from stakeholders by providing appropriate infrastructure and management and establishing communication responsibilities throughout the consortium
- **Stimulate external feedback** and integrate it within the consortium in addition to the External Advisory Board
- **Manage and include stakeholder expectations** to build a qualitative reputation
- Fine-tuning identified target groups and agree on the most **suitable communication channels** (including mass media channels for communication activities)
- Identify and involve collaboration partners (businesses and research networks) by disseminating results to the target groups summarised in Table 9.

We will organise **three dedicated workshops** at the demonstrator sites (SUEZ, Fraunhofer, UGENT/AMCOR-G, and NESTLE) during the project, open to students, researchers, the packaging industry, recyclers, and users of recycled materials, including extra sessions for training, helping to convince industry partners of the market relevance of the CIRCULAR FoodPack technologies. Virtual alternatives may be necessary due to COVID-19. The consortium partners have strong (joint) publication records and have often served as editors/reviewers of leading scientific journals or as advisors to national or international agencies. The project results will be made available to the larger scientific community via **peer-reviewed joint publications in scientific journals**, favouring the open access model to reach the widest possible audience. CIRCULAR FoodPack represents a significant step forward in the translation of state-of-the-art recycling research into efficient modern methods, including sorting, pre-treatment, material recycling, packaging production and compliance testing, which could be adopted by the industry. Therefore, we will publish a **policy brief** setting out the project results and new opportunities arising from them.

Communication activities and tools

CIRCULAR FoodPack will initiate **communication activities from the very beginning of the project** and the strategies will be reviewed and revised on a regular basis, if necessary. We will make target stakeholders aware of

our goals, research methods, achievements and innovations. Fraunhofer, BayFOR and other partners' press officers will be responsible for **public relations** and will support the Dissemination and Communication Manager (DCM) to identify 'newsworthy' items and appropriate media for the dissemination of this information. A **logo** and **corporate identity** will be developed to 'brand' the project and will be used on all public outputs. CIRCULAR FoodPack partners are committed to raise public awareness while simultaneously ensuring that information is disseminated in a responsible and ethical manner. Ultimate responsibility for public disclosure of confidential knowledge generated by the project will lie with the Coordinator and the DCM, both experts in the communication of science to the public.

The **project website** will be launched in month 4. Electronic versions of the newsletters will be accessible on the website. Various **promotional materials** will be produced during the launch phase of the project, and distributed by the project partners at relevant events. **Social media** will allow European research projects to reach a wider audience, but it is difficult to gain significant user numbers for social media channels specifically set up for research projects, so the consortium will adopt a dual approach by communicating through CIRCULAR FoodPack accounts while also making use of the project partners' existing popular channels. We will create project-specific brands (e.g. hashtags like #CircularFoodPackaging) that will be used to highlight the project on social media. Communication activities will be reviewed, evaluated and refined throughout the project under the leadership of Fraunhofer and BayFOR. A brief draft dissemination and communication plan is presented in Table 9, and this will be developed into a formal document at the beginning of the project for submission as D8.2.

Table 9: Draft dissemination and communication plan of CIRCULAR FoodPack.

Timing	Activity	Objective	Target groups	Accomplishments and expected impact	
Start M4, contd.	Interactive project website	Regular updates, news section, partner login	Waste collectors, sorters, recyclers, material suppliers, converters, research	Structure the homepage; formulate the content	Communication platform, continuous engagement.
Plan: M12, M24 M36	Workshop, webinars to selected stakeholder groups, training on site, videos	Minimum three workshops to three types of audiences at the partners in Belgium, France and Germany, at demonstrator sites	Industry, inverters, stakeholders, research, policy-makers, members of CEFLEX, IVLV, ZLV, Plastics Recyclers Europe	Reaching a broad audience, selection of speakers; dissemination through web; training the future staff of commercial plants	Public awareness, feedback (interest in buying packaging with recyclates), recycling instead of waste disposal, life cycle management, market research results
Operational by M4, cont.	Digital newsletter, leaflet, press releases, interviews, partner websites	Public dissemination, 2000 prints to be distributed	Stakeholders, policymakers, scientific community, industry	High-quality documents prepared by the experts	Central source of project information, communication materials, raising awareness, attract audience to engage
Plan: M18 M30 M42	Feedback on roadmaps, exchange webinars (online), and joint events (e.g. project day).	Guidelines on: - Recycling - Effective sorting - Management of packaging after use - Industrial take-up to support circular use of PE PCRs	Research groups, all stakeholders, regulatory bodies, alignment with other H2020 projects, industrial partners, policy makers	Policy brief recommendations for future policy implementations; project updates; industry testimonials and interviews; contribution to industry clusters, roadmap objectives	Engage with research community and policy-makers, provide evidence of potential, publicity for mechanical recycling with CreaSolv® process with advanced pre-treatment and purification for food packaging
Cont.	Social media (YouTube, Twitter, LinkedIn, ResearchGate)	Regular posts (monthly) targeting followers	Industry, science, technologists, investors in waste collection and sorting, recycling	Engagement and interaction with followers and readers	Rising of awareness, spreading news quickly
Cont. M18 M30 M42	Publications in journals, technical articles, some examples already identified by project partners	J. Mater. Chem., Adv. Optical Mater., Resources, Conserv. Recyc., J. Cleaner Prod., Appl. Sci., IEEE Photon. J., Recycling, Food Additives and	All scientific community	At least five papers: - Packaging products containing recyclates - SBS of different packaging waste fractions for better recyclate quality	Communicate scientific contributions, guidelines for EU policymakers, contribution to science, migration behaviour of polyolefin recyclates in packaging, migration challenge test results,

		Contaminants, Thin Solid Films		- Synthesis and optical properties of fluorescence markers	food compliance with recyclates in packaging
Cont. M18 M30 M42	Presentation at industry events, trade shows, scientific conferences, (online and face-to-face), Innoform, SOLPACK, multi-layer packaging conference, IAPRI, ILSI	At least one trade fair, and three conferences per year: ISWA World Congress, Odour and Emission, Plastics Recycling Conference, K-shows, SAMPE conference, JEC conference, Anuga, Extrusion	Policymakers, industry, scientific community: European Circular Economy Stakeholder Platform (https://circulareconomy.europa.eu/platform/)	- Project results at targeted trade fairs (e.g. joint booth with BayFOR at IFAT, Munich, leading trade fair for environmental technologies; packaging fairs, e.g. Fachpack, Interpack and K fair, Biofach, the leading fair for sustainable concepts in food)	Present CIRCULAR FoodPack as innovative process in the section 'good practice' (regular updates); knowledge exchange to increase impact of project results; understand resource efficiency of the increased use of recycled packaging systems; market-uptake increase; engage, promote, inform about the potential cost reduction for waste owners

Plan for exploitation

The exploitation plan and the identification of business opportunities will be built on a dialogue with packaging stakeholders by investigating industry and consumer expectations. The results of expert and consumer interviews will be iteratively included in the project's innovations, ensuring the market acceptance of our technologies. **Commercial exploitation and IPR management** will to some extent align with practices already used by the consortium partners, and their well-established exploitation strategies will be used to commercialise our research. The **industrial partners will be involved in the project from the beginning**, covering the value chain from materials to application, thereby encouraging the exploitation of our results. Most partners already have excellent legal support teams and technology transfer offices (TTOs) that specialise in the identification, protection and marketing of technologies through licensing and spin-off companies. The TTOs of each partner will help the Exploitation and Innovation Manager (EIM) to develop protection and exploitation strategies.

The exploitation plan closely correlates with our communication and dissemination activities. A well-defined plan for the simultaneous protection and promotion of knowledge will be conceived and regularly updated in the PDEC (D8.2, D8.8). The fast and smooth transfer of project know-how and technology is essential for the economic success of the partners. All non-confidential products directly resulting from the project, such as guidelines for the sorting of food packaging items, or the recycling of FCM flexible packaging materials, will be made accessible to the public.

2.2.2 Initial business plans

Potential Markets

Circularity in the flexible packaging value chain will create new markets for the waste management and recycling industry, the European plastics industry, the (flexible) packaging industry, the food and drink industry, the HPC packaging industry, and manufacturers of sustainable packaging.

Waste management: The separate collection of municipal waste[96], stimulated by ambitious EU legislation, is steadily increasing over time. The municipal waste recycling rate in 2018 was 43%.[90] Market growth can be attributed to increasing awareness of sustainable plastic waste management practices, the rise in infrastructural development (including recycling plants) and urbanisation. In Europe, 9.4 million tonnes of plastic post-consumer waste was collected for recycling in 2018.[90]

Recycling industry: In 2018, the recycling rate for PE films in Europe was 23%, with only 13% being used again in films (60% of all flexible films are made from PE). The utilisation of recycling plant capacities for flexible films was estimated at 90% in 2018.[97] Furthermore, 29.1 million tonnes of post-consumer waste was collected in 2018, of which 32.5% was sent for recycling, 42.6% was used for energy recovery (incineration) and 24.9% ended up in landfill. Exports to non-EU states declined by 39% from 2016 to 2018.[93] In 2018, the total amount of worldwide recycled plastics was 29.4 million tonnes, with the market valued at €35.19 billion. The plastics recycling industry is predicted to grow by 6.6% from 2020 to 2027.[98]

Plastics recycling industry: The value of the global recycled plastics market was an estimated €31.14 billion in 2017, projected to reach €42.47 billion by 2022. Packaging is the fastest growing segment, PET being the fastest growing type, and North America being the fastest growing region. The largest market is the Asian-Pacific market.[99]

96 Source: European Environment Agency; Jul. 2020. Plastics Europe, 2019. Plastics – the facts 2019.
97 Plastics Recyclers Europe, 2020. Flexible Films Market in Europe – State of Play.
98 Research and Markets, 2020. Global Plastic Recycling Market Trends Report 2020. (Global-Plastic-Recycling-Market-Trends-Report-2020.html)
99 MarketsandMarkets, 2018. Recycled Plastics Market. https://www.marketsandmarkets.com/Market-Reports/recycled-plastic-market-115486722.html

European plastics industry: The turnover was €360 billion in 2016, with more than 1.6 million people employed by 60,000 companies, mostly SMEs. Packaging is the largest end-use plastics market in Europe, accounting for 39.9% of the 51.2 million tonnes of European plastics demanded by converters in 2018. The European countries with the largest demand in 2018 were Germany (24.6%), Italy (13.9%), France (9.4%), and Spain (7.6%).[100]

European flexible packaging market or packaging industry: This is expected to reach €16 billion by 2023,[101] compared to €14 billion in 2017 (annual growth rate = 2%).[102]

Sustainable packaging: The global market was valued at €200.78 billion in 2019, with a growth rate of 5.7% predicted until 2025.[103]

European food and drinks industry: This manufacturing sector had a turnover of €1,192 billion in 2017, with 2.1% of EU gross value added in 2016, and is a leading employer (4.72 million employees in 2016). There were 294,000 companies active in 2016 (99% SMEs), creating 47.5% of the turnover and employing 60.8% of all employers in the EU food and drinks industry.[104]

European HPC packaging market: The total demand in Europe for HPC products is projected reach €252 million by 2026, while the global market is projected to reach €38 billion in 2025, representing a growth rate of 4.7%.[105]

CIRCULAR FoodPack solutions can improve the competitiveness and innovation capacity of all these markets, with the largest impact on the PE flexible food packaging plastics and recycling industry. In 2018, demand for PE flexible films was an estimated 8.5–9 million tonnes, of which 1.2–1.3 million tonnes was met by recyclates produced within the EU. PE flexible films have a wide variety of uses, the majority (81%) of which are packaging applications (23% food packaging, 41% non-food packaging films, and sacks and bags used for packaging). The amount of film packaging consumed is expected to equal the amount of packaging requiring collection, sorting and recycling within the same year, because of the short lifetime of film packaging. In 2018, ~1.8 million tonnes of the 2.7 million tonnes of PE flexible films sent for recycling were sent to recyclers within the EU (with an addition of 0.5 million tonnes production scrap). The total input of PE flexible films to EU recyclers was therefore 2.2–2.3 million tonnes. The average yield of recyclate from post-consumer PE flexible films within the EU is ~70% of input, suggesting ~1.8 million tonnes of recyclate output was produced in the EU (1.3 million tonnes from post-consumer flexible films and 0.5 million tonnes from production scrap).[106]
The PE film product sector could incorporate 36% recycled content overall. In the EU, demand for PE could require 3.6–5 million tonnes of PE recyclate by 2030 compared to only 1.8 million tonnes in 2018. Due to high levels of non-PE films (PP films, multi-layers) in household waste, output yields are similarly low at ~50%.[107]
The largest end-use market for flexible recyclate is non-food film. The applications of recycled flexible films are limited by quality (primarily the colour, additives and odour). An economically viable business model for higher quality, light-coloured output from household films is needed, in order to develop a product that meets the requirements of the largest non-food flexible film sectors, aiming to displace virgin raw material. In contrast, PE recyclates are not used in flexible food packaging even though food packaging is one of the largest markets for PE. This reflects both policy and R&D gaps, which need to be tackled simultaneously by policymakers and industry, to ensure that larger quantities of collected films can be processed into higher-quality recyclate for the large-scale manufacture of films. CIRCULAR FoodPack will help to fill in these gaps by increasing the recycling output yield for PE rich flexible films to >90% referring to current SBS technologies and providing food-grade PE PCR with 99% purity referring to new TBS approach (SO1), creating a new market segment worth more than €5 billion within the next 10 years (assuming an average PE PCR cost of €1/kg). To achieve this goal, we need the market pull of the large companies in CIRCULAR FoodPack (NESTLE, AMCOR) and the EAB members (CEFLEX, FPE, IVLV, ZLV, STEINERT, ISWA, PRE), especially the leveraging effect of the associations and networks. These will help to motivate companies outside CIRCULAR FoodPack to adopt our technologies, which include novel technologies for the sorting, recycling, material and converting industries. CIRCULAR FoodPack will provide continual improvements in innovative packaging design, solving recyclability challenges, and opening the potential for step changes in how collection, sorting and recycling is optimised for a circular economy.

Identification of key results and preliminary business plan

The highly challenging overall target of CFP, i.e. circularity of multi-layered food packaging, will only succeed if the major sorting and recycling technologies as well as packaging designs are implemented in the market jointly. However, each single building block of our innovative value chain has a great potential to support current market demands for higher qualities and yields of PCR polymers and better recyclable food and HPC packaging. Thus, the expected key exploitable results of each partners prepared during the proposal phase are described in Table 10.

100 Statista, 2018. Distribution of plastics demand in Europe in 2018, by country.
101 https://www.woodmac.com/press-releases/Europes-flexible-packaging-market-will-be-worth-over-16-billion-by-2023/
102 Flexible Packaging Europe, 2018: Strong demand for flexible packaging industry in Europe (/files/FPE/pressreleases/FPE-MarketOutlook2018-GB.pdf)
103 Mordor trends, 2019. Green Packaging Market - Growth, Trends, and Forecast (2025). mordorintelligence.com/industry-reports/green-packaging-market
104 FoodDrinkEurope 2019: Data & Trends of the European Food and Drink Industry 2019.
105 Research and Markets, 2020. Global Personal Care Packaging Industry. (/0/en/Global-Personal-Care-Packaging-Industry.html)
106 PRE market expert.
107 Based on market expert information and the ratio of PE to other materials in food packaging, Axion Consulting (2016) Reflex Project Summary Report.

Table 10: CircularFoodPack key exploitable results.

WP	KER	Owner(s)	Envisaged IPR	Type of exploitation	Target end-users	Timetable
2	Tracers for sorting food, and non-food packaging waste (TBS)	**KIT, POLY**	11 existing patents and trade secrets, potent. new patent appl.	Commercialisation	Tracers: packaging manufacturers; TBS equipment: utilities, sorting factories	3 years after the project
2	Sensor based specification (SBS)	**IRIS**	Trade secret; two patents under assessment	Commercialisation, consultancy services, licensing	Waste managers: Sorters, photonics solution providers	3–4 years after the project
3	Conditions for recovering different fractions using the CreaSolv® process	**Fraunhofer**	5 existing patents and trade secrets, potent. new patent appl.	Know-how transfer and licensing to new companies; services for adjusted process for new materials.	Recyclers, converter, virgin polymer producers, end users, waste managers	At end of the project
3	Insight in application range of de-inking procedures. Design for delamination and de-inking. De-inking as pre-treatment.	**UGENT**	(Existing) patents and confidential know-how	Know-how transfer and licensing to new companies; services for adjusted process for new materials.	Recyclers, converters to recycle production scrap, end users, waste managers, polymer and ink producers	3 years after the project
3	New recycling technology to remove odour from plastic using infrared technology.	**KREYEN**	Patent under assessment, and a novel patent	Commercialisation of odour removal process	Recyclers, original equipment manufacturer (OEM), extruder manufacturers	3 years after the project
3	Inks with tracers for deinking, delamination primers&coatings for barrier layers	**SIEGWERK**	Trade secrets, patents	Commercialisation	Converters, recyclers	3 years after the project
4	Migration models, test methodologies for design of functional barriers	**Fraunhofer**	Methodology (models and analytics) developments owner	Analytical services of novel migration tests	Customers from the industry, packaging companies, material suppliers, converters	At the end of project
5	Design rules for recyclable packaging structures	**Fraunhofer**	Existing barrier patent, additional patents, confidential know-how.	New research projects, know-how transfer and licencing to companies aiming at researching for these new materials	Co-packers, packaging material producers from Germany, Europe, USA and Asia.	At the end of the project
5	Multi-layer films and packaging with recycled PE (PE PCR)	**AMCOR-K AMCOR-G**	Design patents on Polyolefine-PCR/SiO_x packaging	Production of said packaging at industrial scale, protecting these new developments from direct competition/licensing the IP when strategically opportune.	Fast-moving consumer goods (FMCGs), co-packers and convertors	At the end of the project.
6	Sortable packages with PCRs	**NESTLE**	Patents/use of IP	Commercialisation	Consumer	1 year after project
7	Results of LCA and LCC	**UGENT**	Methodology belongs to UGENT	Scientific communication in open access journals, services to companies	Companies spanning the value chain, government	Directly at the end of the project
8	Report on market data and marketing prospects	**ECOZEPT**		Author rights	Converters, food industry, retailers, policy makers	At the end of the project.
8	Business and exploitation plan	**NTUA**	Confidential know-how	Services for companies that implement envisaged technologies in commercial scale	Recyclers, converters to recycle production scrap, end users, waste managers	At the end of the project.

All industrial partners expect to achieve significant increases in knowledge, capacity building and also in **financial benefits** from the exploitation of project results by **increasing their direct sales or market share, or by savings in**

materials and energy reflecting the future use of PCRs. The TRL will reach 5–6 by the end of the project. The industrial companies will see an increase in turnover after ~5 years. Revenues mainly depend on the final quality of PE PCR and the willingness of customers to cover the cost and eventually pay a premium for PE PCR (see Section 2.1.3). During the proposal stage, each partner has communicated their exploitation expectations in line with their business activity and positioning along the value chain (Table 11).

Table 11: Summary of exploitation intentions of the CircularFoodPack partners.

No.	Partner	Core business	Intention for use of project results
1	Fraunhofer	Research and development	At least two technical papers and three industrial press articles. New knowledge about recyclable multi-layer structures with PCRs for food packaging, recycling technologies, migration models and analytics will be shared with companies and translated into industrial applications. Knowledge will be also leveraged to generate additional complementary research projects.
2	BayFOR	Consulting, projects	New partnerships, new customer segments, greater visibility in Europe to promote EU projects and encourage RTOs, SMEs, and industry to participate in EU activities.
3&4	AMCOR-K AMCOR-G	Flexible packaging	New food-grade packaging solutions that meet new regulations and requirements, further differentiating ourselves in the food packaging market.
5	UGENT	Research and teaching	Spin-offs, licensing, research projects, services to industry. Exploitation of the LCSA results via further research projects and services offered to the consortium partners and other companies. The recycling technologies fit with our IP portfolio (patents, databases, software), which is commercialised via industrial collaborations using an open innovation approach. The patents are commercialised via a licensing strategy.
6	IRIS	Photonics and digitisation	HSI-NIR products are already commercialised. We will use established B2B channels, distributors, and OEM channels. The digital tool will be offered as part of our digital solutions portfolio and licensed or commercialised with *ad hoc* adaptations (6–12 monitoring devices in 4 post-project years; projected sales of €500,000–€1.12 million).
7	POLY	Tracers, TBS detection equipment and sorting machinery	Increase in sales volume and global expansion of products. Further cooperation with larger industry players and/or strategic investors. We will continue to manufacture the markers but will use new knowledge to continue R&D on optical detection and sorting machines. TBS could achieve efficient, reliable sorting into ~40 well-specified packaging waste fractions at a total sorting cost (including tracer) of ~€200/tonne of input waste, valorising ~150 million tonnes of plastic packaging that requires better sorting. The budget for a good sorting service is at least €200/tonne, bringing the global market potential to at least €30 billion/year as part of a circular plastic economy. We could also liaise with a large utility and offer sorting as a service, especially in countries without sorting and recycling infrastructure. We predict three digit sales levels and the need for a substantial number of high-quality jobs.
8	KREYEN	Equipment manufacturer (deodourization)	Entering a new market segment (odour removal from plastic). An infrared dryer costs €250,000–300,000 for a 1000 kg/h plant. We develop, design and manufacture products on location. If there is a market for odour removal from plastic, and the process is successful, we will publicise and market this through our network, contacts in the recycling market, OEM extruder manufacturers, trade fairs and congresses.
9	KIT	Research and development	Novel patents, spin-offs, licensing, research projects and services to industry. Novel NIR-emitting tracers that are ideally compatible with existing NIR plastic machines will be patented and then licensed. Target end-users include tracer-based sorting companies. We estimate that a patent application can be lodged in year 3 of the project followed by scientific publications to strengthen future grant applications. The development and licensing of new patent-protected luminescent markers for plastic sorting will provide an additional budget to strengthen the quality of education and research infrastructure. We will recruit PhD students and postdoctoral researchers who will contribute to the economic prosperity of the European and global communities.
10	NESTLE	Food company, end user for food packaging, brand owner	Results from the safety evaluation published and made available to the EC and EFSA for future guidance in the recycling of polyolefins. We envision press releases related to the introduction of these laminates with recycled content. NESTLE expects to reduce its use of virgin plastics by developing innovative packaging solutions containing recycled content. Introducing these laminates earlier than competitors may give a competitive advantage.
11	SIEGWERK	Ink manufacturer, printing	Knowledge concerning the formulation of primers and coatings yet to reach the market and the commercialisation of these products to generate licensing revenue from the IP. A new series of inks and coatings containing TBS markers will be developed, as well as primers and coatings to protect the barrier layers, and to facilitate de-inking and delamination. These new product ranges have a strong market potential given the addressable market of corresponding packaging materials in Europe.

12	ECOZEPT	Consulting, knowledge transfer	Providing services based on the results of market research and SWOT analysis. Target groups: packaging makers and converters, food industry, retailers, policymakers. Development of new international partnerships and networks.
13	SUEZ	Waste management: collection, sorting and recycling of plastic	IP covering sorting and recycling of post-consumer plastic packaging waste, and exploitation via licensing. Analysis of the composition of post-consumer flexible packaging waste in different EU countries will help to improve current sorting steps and to develop new sorting strategies leading to an increase in sorting and recycling rates. High-quality recyclates with virgin-like properties. Our operation of more than 150 sorting plants in Europe and worldwide implementation will allow us to harness TBS and SBS technologies to sort more plastic waste from packaging and to select food packaging answering the first requirement for food-grade PCRs. Increasing the production of high-quality PCRs in Europe requires new and improved de-inking and washing technologies for packaging. The technologies developed in this project will be of interest to recyclers and will be welcomed at recycling plants. A tangible action would be the design and implementation of a prototype of mechanical recycling for PE flexible packaging at a current recycling plant.
14	NTUA	Research and teaching	Cooperation with engineering partners and stakeholders following commercialisation, new partnerships, further EU research projects, networking, creation of new sectors of economic and technological activities reflecting the combination of know-how from various industrial sectors and their innovation systems.
15	UM	Research and teaching	Transfer of project results to the BSc Circular Engineering. Research projects, services to industry.

A preliminary **business model canvas** (Table 12) has already been developed to highlight market potential. It will be developed and regularly updated within the PDEC to keep the focus on most relevant exploitable results for all project partners.

Table 12: Preliminary business model canvas foreseen to be applied to each key exploitable result.

Preliminary business model canvas	
Key partners	Material producers, film producers, packaging converters, waste management bodies, plastics recyclers, sorting technology providers, plastics machinery manufacturers, composites manufacturers, R&D partners, strategic alliance between partners, dealers and distributors, joint ventures.
Key activities	Novel material design, recycling processes, sorting technology integration, networking with customers, value chain management, sales and distribution, IPR and exploitation management of technologies.
Key resources	Experience in waste collection, sorting, processing (pre-treatment, mechanical and solvent-based recycling), package production, end-users, software and hardware, IPR, human and financial resources.
Value propositions	Consumers perceive environmentally-friendly packaging positively and are willing to pay more. Innovations will be perceived as an added-value product assuming that consumers are clearly informed that the packaging consists of recycled materials and can therefore replace virgin plastics. We will empirically determine how consumers and industrial operators evaluate the innovations and whether an added value is actually perceived. This will indicate the most effective messages for communication to consumer groups also including the results of LCA.
Customers	Large multinational food companies are becoming more committed to setting up their own plastic strategies (e.g. to use only recyclable or reusable plastics). Consequently, a high market potential can be assumed. The identification of target customers will be verified empirically through interviews with experts and consumers, and the results will be incorporated into the development of the innovations.
Customer relationships	During planned interviews with packaging experts, the most effective measures will be identified to establish long-term relationships between suppliers of project innovations and their customers. We will determine how users can be best supported in adopting the new technologies.
Channels	The exploitation plan will give insights into the most suitable logistics and organisation of the value chain to ensure the successful introduction of project innovations onto the market. These results will be based on feedback from experts representing the entire packaging value chain, from brand owners and retailers, to sorting and recycling companies.
Cost structure	Purchasing all needed raw materials and components, IPR protection, patents, sales management and distribution costs, upgrading and modification of existing facilities.
Revenue streams	The outcome of the exploitation plan based on the incorporation of packaging experts' and consumers' feedback will determine suitable pricing strategies for the CIRCULAR FoodPack innovations.

SWOT analysis for CIRCULAR FoodPack

In order to enhance the strengths of the project, mitigate weaknesses, seize opportunities and neutralise threats, we will conduct a thorough SWOT plus A (action) analysis in Task 8.1 supported by input from multi-stage interviews with experts and consumers. A preliminary version is presented in Table 13, will be periodically revised and finalised as D8.7.

Table 13: SWOT analysis for CircularFoodPack

	Helpful to achieving the objective	*Harmful to achieving the objective*
Internal origin (attributes of the system)	***Strengths*** • Ability to separate food and non-food plastic waste during sorting by TBS • CreaSolv® and mechanical recycling process cascades enabling sufficient reduction of contaminants in EoL packaging waste • Novel design of packaging laminates with PCRs to ensure a closed loop circular economy • Creating new value chains based on recycled polymers, targeting the food packaging market	***Weaknesses*** • Input waste stream is rather mixed and contains multiple foreign materials making cleaning highly challenging • Close co-operation is required between parties in the value chain.
External origin (attributes of the environment)	***Opportunities*** • Synergistic collaboration among partners with complementary skills, ultimately achieving integrated processes along the whole value chain • Legislation is becoming more demanding in terms of recycling quotas • No direct competition as we tap into currently non-recyclable materials and utilisation in food packaging • Required technologies and know-how available in the consortium • Extremely high market potential	***Threats*** • Regulatory approval processes; strict regulation/legislation framework for FCMs • Resistance to change in well-established procedures of waste management • Difficulty to position vs virgin polymers, if value chain does not value environmental aspects, leading to mere price competition making market entry difficult

2.2.3 Strategy for knowledge management and protection (IPR)

Background IP brought into the project by the partners is summarised in Table 14. The Project Steering Committee (PSC) will oversee the dissemination and IPR policy by taking on the following responsibilities:

- Supervise access to background IP and determine which background IP is essential for the project.
- Adjudicate on ownership of foreground IP. Each partner will make an application for ownership and this claim will be distributed to all confidentiality-bound partners in case of any counterclaim. Resolutions include joint ownership agreed between partners with exploitation rights regulated by the PSC, or sole ownership granted to the partners most likely to exploit results and free research licenses distributed to other relevant partners.
- Decide which information may be published and which should be withheld pending protection or exploitation. This will ensure that results are managed in a way that ensures the most suitable exploitation pathway.
- Approve further grant applications where background IP or results (foreground IP) are to be used. This will ensure no use without the owner's permission. All partners will sign a Consortium Agreement (CA) before the Grant Signature, which will include detailed procedures for IP ownership and conditions of access for both academic and industry partners, thus ensuring optimal cooperation. The CA will be based on the DESCA model.

Table 14: Background IP and expertise brought in by project partners (AMCOR-K and AMCOR-G are considered, under AMCOR).

No.	Partner	Background
1	Fraunhofer	1) Recycling of ABS, PS, PA, PBT, PET, LDPE, HDPE, PP, PVC, EPS and other polymers from plastic waste using CreaSolv® solvents. Recovery of these polymers, removal of soluble and insoluble contaminants from polymer solutions. Scalable hydrogenation processes. 2) Compliance and migration testing, migration modelling, analytical chemistry 3) Development and characterisation of high-barrier materials for food and non-food applications.
2	BayFOR	Offers a broad array of professional services in the fields of management, consulting, networking, PR and administration to entities applying for and managing EU grants (Horizon 2020) and provides profound knowledge in the areas of circular economy and stakeholder engagement.
3&4	AMCOR-K & AMCOR-G	Flexible film convertor. AMCOR-G: industrial production of plastic flexible packaging, extrusion and conversion, PE, PP, film (co)extrusion (blown and cast) with EVOH and PA and laminating with PS, PET, aluminium, printing and slitting. AMCOR-K: electron beam evaporation process.
5	UGENT	1) Delamination and de-inking technology for multi-layer packaging materials. 2) Extensive experience in the context of LCSA

6	IRIS	NIR devices for in-line monitoring, also based on hyperspectral imaging (HSI) in SWIR region; chemometric machine learning modelling, data mining for process optimisation. Two patents are under assessment covering data-driven machine learning techniques for quality and process optimisation.
7	POLY	Develop, optimise, synthesise and process fluorescent particles (11 patents; Section 1.4). A new patent under assessment covering a one-step sorting procedure of packaging waste using TBS. Optical physics, measurement techniques, electronics, automation, machine building to develop and manufacture new detection systems, sorting machines and sorting lines, and to adopt existing NIR sorting machines together with the manufacturers of these machines.
8	KREYEN	Background on deodourization process (based on infrared dryer), patent in registration stage.
9	KIT	Synthesis and optical characterisation of luminescent lanthanide-based materials; synthesis of luminescent materials with exceptionally high luminescence efficiency (PLQY – photoluminescence quantum yield) using approaches such as solid-state and hydrothermal chemical processes; synthetic expertise; measurement of absolute PLQY in single crystals; microcrystalline powders and nanocrystalline materials; knowledge for the correct evaluation of the performance and applicability of luminescent markers.
10	NESTLE	Analytical chemistry (LC-HRMS/GC-HRMS); toxicological assessments (including biological assays); risk assessment of (recycled) packaging materials; various packaging materials for all food types; development of laminates for most food types; new knowledge gained from the performance of recycled polyolefins and to a limited extend the manufacturing of pouches with recycled content. As the biggest food company, NESTLE packages foods in pouches and has all facilities to perform shelf life studies and investigations of the barrier properties of packaging materials.
11	SIEGWERK	Formulation know-how for all kind of inks (solvent-based and aqueous, UV) for packaging structures and substrates reflecting the demand from the packaging market, delivered by more than 60, globally distributed production and blending sites. Experience in de-inking and delamination technology and the formulation of primers, inks and coatings to ensure proper deinking and delamination during recycling.
12	ECOZEPT	Sustainable agro-food markets; market intelligence systems and marketing approaches for niche and innovative markets in the agro-food industry and circular economy (organic products, regional, typical products, sustainable packaging). Practical insights in value-creating supply chains, contribution by combining practitioners' views with analytical approaches for understanding sustainable agro-food value chains and the behaviour of economic actors, from farmers to consumers.
13	SUEZ	Waste management activities in Europe, including collection, sorting (with more than 150 sorting plants) and recycling of plastic waste (nine recycling plants in EU). Sampling and characterisation methods dedicated to waste streams and methods to assess the operational performance of process lines.
14	NTUA	Exploration of socioeconomic and development effects of technology and innovation, focusing on the analysis of business strategies and related public policies at national and EU levels; promotion of innovative and environmentally friendly technologies and energy saving measures; qualitative (e.g. case studies, panels of industry experts/academics/policymakers to study specific value chains and innovative processes and business activities) and quantitative techniques (e.g. multivariate statistical analysis, social network analysis) and empirical methods (e.g. large-scale surveys).
15	UM	RQ, a 'quality factor' tool that predicts the quality of mixed recycled polymers in terms of the composition and miscibility (based in interfacial tension) of the components. Polymer science for mechanical recycling of plastics.

2.2.4 Management of research data

The CIRCULAR FoodPack project will participate in the Open Research Data (ORD) Pilot, which is an extremely effective method for disseminating scientific results and making them available to the public. Participation in the ORD Pilot will be based on the underlying principle 'as open as possible, as closed as necessary'. A Data Management Plan (DMP) will be developed by month 6 (D1.3) and updated at the end of the project. CIRCULAR FoodPack partners will implement the DMP according to FAIR (Findability, Accessibility, Interoperability, and Reusability) principles, taking into account exploitation activities driven by the industry partners. The DMP will contain all necessary information regarding the handling of research data, including the definition of data that will be shared openly or restricted, types of collected and processed data, methods and standards applied to data generation and processing, and information on how data will be stored, accessed and analysed during and after the project. In line with the DMP, the Coordinator will ensure that project data are deposited in a trustworthy digital repository, such as Zenodo. Information concerning the tools and instruments used to validate the results will also be provided and all open-access publications will be deposited on the project website.

2.2.5 Communication activities

Some dissemination activities can include communication and vice versa (e.g., trade fairs). Our communication strategy and its objectives are therefore included in the Plan for Dissemination and Exploitation of Results and Communication Activities (PDEC; D8.2). The PDEC is a comprehensive plan that comprises the dissemination, exploitation and communication activities of CIRCULAR FoodPack and the communication strategy is described in detail in Section 2.2.1.

3. Implementation

3.1 Work plan

The overall structure of the work plan: The workplan foresees 8 **WP**s (Fig 19).

WP 1 deals with project coordination and management (**Fraunhofer**). **WP 2** will perform tracer development (**KIT**), and Tracer Based Sorting (TBS) trials at the already existing pilot sorting line of POLY. VIS/NIR-emitting tracers will be used to test-wise mark food packaging items as mono- and multi-layered (MLC) food packaging waste. **KIT** develops specialized markers, with regards to excitation and emission wavelength, quantum yield, particle size, luminescence lifetime, optical performance, and concentration. The integration of tracers will primarily take place via printing inks on packaging items. Likewise, the tracers will be well washable from the packaging by a deinking process. Routes for tracer recycling are examined. In the first part of the project, until the packagings with integrated markers are available, **IRIS** will apply Sensor-Based-Specification (SBS) as an off-line characterisation approach for the current un-traced MLC to be delivered by **SUEZ,** a company operating processes for collection, sorting and recycling of waste packaging. These approaches will provide substantial information on the presence of different polymers and other constituents and its composites and will be used to suggest suitable recycling process parameters. This will also be applicable for the case of the non-marked post-industrial (pi) and post-consumer (pc) waste in the future and guide recovery of currently unused EoL packaging fractions. EoL packaging waste based on the novel food packaging concept of WP5 will be mixed into real packaging waste streams provided by **SUEZ** and further sorted by **POLY** with novel TBS sorting devices in order to separate specific plastic packaging fractions stream consisting of multi-layer food packaging only. **SUEZ,** specialized on collection and valorization of waste plastic packaging, will source packaging waste samples from their existing sorting and plastic recycling lines check market acceptance of food packaging with secondary raw materials.

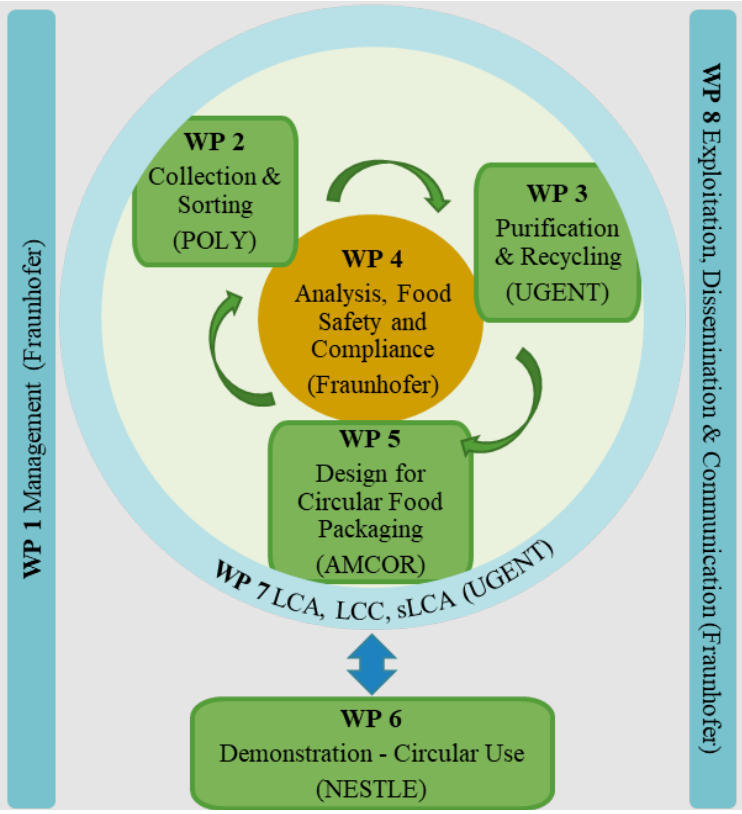

Fig 19. CIRCULAR FoodPack - Pert diagram (WP Leader)

In **WP 3** a "food-grade only" plastic waste stream separated in WP 2, and consisting of different mono- and multi-layer materials, will be subjected to innovative post-sorting processes incl. decoloration and delamination (**UGENT**) and deodourization (**KREYEN**). Downstream, **UM** and **Fraunhofer** will perform mechanical recycling (melt filtration and re-compounding) if applicable, and solvent-based physical recycling, if multi-material MLC are present or higher purification (CreaSolv®) is required. Tracer marked packaging items from TBS sorting are well defined and their recovery processes can be adjusted accordingly, with respect to no-marked items results of SBS will guide the adjustment of suitable recycling process parameters. Finally, combinations of pre-treatment and recycling processes will be tested in different set-ups to validate economic viability and the required removal of NIAS and IAS.

WP 4 will perform food safety assessments based on analytical and experimental migration testing and mathematical migration modelling at **Fraunhofer** with the recovered materials in WP 3 and selecting materials suitable for developments in WP 5. Materials not suitable for food packaging will be separated and tested for non-food flexible packaging.

In **WP 5 AMCOR, Fraunhofer** will focus on modern circular design ("Design4Circular food packaging") approaches to design a new flexible packaging structure based on a mono-material multi-layer (PE based) with PE PCRs, and on recyclable functional barrier systems (e.g. SiO_x). Printing of tracer-inks will be included. Deinking tests of WP 3 will result in proposals for new ink designs that are easier deinked and/or delaminated if needed (**UGENT, SW**). This allows production of cleaner recyclates ready for closed loop recycling.

WP 6 demonstrates: **(1)** Beneficial effects of advanced sorting, purification and recycling technologies of WP2, WP3 on PE PCR yield and purity qualities, **(2)** applicability of such PE PCRs in mono-material based flexible packagings, as designed in WP5 based on migration models of WP4, **(3)** beneficial effects of these new mono-material based laminates on the performance of sorting, purification & recycling processes, on resulting PCR quality and circular use. Demonstration activities will be organised in **3** Loops, starting from Month 13.

Loop 1 (Fig 20) will receive real flexible packaging waste of about 1000 kg, not containing tracer marked packaging yet (**SUEZ**). By NIR-separation of flexible packaging low in PE content, SUEZ will generate a fraction of flexibles with PE and other polymers, thus a multi-layer flexible fraction without differentiation of food and non-food packaging items. These MLC films will be characterised by an off-line SBS tool (**IRIS**). The application here would be home & personal care (**HPC**) products, since a PCR without definition of food grade cannot be used for food packaging.

In ***Loop 2*** (Fig 21), SUEZ will collect the input waste of **NESTLE** customers in a Business-to-Business (B2B) material recovery approach (~ 200 kg in total) that returns empty to NESTLE food packs to a NESTLE factory after use, guaranteeing 99% content of food packages. This material is converted into food grade PE PCR and considered for new food packaging.

Loop 3 (Fig 22) will demonstrate food pack sortings by TBS (**POLY**); e.g. 350 kg of tracer marked food packs sorted out from a total of 1400 kg packaging waste (1000 kg is real waste, 400 kg is tracer marked). The sorted waste will enter into recycling processes, in each **Cycle**. PE PCRs are applicable in laminates, if they comply with the envisaged maximum residual NIAS/IAS levels (WP4). **NESTLE** will test-wise apply the new material in food packaging systems and demonstrate technical feasibility for sensitive dry food. Packed items will be used for shelf life, health safety studies at **Fraunhofer**, **NESTLE**. ***Loop 3*** will undergo two **Cycles**; **Cycle 2** with the food packagings of novel mono-material laminates: **NESTLE** products (with tracers) will undergo TBS, recycling, PCR characterisation, re-use: **Circular use in food packaging will be demonstrated**.

WP 7 will address life cycle assessment (LCA), and costing (LCC), including aspects such as recyclate quality, market size and balance between food loss and packaging design. The assessment of new technologies in function of a sustainable development as exemplified in EU policies will be performed. Detailed assessment of environmental (full LCA) aspects for food packaging recyclates will be initiated, following EU guidelines for circular economy assessment and consumer safety aspects for development of sustainable and safe flexible packaging, measuring contribution to circular economy will be included by **UGENT, ECOZEPT, IRIS, NTUA**.

WP 8 will cover innovation management, exploitation, economic assessment (full cost of ownership, business plans), communication to promote public awareness of the circular economy concept and focus on interaction with wider stakeholders and EAB (**Fraunhofer**, **BayFOR**). It will care about dissemination to raise awareness from early on to maximize wide uptake of results. Based on desk research and multi-stage interviews with packaging experts and consumers, market research and a SWOT analsysis will be performed by **ECOZEPT,** following by exploitation plans developed using the Business Model Canvas by **NTUA** to identify feasible business opportunities and marketing strategies to ensure a market acceptance of the CIRCULAR FoodPack technologies.

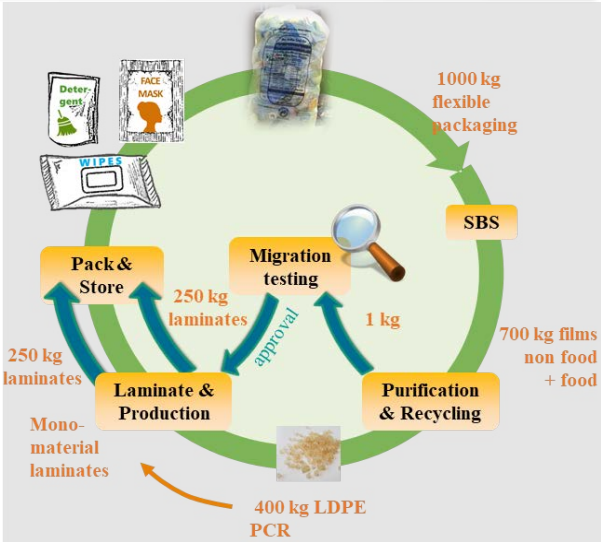

*Fig 20. Loop 1 – Circular Use in **HPC***

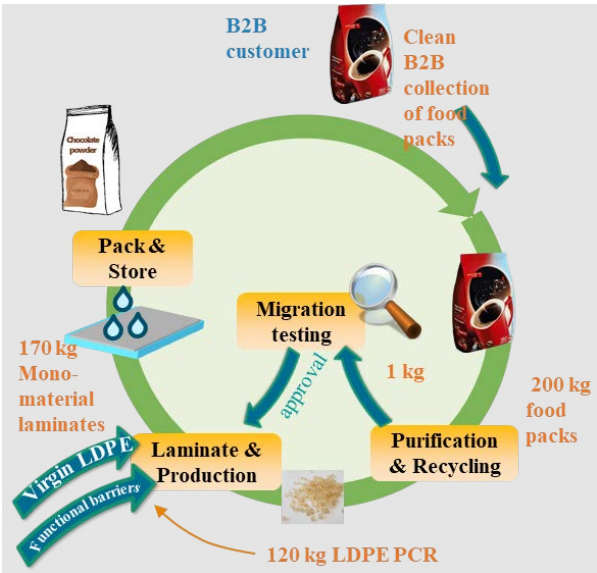

*Fig 21. Loop 2 - food packaging, **B2B collection***

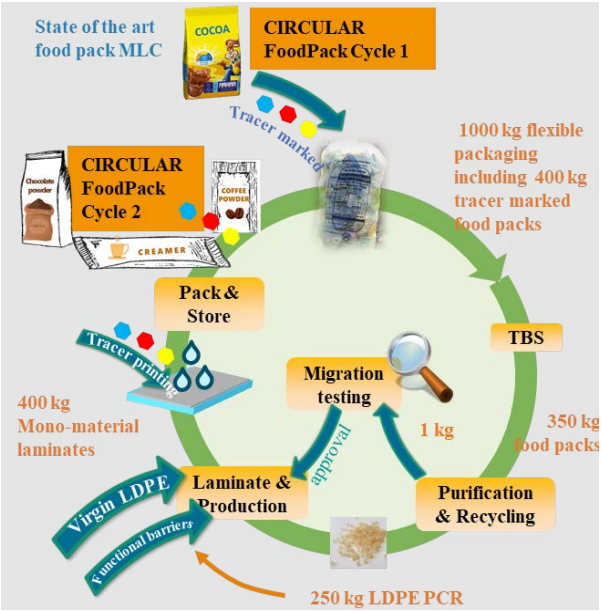

*Fig 22. Loop 3 - **CIRCULAR FoodPack** technologies*

Timing of the different work Packages and main tasks - Gantt chart

		WP Leader	Task participants	Year 1												Year 2												Year 3												Year 4					
				1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42
WP1	**Project Management and Coordination**	**Fraunhofer**																																											
T1.1	Implementation of efficient management and support structures	BayFOR	Fraunhofer	D																																									
T1.2	Day-to-day project management and coordination	Fraunhofer	All			D																																							
T1.3	Reporting, financial management & controlling	Fraunhofer	All																																										
T1.4	Monitoring & ensuring progress as well as risk management through scientific coordination	Fraunhofer	All																																										
T1.5	Data Management	Fraunhofer	All						D																																			D	
WP2	**Collection & Sorting**	**POLY**																																											
T2.1	Analysis of SOA ML food packaging in EU waste, collection, sorting, recycling, mat. sourcing	SUEZ	POLY, IRIS									D	D																																
T2.2	TBS tracer development	KIT	POLY												D																														
T2.3	Optimisaton of TBS tracers and their integration in ML food packs	POLY	KIT, AMCOR															D																											
T2.4	Development of TBS sorting machine for conventional NIR sorters and TBS trials	POLY	SUEZ, KIT																D		D																								
T2.5	Design, optimisation and building of HSI monitoring solution for ML identification	IRIS	POLY, SUEZ, KIT																								D						D												
WP3	**Purification and Recycling of Flexible Packaging**	**UGENT**																																											
T3.1	Characterisation and qualification of waste plastics	Fraunhofer	AMCOR-G, UGENT, KREYEN, SIEGWERK, SUEZ												D																														
T3.2	Washing and deodorization	KREYEN	UGENT, Fraunhofer															D																											
T3.3	Delamination and deinking	UGENT	SIEGWERK, AMCOR-G, Fraunhofer																		D																								
T3.4	Dissolution based purification: CreaSolv® process	Fraunhofer	UGENT, KREYEN																								D																		
T3.5	Mechnical recycling tests and application development	UM	Fraunhofer, SUEZ																														D						D						
T3.6	Process chain design	UGENT	Fraunhofer, KREYEN																																								D		
T3.7	Design for sorting and recycling guidelines	UM	Fraunhofer, SIEGWERK, KREYEN, AMCOR-G																																									D	
WP4	**Analysis, Safety, Compliance Evaluation of Polyethylene Recyclates and Packaging**	**Fraunhofer**																																											
T4.1	Regulatory assessment and compliance evaluation of the used raw materials	Fraunhofer	NESTLE, AMCOR, POLY, KREYEN, SIEGWERK								D																																		
T4.2	Analytical characterization and monitoring of recycling process quality and recycling products	Fraunhofer	NESTLE																																				D						
T4.3	Challenge test for demonstrating cleaning efficiency	Fraunhofer	WP3-participants																																				D						
T4.4	Development of analytical databased tools to demonstrate safety	NESTLE	Fraunhofer																																								D		
T4.5	Design and validation of functional barrier concepts for the use of PCR in food packaging	Fraunhofer	NESTLE, AMCOR																		D																							D	
WP5	**Design for Circular Food Packaging**	**AMCOR-K**																																											
T5.1	Definition of specifications for the targeted products and a packaging concept	NESTLE	AMCOR, Fraunhofer, UM												D																														
T5.2	Functional barrier layer combinations – Assessments with model contaminants	Fraunhofer	AMCOR, SIEGWERK										D																																
T5.3	Film extrusion of PE-PCRs coming from WP3 and characterisation	AMCOR-G	Fraunhofer, UM																		D																								
T5.4	SiO_x deposition and deinkable printing as functional barriers on PE-PCR films	AMCOR-K	AMCOR-G, Fraunhofer, SIEGWERK																								D																		
T5.5	Printing of registered tracers to enable identification and managements after use	SIEGWERK	POLY, AMCOR, KIT, Fraunhofer																																				D						
T5.6	Machinability of novel tracer prtinted mono-material laminates with PE PCRs during packing	AMCOR-G	NESTLE, AMCOR-K, Fraunhofer																																										
WP6	**Demonstration - Circular Use**	**Fraunhofer**																																											
T6.1	Eco-design of prototypes in view of project technological developments	AMCOR	WP6-participants																														D												
T6.2	Loop 1: PE PCRs from real pc-flexible waste, circular use in Home Personal Care (HPC)	SUEZ	WP6-participants																																							D			
T6.3	Loop 2: PE PCRs from B2B collected food packs, circular use in flexible food packaging	NESTLE	WP6-participants																																				D						
T6.4	Loop 3: PE PCRs from TBS-Sorted Waste, circular use in flexible food packaging	POLY	WP6-participants																																									D	
T6.5	Implementation by packing and lifetime tests, evaluation and guidelines for industrial take-up	NESTLE	AMCOR, Fraunhofer, UGENT, UM, ECOZEPT																																									D	
WP7	**Comprehensive Sustainability Assessment and Decision-Support Tools**	**UGENT**																																											
T7.1	Scenario definition and process flow schemes	UGENT	WP7-participants																		D																		D						
T7.2	Mass and energy balances	UGENT	WP7-participants																																										
T7.3	Comprehensive sustainability assessment: framework development and application	UGENT	WP7-participants																								D																	D	
T7.4	Quality Assurance System (QAS) database and Decision Support System (DSS) tool	IRIS	WP7-participants																														D											D	
WP8	**Market Analysis, Exploitation, Dissemination and Communication**	**NTUA**																																											
T8.1	Market analysis and identification of stakeholders' preferences, acceptances and expectations	ECOZEPT	NTUA, all																		D																								
T8.2	Exploitation of Results, Innovation Management and Technology Transfer	NTUA	ECOZEPT, all																																				DD					DD	
T8.3	Dissemination and communication and development of CIRCULAR FoodPack's brand identity	BayFOR	All			D	D																																					D	
T8.4	Continuous dissemination and interaction with the scientific and industrial community	BayFOR	Fraunhofer, all																																										

3.2 Management structure, milestones and procedures

3.2.1. Organisational structure

In order to ensure a smooth decision-making process and collaboration between the 14 partners, the organizational structures are kept as straightforward as possible. It is of high importance not to neglect the necessary management elements to avoid conflicts or uncertainty about responsibilities and obligations. Figure 23 shows the structure of the project organization. The Coordinator (CO) will report directly to the EC Project Officer, supported by the project manager (PM). The Project Steering Committee (PSC) will interact with the Coordinator who maintains an overview of the project. The WP leaders (WPLs) will oversee the work packages and report progress to Coordinator and PSC. The main goals of the project coordination are: (i) organize the information flow in timely manner and accurately inside the consortium to meet project objectives; (ii) take the appropriate decisions to reach the project objectives; (iii) provide the EU with reports and organize reviews for the suitable tracking of the project progress.

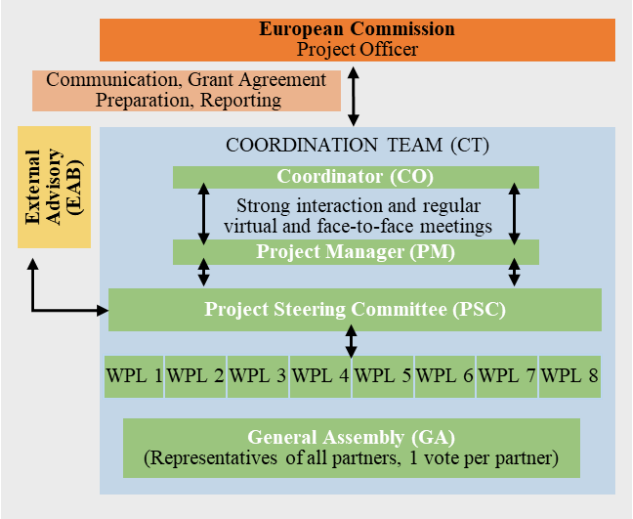

Fig 23. CIRCULAR FoodPack Management structure

Coordination Team (CT) with the Coordinator (CO) and the Project Manager (PM)

The project will be coordinated by Fraunhofer that has participated in various European projects in FP6, FP7 and Horizon2020. On behalf of Fraunhofer, Dr. Esra Kucukpinar will act as the CIRCULAR FoodPack coordinator responsible for the overall coordination of activities. Dr. Kucukpinar has gained experience in the management of collaborative research projects through more than 20 national and European collaborative research activities and projects, whereas she acted and still acts as the technical manager in six projects. With respect to the recyclable packaging material developments, Dr. Kucukpinar and her team successfully designed and developed recyclable mono-material based multi-layered laminates for flexible drink pouches. She has the experience and the competence on how to design and develop flexible packaging materials for various types of packaging applications including sensitive food products, vacuum insulation panels, flexible photovoltaics and displays. Dr. Kucukpinar will be supported through Fraunhofer colleagues, by Dr. Martin Schlummer for the topic of the purification and recycling of flexible packaging by the CreaSolv® technology and by Dr. Diana Kemmer for the area compliance for food packaging. Dr. Kucukpinar has access to a strong backbone of administration personnel dedicated to managing European funded projects within the Fraunhofer headquarters. As Europe's leading organization for applied research and as project partner in many EU-projects, Fraunhofer fulfils these principles. There is thus extensive expertise in the management of EU projects and the PSC members are also experienced with project management of European as well as national projects.

The CO will rely on the WPLs for monitoring the technical implementation of the project. He can take urgent low-level decisions, if necessary and after discussion with the WPLs. He can propose solutions and contingency plans to the partners in case of delay, discrepancies or when a technical choice has to be made. Furthermore, the CO will be assisted in all administrative and financial management tasks by the PM located at partner BayFOR, by WPLs and by the General Assembly, which will be empowered to make high-level decisions on every aspect of the project. **Melanie Schulte**, a well-experienced EU project manager with a scientific degree in forest science and resource management is going to fulfil the role of the PM. Both the CO and the PM together build a highly skilled Coordination Team. The organizational structure and the rights and duties of the operational bodies responsible for the decision-making in the project will be explained and described in detail in the Consortium Agreement.

Project Steering Committee (PSC)

A Project Steering Committee, led by the CO, will be installed as a supervisory body supporting the project's normal operation. The PSC will supervise crucial decisions related to all relevant administrative management and scientific matters and will be responsible of the operation of the technical work programmes. More specifically, PSC has a double role:

- It is the governing body that prepares the decisions for the General Assembly. It is responsible for proposing redistributions of funds if needed and for the management of the joint funds. Other tasks include settling potential conflicts among partners.
- It will assess the work within the WPs and ensure alignment of the individual WPs with the general objectives of CIRCULAR FoodPack. The PSC is further responsible for monitoring the budget and the overall management of the project.

The PSC shall meet every six months alongside the General Assembly meeting, unless interests of the project require intermediate meetings or ad hoc upon written request by any PSC member. Its members can vary as the project needs, but will consist at least of CO and PM, Quality Manager (QM), Research and Technology Manager (RTM), Exploitation and Innovation Manager (EIM), and Dissemination and Communication Manager (DCM):

Quality Manager (QM): The QM **(Prof. Steven Demeester**, UGENT) will be in charge of disseminating the quality procedures, encouraging their use and monitoring with adequate metrics. This will include setting up a qualitative review process to ensure proper quality of all deliverables and contribute to reporting. Steven Demeester wide experience in the field of Quality Management will guarantee high-level project deliverables through establishing quality assurance. All deliverables generated by the persons in charge (see list of deliverables) will be checked and controlled in this sequence: WP Leader → PM → CO and QM → submission to EC. The QM will conduct the internal quality assessment of milestones and deliverables that are based on the quantified objectives detailed in section 1.1 and within the WP descriptions and give status updates within the PSC.
Research and Technology Manager (RTM): The RTM (**Pieter Cools**, AMCOR-G) will have an advisory role on the R&D decisions within CircularFoodPack and will ensure that project milestones and goals are achieved. The responsibilities of the RTM include: determining, implementing and monitoring research and development strategies, policies and plans. This role is appointed to Pieter Cools who has extensive R&D experience both as a coordinator of EU projects ad partner in European and national initiatives, dealing with high impact technological advancements.
Exploitation and Innovation Manager (EIM): The EIM (**Prof. Aggelos Tsakanikas**, NTUA) will be in charge of the continuous assessment of the market potential of the developed know-how. The determination of actual market trends is imperative to realize the full commercial potential of project results. The EIM will lead the preparations of exploitation strategies in cooperation with WP8 members. The EIM will further identify potential users outside the consortium and act as the main contact point for business related issues while reporting these activities to the PSC and General Assembly. The EIM will capture and manage research results generated by the project in order to exploit them in the most effective manner.
Dissemination and Communication Manager (DCM): During the project's duration, several dissemination and communication activities will be undertaken (cf. chapter 2.2. and WP8) to successfully spread important information and project results. **Melanie Schulte** (BayFOR) will act as the DCM and has the responsibility to keep track of all communication and dissemination activities within CIRCULAR FoodPack. The DCM, in close contact with the CO will ensure the dissemination of project results to relevant stakeholders, governmental bodies, research communities and others. The DCM will regularly present all related activities in PSC and GA meetings. Detailed communication and approval procedures will be set out in the Consortium Agreement.

The General Assembly (GA):
The GA is composed of one representative from each partner and chaired by the CO. The GA will represent the highest level of decision-making within the project and will decide the overall project strategy, including contractual matters and exploitation and will also supervise the achievement of the milestones. The GA will convene biannually (at least once per year in a face-to-face meeting). Additional meetings via digital conference tools will be held on any extraordinary occasions requiring a decision by the GA. The main mission and tasks of the GA are: Assessment of work carried out; decisions about project strategy and technical issues; analysing technical conflicts between partners and proposing agreements; defaulting party management; studying the evaluation reports on all activities of all partners; deciding by confidential vote about allocation of finances, adding new members to the consortium, reviewing or amending the EU contract, change or exchange of WPs.

Work Package Leaders (WPLs): To facilitate the organization and management, the project is structured in 8 WPs. Each WP is carried out by a varying number of involved parties and will be headed and coordinated by an experienced principal investigator as WPL to manage and ensure success of the WP. The WPL supervises and adjusts the process flow and works closely with the CO. The WPL has an integrating function and is responsible for engaging and communicating with all participants in the WP. (Virtual) WP meetings will be arranged on a regular basis. The WPL will report on the progress in relation to the deliverables and milestones achieved and delays. To ensure and document that this is being achieved, the WPL will biannually send an internal Interim Report to the PSC and report to the GA. In the unlikely event of disagreements within a working group that cannot be resolved simply by the WPL, the issue should be reported to the CO, who will ascertain the facts and present them to the GA, which will take a decision for the benefit of CIRCULAR FoodPack as a whole. WPLs will also be the main contact point for the EAB.

External Advisory Board (EAB): To ensure quality and societal relevance beyond the consortium, CIRCULAR FoodPack will establish an EAB until the project starts. It consists of international experts of industrial players (Table 14), who will participate as non-funded partners to complement and enhance the expertise of the consortium. These large well-known enterprises not only function as advisors but also as stakeholders, which can commercialize the CIRCULAR FoodPack ideas globally. With such an involvement, it will be possible to eliminate contaminants from

"dirty" flexible food packaging waste and lead post-consumer polyolefin recyclates successfully into a flexible packaging laminate also for food packing, leading to circular economy. The suggested experts have been preliminarily contacted and informed about the project concept and objectives. All stated members of the EAB agreed to be part of this and showed their large interest by an expression of interest letter (attached as an Annex). During project, the consortium will appoint additional members, whenever the results or missing expertise of the consortium makes it necessary. The EAB ensures a high standard of the project and will interact regularly with the consortium and, where and when relevant, the EAB or selected members will be invited to GA meetings or technical meetings. The EAB has hereby no voting rights.

Table 14: CircularFoodPack confirmed External Advisory Board Members

Name	Value for CIRCULAR FoodPack
Steinert	A Leading developer and manufacturer of NIR and other sorting machinery equipment with international sales & service, located in Germany
UBESOL SL	UBESOL in Spain, will consult non-food packaging. They will assess the processability of the laminates for wet-wipes in flow-wraps, cosmetics in sachets and detergents' packaging.
CEFLEX	CEFLEX represents the entire value chain of flexible packaging, will act as multiplier to get access to different companies of the value chain, to receive feedback on defined questions. AMCOR, NESTLE, SUEZ, SIEG-WERK are members.
ISWA	A non-profit association in Netherlands, consists of international waste management experts and relevant stakeholders. It organizes international networking events, conducts educational workshops on waste management, engages in developing economies to establish sustainable waste management practices.
FPE	Flexible Packaging Europe Assocication in EU supports in identifying the EU-wide needs in terms of flexible packaging recycling and disseminate the project results among their members as potential users of recyclates
IVLV	IVLV is an Association for Food Technology and Packaging, in Germany. It initiates, supports application-oriented projects. It can give access to food packers to access their aspirations and concerns in terms of recyclability of packaging materials and use of recycled materials, respectively
ZLV	ZLV is an association focusing to added value along the entire value chain for packed food stuff. Give access to food packer to access their aspirations and concerns in terms of recyclability of packaging materials and use of recycled materials, respectively.
PRE	Plastics Recyclers Europe represents the voice of EU plastics recyclers, so will promotes the use of quality plastic recyclates and give concrete advice on developing innovative, recyclable products and packaging.
JRC	Joint Research Centre, EU science and knowledge service providing independet scientific research-based advice and support to EFSA and EU policy, committed to be in EAB, but is formally not able to sign the LOI.

3.2.2. Decision making and conflict management

The **General Assembly** (GA) is the highest decision-making body within CIRCULAR FoodPack. It takes *high-level decisions* to achieve the goals of the project, wherefore it is crucial to achieve consensus among the project partners. All major decision making procedures will be refined and detailed in the Consortium Agreement such as the interaction between partners, distribution of EC funds and voting rights. Before each GA meeting, partners will have to prepare a short report to be uploaded on the collaboration platform. These reports will include the main results within each WP, difficulties encountered, dissemination and communication activities, exploitation routes, use of resources and deviations from Annex 1 and 2 of the Grant Agreement. Such issues will be discussed by the GA and decisions taken accordingly.

The **PSC** can take *medium-level decisions* with the agreement of GA. The GA itself is responsible for all important management issues and thus takes the high-level decisions. This overall decision-making structure is supported by well-defined deliverables and clear milestones, which will be closely tracked during the proposal and serve as the basis for decision points (see Table 3.2a). Besides the GA meetings, there will also be (online) **technical meetings** attended by a limited number of partners and the respective **WPLs** to address particular WPs. Additionally, contact among partners will be maintained by e-mail and telephone, with regular written progress reports circulated to all partners. Such technical meetings will be strongly encouraged to promote technical exchange and multidisciplinary cross training among partners. This will allow decision-making on a practical day-to-day level, with *low-level decisions* of minor scope taken by the WPLs respectively.

The CO assisted by the PSC will propose corrective actions if necessary, following consultation with GA members to allow for proper **conflict management**. If for any reason, the CO and PSC cannot resolve the conflict, the GA will conclude on the conflict by majority vote (in case of draw, the CO has the decisive vote). For serious and urgent matters, an extraordinary GA meeting can be called. Clear strategies will be laid out in the **Consortium Agreement (CA).** The CA will be signed prior to the Grant Agreement and will define the roles and obligations of each partner, rules for partners joining or leaving the consortium, internal organization of the consortium, its governance structure and voting mechanisms, criteria for evaluation of the activities (referring to the quality assurance procedures), provisions for the settlement of disputes and exploitation.

3.2.3. Innovation management

The implementation of CIRCULAR FoodPack follows a targeted innovation management. It is pulled by end-user needs with clearly defined targets and corresponding methodology to enable a circular economy of flexible food packaging. The identification of the market opportunity has already started that exists for reprocessing and upgrading of PE PCRs before they enter a new production cycle of packaging.

Innovation management includes the translation of the existing knowledge into CIRCULAR FoodPack concept with the support of a preliminary market analysis in WP8 performed for the related market sectors. These sectors will also be represented by the relevant EAB (Table 14). Within WP2 -6, the most promising results and technologies will be defined according to their transferability in industrial scenarios, their final cost for industrial take-up and exploitation (investment costs, operational costs) and their LCSA. The project management approach proposed provides mechanisms to identify and resolve potential risks. Risk mitigation measures (cf section 3.2.4) have been and will be selected according to their impact on the exploitation of the project results.

Several innovation management tools and processes are planned, e.g. fast innovation cycles during specification, end-user involvement during validation phase through the end-user NESTLE, customers of AMCOR, and partners of EAB, periodic IPR assessment, result-focused project management(incl. phasing of project – laboratory & industrial, focus on manufacturing). CIRCULAR FoodPack will actively reach out to recyclers and standardization bodies (e.g. the reuse of PE PCRs in flexible food packaging), to adopt and share best practices for higher-TRL technology development and dissemination. Two specific aspects of the innovation management we want to highlight are (i) introducing a system innovation and (ii) following a market pull and technology push.

By appointing a dedicated Research and Technology Manager (RTM), Exploitation and Innovation Manager (EIM), it is ensured that the innovations are detected and handled in time and in accordance with the Consortium Agreement.

3.2.4 Overview of risks, their potential impact and proposed contingency

None of the foreseen risks are within the highest critical region for the project. As will be further defined in the CA, in case of a significant unexpected risk requiring a quick decision, an emergency/ extraordinary meeting of the relevant WP will be called by its responsible WPL with the CO. Following the decision procedures of chapter 3.2.2, low-level decisions can be taken by WPL in accordance with CO. Decisions of strategic importance will be the mandate of the GA together with the PSC.

3.3 Consortium as a whole

Due to the global network in trading of products, it is not possible to implement a circular economy by only a single country (region). For that reason CIRCULAR FoodPack sets up a European consortium of different organizations to link end users (NESTLE), material converters for flexible packaging (AMCOR & customers through the EAB), waste managers (SUEZ), sorting company (POLY), separation technologies (SIEGWERK), deodourization (KREYEN), technology provider (IRIS) with researchers for marker development (KIT), pre-treatment and recycling processes (Fraunhofer, UGENT, UM). Thus, the complete value chain of flexible packaging flow can be covered. All partners in this consortium have been selected in view of their individual expertise and complementarities. For CIRCULAR FoodPack waste manager (SUEZ) is providing the consortium with market data of waste generation and flexible packaging waste. The research institutes (Fraunhofer, UGENT, KIT, UM) develop the novel markers for sorting of food packaging waste from non-food and corresponding printing technologies (KIT), purification and recycling processes for MLC (UGENT, UM, Fraunhofer) and deinking and deodourization processes together with the industrial partners SIEGWERK and KREYEN, respectively. The PE based recyclates will undermine an analytical process for contamination and migration challenge tests, which will be performed at the accredited laboratories of the research institute (Fraunhofer) for the novel functional migration barrier developments at AMCOR-K and AMCOR-G. Material converter (AMCOR) have the facility to analyse the usability of PE PCR for food and HPC flexible packaging and give advices for the novel mono-material based laminate design and production with PE PCRs including functional barriers and the novel tracers in deinkables. Environmental aspects, e.g. risks for environment originated from PE PCR, LCA for PE PCR, social impacts, gender aspects and economic aspects (market analysis. business plans, marketing strategies) are screened through experts (UGENT, IRIS, NTUA and ECOZEPT), which have many years of experience in these fields. ECOZEPT as a private market research and marketing consulting agency will conduct extensive market analysis to iteratively integrate consumers' and experts' perceptions into the development of the CIRCULAR FoodPack products.Finally, promising results are disseminated and communicated to public and relevant organizations by non-profit-partner BayFOR. CIRCULAR FoodPack will be managed by a research institute (Fraunhofer), supported by BayFOR, which both plenty of practice in handling of EU funded projects. Our multinational cooperation is essential in order to reach the intended objectives.

The key personnel of Fraunhofer, IRIS, UGENT, SUEZ, BayFOR, ECOZEPT, UGENT, UM will be female (ca. 40%). Key decision-making positions within the project will be equally filled by men (RTM, EIM, QM) and women

(CO, PM, DCM). Gender balance in recruitment for the project will be overseen by CO to ensure a reasonable level of balance.

Industrial/commercial involvement in the project.

Industrial or commercial occurs through industrial participation: 5 out of 14 partners are SMEs and 5 are large industry partners. The industrial partners consist of flexible packaging material converters, a brand owner and end user, waste manager, one technology provider and consultants and they are essential for the success of the project. The waste manager (SUEZ) is collecting the contaminated flexible packaging from different available sorting sites in France or Belgium and ensure the safe handling of that. The waste will go to the sorting line advanced with the Tracer-Based-Sorting technology of POLY, and potentially in addition with Sensor-Based Specification technology of IRIS. These will guarantee the separation of flexible MLC food packaging waste from the non-food flexibles with the rich in PE content. Thus, the input flow is guaranteed, this input then goes into purification and recycling plants (CreaSolv®). Thermally assisted deodourization by KREYEN, deinking technology with the novel ink formulations of SIEGWERK will ensure the higher purification levels before the recycling processes. Flexible packaging producers (AMCOR-K and AMCOR-G) and the end user brand owner (NESTLE), which bring products on the European market, must adhere on EU guidelines, recommendations and Regulations by EFSA. The PE PCRs recovered from food packaging waste, and free from hazard contaminants, implemented in mono-material based laminates will be commercialized by these experts. They are establishing relationships to the customers/public. With product stewardships, in where the producer guarantees an environment friendly product, they win the customer's confidence. In this way, the image of PCRs in flexible packaging applications in public can be improved. Product stewardships are based on analytical determinations, in where the environmental risks and benefits from product and process are evaluated. Therefore, experts (ECOZEPT, IRIS,) compile a risk assessment and LCA as well as conducting market research to identify business plans and marketing strategies for the product.